Minimum Wage, Trade Policy, and Informality: Evidence from Colombia

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Abstract

Many developing countries have lowered trade barriers and raised real minimum wages while informality has remained widespread. This paper studies how the two policies interact and jointly determine informality and welfare. Colombia has one national minimum wage but large regional differences in pay, so it binds more tightly in some local labor markets than in others. Combining industry-level tariff cuts from the 2010–2011 reform with pre-reform differences in minimum-wage exposure, I find that a one-percentage-point larger tariff reduction raises informality by 0.8 percentage points where the minimum wage is more binding, but has no detectable effect where it is less binding. I then develop and estimate a spatial general-equilibrium model with firm heterogeneity, employer wage-setting power, and endogenous choices of formality and exporting to study the welfare effects of international trade and the minimum wage. In the estimated model, worker welfare is maximized at a minimum wage of about 56 percent of the current level. The optimal minimum wage relative to average earnings barely changes with trade policy.

Keywords: Minimum wage, trade liberalization, informality, monopsony, Colombia.
JEL Codes: F13, F16, J38, J42, O17.

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1 Introduction

Over the past decades, many developing countries have lowered trade barriers and raised real minimum wages, while informal employment has remained widespread. Trade liberalization expands firms' access to foreign markets and intensifies product-market competition. Minimum wages seek to raise earnings and redistribute income and, in imperfectly competitive labor markets, may also improve efficiency by limiting employers' wage-setting power. Separate empirical literatures provide evidence that trade reforms reshape formal and informal employment and that minimum wages affect informal employment and wages.[1] These findings differ across countries, and they are conditioned on the labor-market institutions in place. In much of Latin America, one such institution is a high minimum wage.

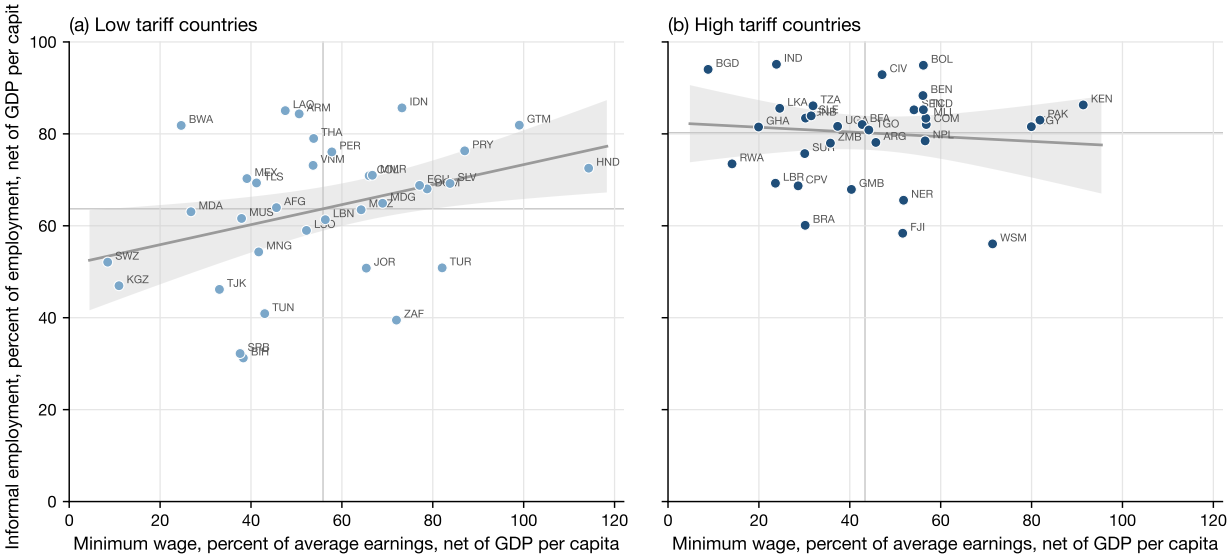
Yet research and policy debates typically treat the two policies as separate instruments. Trade policy and minimum-wage policy affect the same firms and workers through different economic margins. A tariff cut changes firms' sales and labor demand, while a binding minimum wage constrains their ability to adjust wages. By affecting labor costs, the minimum wage can also alter the competitiveness of domestic firms in product markets. When import competition reduces demand, firms that cannot adjust wages may instead cut formal employment or operate informally, while displaced workers may move into informal employment or self-employment. Conversely, changes in access to foreign markets can move firms into or out of the range in which the minimum wage binds. The minimum wage can therefore change who gains and loses from trade liberalization and how large the aggregate gains are. Across developing countries, the relationship between the minimum wage and informality differs with trade exposure. Figure 1 splits 67 low and middle income countries at their median applied tariff and plots informal employment against the minimum wage relative to average earnings, conditioning on income per capita. Among the more open economies, those with a higher minimum wage have substantially more informal employment, while among the more protected economies the relationship is absent.[2]

This paper asks how the effect of trade liberalization on informality depends on the minimum wage policy, and how the trade policy and minimum wage policy jointly determine aggregate outcomes, welfare, and optimal policies in developing countries. My empirical setting is Colombia, where a uniform and high national minimum wage binds much more

[1] On trade and informality, see Goldberg and Pavcnik (2003), McCaig and Pavcnik (2018), Dix-Carneiro and Kovak (2019), Ponczek and Ulyssea (2022), and Dix-Carneiro et al. (2026). On minimum wages and informality, see Mora and Muro (2017), Arango and Flórez (2021), Corcuera (2024), Derenoncourt et al. (2025), and Parente, Brotherhood, and Iachan (2026).

[2] The difference between the two slopes is significant at conventional levels conditional on income per capita, though not once region fixed effects are included. These are descriptive cross-country associations, not causal estimates.

Figure 1: Informal employment and the minimum wage, by trade exposure



Notes: Each point is one of 67 low and middle income countries, split at the median applied tariff of 9.3 percent, with 34 countries with tariffs of 1.3 to 9.3 percent in panel (a) and 33 countries with tariffs of 9.5 to 17.9 percent in panel (b). Both axes are net of log GDP per capita within the panel, with the panel mean added back, and the thin gray lines mark the panel means. The line is the least-squares fit and the shaded band its 95 percent confidence interval with heteroskedasticity-robust standard errors. The slope is 0.22 ($p = 0.010$) in panel (a) and -0.05 ($p = 0.603$) in panel (b). All variables are averages over the years available in 2015–2022. *Sources:* ILOSTAT (share of informal employment in total employment, SDG indicator 8.3.1; statutory monthly minimum wage relative to average monthly earnings of employees) and World Bank World Development Indicators (simple mean applied tariff, all products; GDP per capita at purchasing power parity, constant prices; income classification).

tightly in some local labor markets than in others because of large regional differences in pay. Informal employment accounts for roughly 70 percent of employment, with self-employment constituting an important component. Colombia's 2010–2011 trade reform also generated large and heterogeneous tariff cuts across industries. Because the minimum wage is set nationally but binds with different intensity across local labor markets, I can compare the same tariff cuts across labor markets within one country, where trade policy and other national institutions are the same. These features raise two questions. How did the effect of these tariff cuts on informality vary with how tightly the minimum wage bound in each local labor market? And is the Colombian minimum wage set at its optimal level, and how does that level change with trade policy?

I answer these questions in three steps. First, using the household survey, I relate changes in informality around the reform to industry-level tariff reductions and allow the response to vary with pre-reform minimum-wage exposure across regions and education groups. The tariff reductions were driven largely by pre-reform tariff levels as the reform rounded down the existing tariff schedule, rather than by contemporaneous labor-market conditions. I

then examine the entire wage distribution, following Cengiz et al. (2019), to trace where tariff exposure shifts employment. Comparing industries receiving larger and smaller tariff cuts before and after the reform reveals whether employment moves away from formal jobs at the minimum wage and toward informal work below it.

The estimates show that the relationship between tariff reductions and informality differs sharply with initial minimum-wage exposure. A one-percentage-point larger tariff reduction is associated with a 0.8-percentage-point increase in informality in labor markets where the minimum wage is more binding, while the corresponding estimate where it is less binding is statistically indistinguishable from zero. The wage-distribution analysis points to the margin of adjustment. In industries receiving larger tariff cuts, formal employment near the minimum wage declines and informal employment below the minimum wage rises, relative to industries receiving smaller cuts. The increase in informality is concentrated in self-employment rather than informal wage employment. Together, these patterns are consistent with workers remaining economically active but moving outside formal employment when formal wage adjustment is constrained. These estimates recover relative differences across labor markets. They do not determine whether the reform raised or lowered informality nationally.

Second, to interpret these patterns, I develop a tractable small-open-economy framework with heterogeneous firms, employer wage-setting power, endogenous formality, and export selection. In the absence of a minimum wage, firms pay workers a markdown relative to their marginal revenue product and employ too few workers. A binding minimum wage is irrelevant for high-productivity firms that would pay above it, raises wages and employment at intermediate-productivity firms by limiting employer wage-setting power, and reduces employment at low-productivity firms whose labor demand cannot support the mandated wage. Informal production gives firms an alternative to complying with the policy, while export selection gives productive firms access to foreign demand. Trade moves these regimes through two channels, the level of the minimum wage relative to market wages, which depends on how the minimum wage is adjusted, and firms' market access. When the minimum wage keeps pace with wages, liberalization shrinks the range of firms for which it raises employment from the top, as exporters outgrow it, and from the bottom if the weakest formal firms become informal. The framework yields three implications that match these patterns. Minimum-wage exposure can amplify the informality response to import competition, formal employment losses should be concentrated at the minimum wage, and employment should appear below it.

Third, I develop and estimate a spatial general-equilibrium model to evaluate aggregate welfare and policy. The reduced-form estimates do not reveal the reform's national conse-

quences or evaluate alternative minimum wages, and the theoretical framework abstracts from the spatial, sectoral, and distributional margins needed to quantify them. A spatial model is needed because the same national minimum wage binds differently across regions and worker groups, tariff cuts reach regions through their industrial composition, and local responses propagate across regions through domestic trade and prices. The quantitative model links 24 regions through domestic and international trade and introduces tradable and nontradable sectors, college and noncollege workers, and self-employment. Heterogeneous firms retain the choices of whether to export and operate formally (Melitz, 2003; Ulyssea, 2018). Workers choose among formal wage employment, informal wage employment, and self-employment, and housing prices and government transfers determine how nominal income changes translate into welfare. These features map the local responses documented above into national effects and quantify aggregate efficiency and redistribution across workers and places under alternative minimum-wage and tariff policies. I bring the model to Colombian household surveys, customs records, and an interregional input-output table. The simulated-method-of-moments estimation targets regional labor-market, firm, and trade moments in 2015. The differential response of informality to the 2010–2011 tariff reform is not among these targets, so comparing the model's response with the reduced-form evidence provides an untargeted test of its central mechanism.

Simulating the actual 2010–2011 tariff reductions with the estimated model, I find that national informality falls slightly, by 0.08 percentage points, a national effect the reduced-form design cannot identify. Within the tradable sector, the reform lowers informality by less among noncollege workers where the model's minimum-wage bite is higher, the direction of the reduced-form evidence. The relationship is much weaker than in the data, mainly because the model pools all tradable industries in a region into one sector and so cannot reproduce the comparison across industries within a region that identifies the reduced-form estimates.

I next ask whether the minimum wage changes what trade does to workers. Lowering international trade costs by 10 percent raises worker welfare by 0.97 percent without a minimum wage and by 0.92 percent at the current one. Informality matters more. Removing informal firms alone lowers the gain slightly, but removing self-employment as well raises it to 1.30 percent, because the cut lowers the earnings of self-employed workers who produce tradable goods relative to wage earnings. This is the opposite of Dix-Carneiro et al. (2026), who find that informality amplifies the gains from trade because trade moves resources from informal firms, which escape taxes and regulations, toward formal firms.

With the estimated model, I evaluate alternative levels of the national minimum wage, initially holding trade policy fixed. Aggregate worker welfare is maximized at a minimum wage of about 56 percent of the current Colombian minimum wage. At its current level,

the minimum wage lowers welfare by 0.65 percent in consumption-equivalent terms relative to an economy without a minimum wage, and moving to the optimum raises welfare by 0.80 percent. The welfare-maximizing minimum wage is positive because employers have wage-setting power, and redistribution toward noncollege workers sets its level. Relative to no minimum wage, the optimum benefits noncollege workers at the expense of college workers, so the aggregate welfare comparison depends on how gains and losses across the two groups are weighted, highlighting the role of redistribution (Dávila and Schaab, 2025). By contrast, reducing the minimum wage from its current level to the optimum benefits both groups, with larger gains for college workers. Re-solving under free trade, 2009 tariffs, and higher protection, with all general-equilibrium outcomes adjusting, the welfare-maximizing minimum wage is nearly invariant to the tariff regime when measured relative to average earnings, at 0.56 times the current level under free trade and 0.55 under high protection.

Related literature. This paper contributes to three strands of literature. First, it contributes to the literature on trade, informality, and labor-market adjustment in developing countries by showing that the same tariff reduction generates different labor-market adjustment depending on how tightly the national minimum wage binds. Existing studies show that trade reforms can reshape the allocation between formal and informal work, with the direction and margin of adjustment depending on the nature of the trade shock and the institutional setting (Goldberg and Pavcnik, 2003; Paz, 2014; McCaig and Pavcnik, 2018; Dix-Carneiro and Kovak, 2019; Ponczek and Ulyssea, 2022; Dix-Carneiro et al., 2026). They also document persistent and heterogeneous adjustment across regions and workers (Attanasio, Goldberg, and Pavcnik, 2004; Dix-Carneiro, 2014; Dix-Carneiro and Kovak, 2017). A related literature studies how international trade changes employers' wage-setting power and other labor-market distortions (Jha and Rodríguez-López, 2021; Pham, 2023; Felix, 2026). I identify minimum-wage exposure as an important determinant of this adjustment. Tariff reductions raise informality where the minimum wage binds more tightly but have no detectable effect where it binds less tightly. I find that this adjustment is mainly in self-employment, which in the estimated model also lowers workers' gains from trade.

Second, the paper contributes to research on minimum wages in imperfectly competitive labor markets and in economies with substantial informality by showing that trade can change the efficiency rationale for minimum-wage policy, although in the estimated model this effect is small. Theoretical and quantitative work examines how a minimum wage affects employer wage-setting power, redistribution, and employment rationing (Bhaskar and To, 1999; Manning, 2003; Flinn, 2006; Cahuc and Laroque, 2014; Lee and Saez, 2012; Berger, Herkenhoff, and Mongey, 2022, 2025; Ahlfeldt, Roth, and Seidel, 2023). Empirical studies

examine its effects on employment, reallocation, incidence, and the wage distribution (Cengiz et al., 2019; Harasztosi and Lindner, 2019; Derenoncourt and Montialoux, 2021; Dustmann et al., 2022; Engbom and Moser, 2022). A complementary literature on developing countries studies informal employment, workforce composition, and firms' formality choices, and how these margins shape wages and responses to labor regulation (Meghir, Narita, and Robin, 2015; Ulyssea, 2018; Mondragón-Vélez, Peña, and Wills, 2010; Haanwinckel and Soares, 2021; Arango and Flórez, 2021; Parente, Brotherhood, and Iachan, 2026; Derenoncourt et al., 2025). Related work documents labor-market power and the role of self-employment (Amodio and de Roux, 2024; Amodio, Medina, and Morlacco, 2025). I connect these literatures by showing that changes in product-market competition alter the range of firms for which the minimum wage limits wage-setting power without reducing employment. I then estimate an open-economy model in which the welfare-maximizing minimum wage reflects employer wage-setting power, informal adjustment, and redistribution across workers and regions.

Finally, the paper contributes to the literature on trade policy in economies with domestic distortions by quantifying how the minimum wage and trade policy interact in welfare terms. The classical second-best literature shows that the welfare effects of trade policy depend on distortions elsewhere in the economy (Bhagwati and Ramaswami, 1963; Johnson, 1965; Brecher, 1974). More recent work emphasizes labor-market frictions and informality in trade and development (Helpman, Itskhoki, and Redding, 2010; Dix-Carneiro, 2014; Atkin and Donaldson, 2022; Dix-Carneiro et al., 2026). Dix-Carneiro et al. (2026) incorporate an observed minimum wage into a nonspatial quantitative model of trade and informality but do not study its optimal level. In their model, informality amplifies the gains from trade. In mine, it lowers workers' gains, because much of it is tradable self-employment that competes with imports. Ruggieri (2019) studies how minimum wages and other labor-market institutions shape firm selection, job reallocation, unemployment, and the distribution of gains from trade. Both papers model labor markets through search frictions and wage bargaining, under which the minimum wage constrains downward wage adjustment. Neither framework incorporates heterogeneous worker types and therefore neither studies how minimum-wage and trade policies redistribute welfare across worker groups. I instead introduce employer wage-setting power in a spatial economy with heterogeneous worker types. This structure gives a moderate minimum wage a potential efficiency role and allows how tightly it binds, and its incidence, to vary across firms, workers, and regions. The estimated model allows me to vary both policies independently and evaluate their joint implications for aggregate welfare and redistribution.

The remainder of the paper is organized as follows. Section 2 describes the Colombian setting and data and presents the reduced-form evidence. Section 3 develops a theoretical

framework that interprets it. Section 4 develops the quantitative spatial general-equilibrium model. Section 5 estimates the model and presents the quantitative analysis and counterfactuals. Section 6 concludes.

2 Setting, Data, and Reduced-Form Evidence

This section asks whether the informality response to Colombia's 2010–2011 tariff reform depended on how tightly the minimum wage binds. The reform reduced tariffs by different amounts across industries, largely by compressing the inherited tariff schedule. Colombia sets a uniform national minimum wage, but wages differ widely across regions and education groups, so the same minimum wage binds more tightly in some local labor markets than in others. Combining these two sources of variation, I compare the informality response to a given tariff reduction across local labor markets with different pre-reform exposure to the minimum wage. I then ask whether formal employment losses are concentrated near the minimum wage and whether employment moves below it. Subsections 2.1 and 2.2 describe the reform, the data, and the exposure measure. Subsections 2.3–2.6 present the empirical strategy, the results, and their robustness.

2.1 The 2010–2011 Tariff Reform

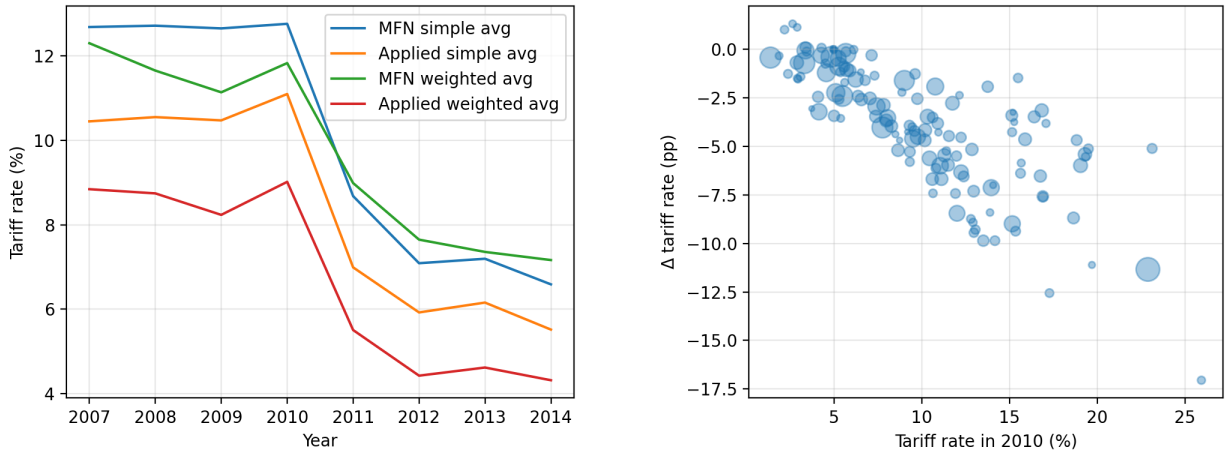
Colombia implemented a major unilateral tariff reform beginning in November 2010 through Decrees 4114 and 4115. The reform sought to reduce tariffs and simplify customs administration. It concentrated previously dispersed rates around 5, 10, and 15 percent and was implemented over a relatively narrow period from late 2010 through 2011.[3] Across the 29 tradable industries in the estimation sample, the average effectively applied tariff fell from 8.5 percent in 2009 to 5.6 percent in 2011, a reduction of 2.9 percentage points. Because the reform took effect between late 2010 and 2011, the empirical analysis treats 2011 onward as the post-reform period.

Tariff data come from UNCTAD-TRAINS at the HS6 product level and are mapped to four-digit ISIC Rev. 3 industries using standard concordances. Panel (a) of Figure 2 shows the timing of the reform, while Panel (b) shows that industries with higher tariffs before the reform generally received larger reductions. This pattern does not by itself establish strict exogeneity because initial protection may be related to persistent industry characteristics.

[3]Colombia's free trade agreements with the EFTA states and Canada entered into force in July and August 2011, and those with the United States and the European Union in May 2012 and August 2013. The effectively applied tariff includes the preferential rates in force in each year; the statutory MFN reduction used as an instrument in Appendix A.6 is unaffected by these agreements.

Reassuringly, the reductions show little systematic relationship with a range of pre-reform industry characteristics, including informality levels and trends.[4]

Figure 2: The 2010–2011 Colombian tariff reform



(a) Tradable-industry tariff rates, 2007–2014

(b) Tariff change vs. pre-reform level

Notes: Panel (a) plots simple and import-weighted MFN and AHS (effectively applied) tariff rates across four-digit ISIC tradable industries, 2007–2014. Panel (b) plots the AHS tariff change between 2010 and 2011 against the 2010 tariff level for 136 four-digit tradable industries. Dots are sized by 2010 imports. The baseline reduced-form treatment uses the 2009–2011 change, aggregated to the two-digit industry level. *Source:* WITS/UNCTAD TRAINS.

2.2 The Labor Market and the Minimum Wage

The worker-level data come from the Gran Encuesta Integrada de Hogares (GEIH), Colombia's monthly household labor-force survey, for 2007–2019. The survey reports employment status, earnings, hours, industry, education, location, and demographic characteristics. I harmonize geographic identifiers into 24 first-level administrative regions, which I refer to as regions throughout. I classify workers into three mutually exclusive employment categories. Formal wage workers are employees whose employers contribute to pension and health insurance on their behalf. Informal wage workers are employees without these employer contributions, and self-employed workers have no employer. The main outcome equals one for informal wage workers and the self-employed and zero for formal wage workers, following the convention in the Colombian literature (Goldberg and Pavcnik, 2003; Arango and Flórez, 2021). The estimation sample consists of employed workers aged 18–65 in 29 two-digit tradable industries spanning agriculture, mining, and manufacturing (ISIC Rev. 3

[4]Appendix A.2 reports the corresponding balance tests.

01–36).[5]

Colombia sets a single national minimum wage, with no variation across regions, industries, occupations, or firm sizes. It is negotiated annually among the government, employers, and unions, and a 1999 Constitutional Court ruling requires each annual adjustment to compensate at least for inflation.[6] The minimum wage is high relative to much of the Colombian wage distribution (Bell, 1997; Maloney and Núñez Mendez, 2003; Arango and Flórez, 2021). In the pre-reform tradable-sector sample, it equals 86 percent of the median formal wage, and 44 percent of formal salaried workers earn no more than 1.10 times the minimum wage. Panel (a) of Figure 3 shows the contrast between formal and informal earnings distributions. Formal earnings display a pronounced mass around the minimum wage, whereas informal earnings extend substantially below it.

I measure how tightly the minimum wage binds in a local labor market using *minimum-wage exposure*, the share of formal salaried workers in each region × education cell, across all sectors, earning no more than 1.10 times the contemporaneous hourly minimum wage, averaged over 2007–2009.[7] Constructing the measure before the tariff reform prevents it from responding mechanically to post-reform employment changes. I define local labor markets by region and education because workers can move across industries within a region, while wage distributions and minimum-wage bindingness differ sharply across skill groups. Keeping industry out of the exposure measure also separates minimum-wage exposure from the tariff treatment, which varies across industries. I restrict the measure to formal salaried workers because it is intended to capture how concentrated compliant jobs are near the statutory minimum wage. Including informal workers would conflate this concentration with the prevalence of informal employment. The 24 regions and four education groups produce 96 cells. The baseline analysis divides these cells at the median exposure.[8]

Table 1 documents substantial variation in labor-market outcomes across both education groups and regions. In the pre-reform tradable-sector sample, 72 percent of workers are informal and 43 percent are self-employed. Informality declines from 88 percent among workers with primary education or less to 29 percent among workers with tertiary education. Considerable geographic variation remains within education groups. Among workers who completed secondary education, for example, regional informality rates range from approximately 41 percent at the 10th percentile to 93 percent at the 90th percentile. Formal workers also earn more than informal workers at every education level.[9]

Minimum-wage exposure varies sharply across education groups and, within education

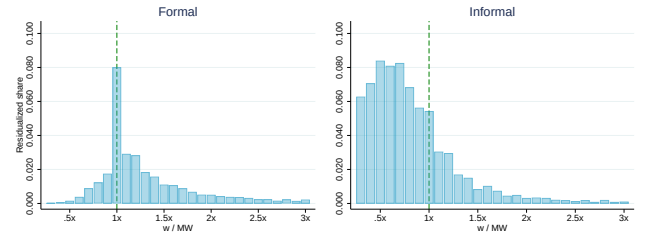
[5] Appendix A.1 details the sample construction.

[6] Appendix Figure A.1 plots the real minimum wage over 2007–2019.

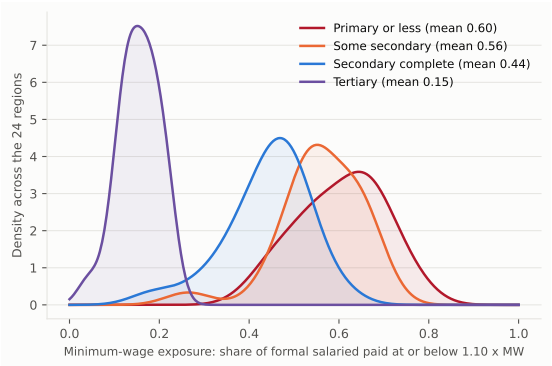
[7] See Appendix A.5.

[8] Appendix A.10 considers continuous exposure, alternative thresholds, and exposure measured at the

Figure 3: The minimum wage and local exposure to it before the tariff reform



(a) Earnings relative to the minimum wage



(b) Exposure across region × education cells

Notes: Panel (a) plots the residualized share of workers in bins of earnings relative to the contemporaneous minimum wage, separately for formal and informal workers. The dashed line marks the minimum wage. Panel (b) plots the distribution across the 24 regions, separately by education, of the pre-reform minimum-wage exposure measure. *Source:* GEIH.

Table 1: Labor-market statistics in tradable industries, pre-reform period

	All	Primary	Some sec.	Sec. complete	Tertiary
Panel A: Worker-level statistics					
Informality rate	0.718	0.876	0.722	0.524	0.286
Self-employment rate	0.426	0.521	0.414	0.306	0.190
Formal-employment share	0.282	0.124	0.278	0.476	0.714
Formal/informal wage ratio	2.19	1.90	1.69	1.68	1.82
Minimum-wage exposure	0.516	0.598	0.551	0.458	0.155
Panel B: 10th–90th percentile range across regions					
Informality rate	[0.37, 0.97]	[0.79, 0.99]	[0.61, 0.97]	[0.41, 0.93]	[0.25, 0.70]
Self-employment rate	[0.19, 0.72]	[0.31, 0.79]	[0.29, 0.68]	[0.19, 0.69]	[0.14, 0.50]
Minimum-wage exposure	[0.14, 0.66]	[0.47, 0.71]	[0.47, 0.65]	[0.33, 0.52]	[0.10, 0.21]

Notes: Both panels use workers aged 18–65 in the 29 two-digit tradable industries of the reduced-form analysis, 2007–2009, excluding January interviews. Panel A reports survey-weighted worker-level statistics. Panel B reports the unweighted 10th–90th percentile range across the 24 regions of survey-weighted region × education cell means, separately within each education group. The "All" column pools the 96 cells. Minimum-wage exposure is the measure used in the regressions, the share of formal salaried workers in the region × education cell, across all sectors, paid at or below 1.10 times the contemporaneous hourly minimum wage, averaged over 2007–2009. Panel A reports its average over workers. *Source:* GEIH.

groups, across regions. It averages 0.60 among cells for workers with primary education or less, compared with 0.15 among tertiary-education cells. Across regions, the 10th–90th percentile range is 0.47–0.71 for primary-education cells, 0.33–0.52 for completed-secondary

industry × region and industry × region × education levels.

[9]Appendix Table A.3 reports corresponding labor-market statistics for workers across all Colombian sectors.

cells, and 0.10–0.21 for tertiary-education cells. Panel (b) of Figure 3 displays these distributions. Although education accounts for most of the overall variation, meaningful regional variation remains within each education group. The baseline analysis uses both dimensions.[10]

2.3 Empirical Strategy

I use two complementary exercises for the reduced-form analysis. The first estimates how the informality response to industry-level tariff reductions varies with pre-reform minimum-wage exposure across local labor markets, to ask whether the response depends on how tightly the minimum wage binds. The second examines changes across the wage distribution, to ask whether formal employment losses are concentrated near the minimum wage and whether employment moves below it.

For the reduced-form analysis, I aggregate the tariff measures to the two-digit industry level to limit sampling noise within industry × region × education cells. I define the tariff reduction in industry j as $\Delta\tau_j \equiv \tau_{j,2009} - \tau_{j,2011}$, so that a positive value denotes liberalization. Across the 29 industries, the reductions have a standard deviation of 2.7 percentage points and range from −0.9 to 11.3 percentage points. Three industries experience small increases in their effectively applied tariffs. Industries with higher tariffs in 2009 generally received larger reductions, producing a correlation of 0.82 between the initial tariff and the subsequent reduction.

I begin with the static interaction specification

$$\begin{aligned}\text{Informality}_{ijdgt} = \; & \beta_L\big(\Delta\tau_j \times \text{Post}_t\big) + \beta_H\big(\Delta\tau_j \times \text{Post}_t \times \text{High}_{dg}\big) \\ & + \mathbf{X}_i'\boldsymbol{\Gamma}_t + \alpha_{jdg} + \delta_{dgt} + \varepsilon_{ijdgt},\end{aligned} \tag{1}$$

where Post_t equals one beginning in 2011. The coefficient β_L is the response to a one-percentage-point tariff reduction in low-exposure cells, $\beta_L + \beta_H$ is the corresponding response in high-exposure cells, and β_H is the difference between them.

The fixed effects determine the comparison underlying these coefficients. Industry × region × education fixed effects, α_{jdg}, absorb permanent differences across these cells. Region × education × year fixed effects, δ_{dgt}, absorb changes common to workers in the same local labor market, including local business cycles. The remaining comparison is across industries within the same region × education × year cell. Demographic controls are interacted with year, the specifications include calendar-month fixed effects, and all estimates use survey weights.

[10] Appendix A.7 examines the regional variation within education groups separately.

The dynamic specification replaces Post_t with year indicators, omits 2010, and allows both the low-exposure response and the high-minus-low differential to vary by year. Interpreting β_H causally requires that no other shocks correlated with tariff reductions differentially affected high-exposure labor markets in industries receiving larger tariff cuts after 2010. Shocks common across exposure levels within an industry do not bias this differential. Shocks that disproportionately affect high-exposure labor markets within that industry may. The pre-reform coefficients provide a diagnostic for this assumption but cannot rule out differential post-reform shocks. Standard errors are two-way clustered by industry and region. Because there are only 29 industries and 24 regions, I also report wild-cluster-bootstrap and permutation-based inference.[11]

2.4 Main Informality Result

Table 2 reports the estimates from equation (1). The relationship between tariff reductions and informality differs sharply according to how strongly the minimum wage initially binds. In high-exposure labor markets, a one-percentage-point larger tariff reduction is associated with a 0.79-percentage-point increase in informality. In low-exposure labor markets, the corresponding estimate is 0.09 percentage points and is statistically indistinguishable from zero. The high-minus-low differential is 0.71 percentage points, with a conventional two-way-clustered standard error of 0.27. At the average tariff reduction of 2.89 percentage points, this differential corresponds to approximately 2.0 percentage points of informality.

Figure 4 plots the event-study estimates. The low-exposure path remains close to zero throughout the sample, whereas the high-exposure path rises after 2010 and remains positive through 2019.[12]

The additional informality response is concentrated in self-employment rather than informal wage employment. The self-employment differential is 0.89 percentage points, while the informal-wage-employment differential is -0.19 and statistically indistinguishable from zero. This pattern is relevant for the spatial model of Section 4 because self-employment does not require another employer to create an informal job.[13]

[11] Appendix A.6 provides the dynamic specification, fixed-effect ladder, instrumented estimates, and inference details.

[12] Both paths fall by about 0.6 percentage points in 2018–2019 relative to 2011–2017, so the high-minus-low differential does not decline late in the sample. Its 2018–2019 average differs from its 2011–2017 average by -0.04 percentage points ($p = 0.857$), and the final two coefficients do not differ jointly from the earlier post-reform coefficients ($p = 0.921$).

[13] Appendix A.8 reports the full decomposition.

Table 2: Informality response to tariff reductions by minimum-wage exposure

	Informality
Tariff cut × Post (β_L)	0.088
	(0.145)
Tariff cut × Post × High exposure (β_H)	0.706**
	(0.273)
High-exposure response ($\beta_L + \beta_H$)	0.795***
	(0.250)
Demographic controls × year	Yes
Observations	689,898
Industry × region × education FE	Yes
Region × education × year FE	Yes

Notes: Worker-level regressions of the informality indicator on the applied 2009–2011 tariff cut of the worker's industry, interacted with Post and with High exposure. Post equals one in 2011–2019. High exposure identifies region × education cells above the median of the pre-reform minimum-wage-exposure distribution. Coefficients are percentage-point changes in informality associated with a one-percentage-point tariff reduction. The sample contains 29 tradable industries. All specifications include calendar-month fixed effects and use survey weights. Standard errors two-way clustered by 29 industries and 24 regions are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. Appendix A.6 reports the full static and event-study estimates and discusses alternative inference procedures.

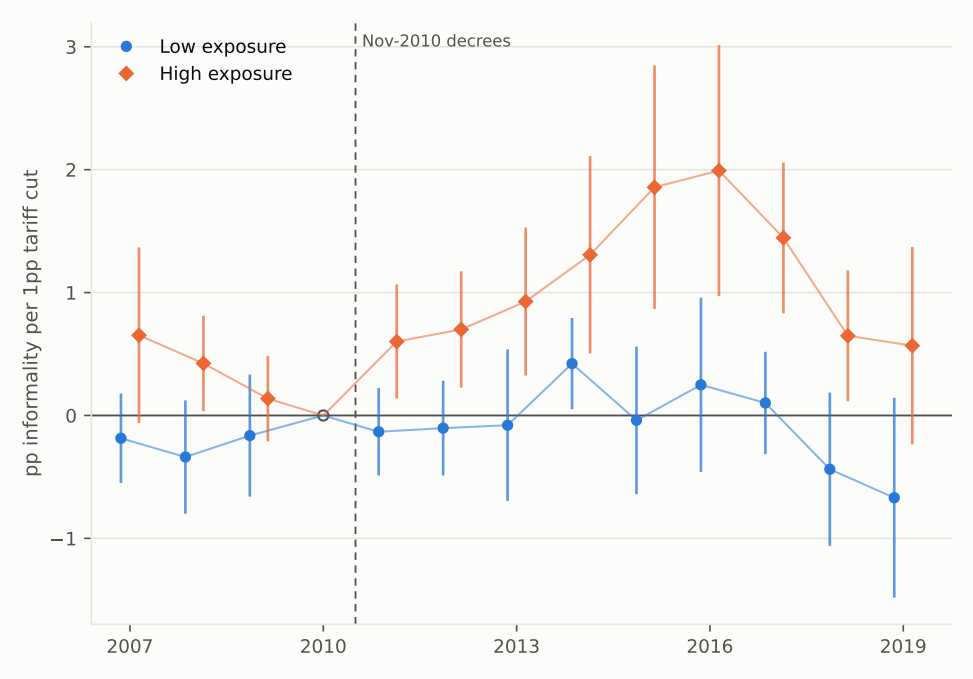
2.5 Adjustment Across the Wage Distribution

The informality result shows where the response is larger, but not which formal jobs disappear. If the minimum wage prevents formal wages from adjusting downward, formal employment losses should be concentrated at and just above the minimum wage, and informal employment, which is not bound by the minimum wage, should rise below it. To examine this, I follow Cengiz et al. (2019) and study how employment changes across the entire wage distribution. I divide workers into 28 bins of earnings, each 0.10 times the contemporaneous minimum wage wide, separately by formal and informal status. For each employment status s and earnings bin k, I estimate

$$Y_{ijdt}^{s,k} = \beta_k^s\big(\Delta\tau_j \times \text{Post}_t\big) + \mathbf{X}_i'\boldsymbol{\Gamma}_{t,k}^s + \alpha_{jd,k}^s + \delta_{dt,k}^s + \mu_{m,k}^s + \varepsilon_{ijdt}^{s,k}, \tag{2}$$

where $Y_{ijdt}^{s,k}$ equals one if worker i is employed in status s and earnings bin k, and zero otherwise. Industry × region fixed effects, $\alpha_{jd,k}^s$, absorb permanent differences across local industries, and region × year fixed effects, $\delta_{dt,k}^s$, absorb local labor-market shocks. The vector $\mathbf{X}_i$ contains the demographic controls of equation (1) and education, all interacted with year, and $\mu_{m,k}^s$ are calendar-month fixed effects. The specification uses survey weights and clusters standard errors by industry. The coefficient β_k^s is the post-reform change in the

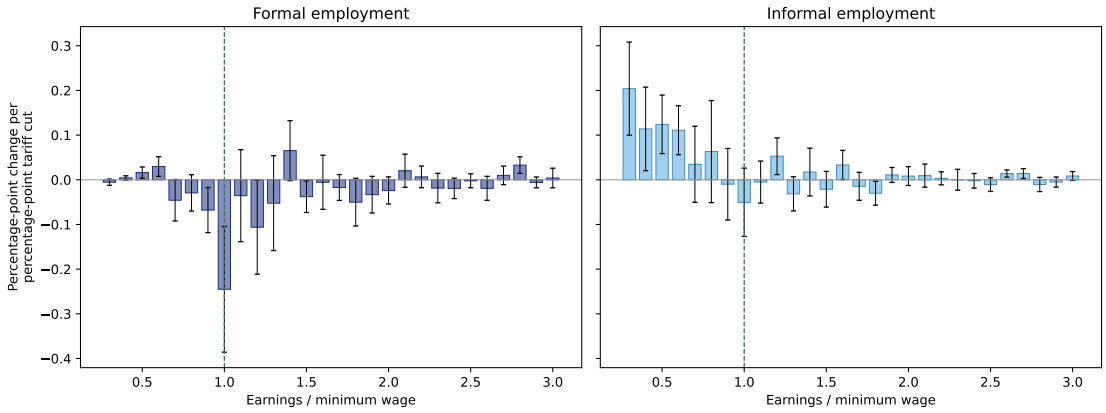
Figure 4: Informality response by minimum-wage exposure



Notes: The figure plots the low-exposure response and the implied high-exposure response to a one-percentage-point applied tariff reduction in each year, with 2010 omitted. Whiskers are 90 percent confidence intervals based on standard errors two-way clustered by industry and region.

share of workers in bin k and status s associated with a one-percentage-point larger tariff reduction.

Figure 5: Differential employment changes across the wage distribution



Notes: Each bar reports $\widehat{\beta}_k^s$ from equation (2). The dashed vertical line marks the minimum wage. Whiskers are 90 percent confidence intervals based on standard errors clustered by industry.

Figure 5 shows that the decline in formal employment is largest at the minimum wage. A one-percentage-point larger tariff reduction lowers the formal employment share by 0.25 per-

centage points in the bin containing the minimum wage, with smaller declines in the adjacent bins, while formal employment well above the minimum wage changes little. Informal employment below the minimum wage rises by 0.64 percentage points in total (wild-bootstrap $p = 0.02$), while informal employment above it is essentially unchanged. The response comes from workers with primary and lower-secondary education.[14] The concentration of formal employment losses around the minimum wage is consistent with limited formal-wage adjustment. Plant-level data from manufacturing point the same way. Plants paying close to the minimum wage cut production employment after larger tariff reductions while their pay is unchanged.[15]

2.6 Robustness and Limits of Interpretation

The high-minus-low differential remains positive when exposure is entered continuously or divided into terciles, when the threshold used to construct exposure is changed, when exposure is replaced by the minimum wage relative to median formal earnings, across specifications with industry × region × education fixed effects, and when the sample is restricted to manufacturing industries. Instrumenting the applied tariff reduction with the statutory reduction also produces a similar point estimate. Appendices A.6, A.10, and A.11 report these results.

The result is weaker under two alternative exposure measures. Including all salaried workers, rather than only formal salaried workers, in the denominator lowers the differential from 0.71 to 0.26 and makes it imprecise. Measuring exposure at the industry × region level also produces an estimate close to zero. I use the baseline measure because informal employment is itself an outcome and because the finer industry-level cells are measured from much smaller pre-reform samples.

Two limitations remain. First, 82 percent of the variation in region × education exposure lies between education groups. The estimates are positive in separate college and noncollege samples, but the design does not fully separate regional variation in minimum-wage bindingness from differences across education groups.[16] Second, the GEIH is a repeated cross-section. The estimates capture changes in employment composition across industries and labor markets. They cannot distinguish workers changing formality status from worker reallocation or changes in the composition of employment.

[14] See Appendix Table A.6.
[15] See Appendix A.12.
[16] Appendix A.7 reports specifications that use only the regional variation.

2.7 Taking Stock

The reduced-form evidence yields two conclusions. First, a one-percentage-point larger tariff reduction is associated with a 0.71-percentage-point larger increase in informality in high-exposure labor markets than in low-exposure labor markets, where the estimated response is close to zero. Second, larger tariff reductions lower formal employment, most at and around the minimum wage, and raise informal employment below it. The estimates therefore document that minimum-wage exposure is associated with different labor-market responses to trade liberalization. Because the comparisons are relative across industries and labor markets, they do not determine the reform's national effect on informality, its welfare consequences, or the optimal minimum wage. Section 3 develops a framework that explains why minimum-wage exposure shapes the response, and the spatial model in Section 4 addresses the aggregate and policy questions.

3 A Simple Theoretical Framework

This section develops a parsimonious framework for interpreting how a minimum wage changes firms' responses to trade liberalization. The framework combines heterogeneous-firm export selection (Melitz, 2003), employer wage-setting power (Manning, 2003) and its role in international trade (Jha and Rodríguez-López, 2021), and an endogenous formal–informal production margin (Ulyssea, 2018). Its minimum-wage mechanism follows the logic that a moderate minimum wage can counteract monopsonistic underemployment (Berger, Herkenhoff, and Mongey, 2025). It describes a single economy without regions or sectors and derives how firms' regimes respond to a change in trade costs, holding aggregate conditions fixed unless stated otherwise. Subsection 3.4 explains how these firm-level predictions map into the comparison across local labor markets in Section 2, and the spatial model of Section 4 adds the margins needed to aggregate firm responses into welfare, redistribution, and an optimal minimum wage. Here the aim is to isolate three mechanisms that explain the evidence in Section 2. First, employer wage-setting power can make a moderate minimum wage raise employment and employer-worker surplus at some firms. Second, product-market access determines which firms lie in this employment-improving region and which instead ration workers or operate informally. Third, trade affects market access asymmetrically. Import competition acts through domestic demand, while exporting gives sufficiently productive firms access to foreign demand.[17] Together, these mechanisms imply that the same

[17] A closed-economy model with an exogenous demand contraction could capture part of the import-competition mechanism, but it could not distinguish the loss of domestic market access from the gain in foreign market access, generate selection into exporting, or explain why the same trade shock moves exporters

trade shock changes informality differently depending on how tightly the minimum wage binds. Three ingredients deliver these mechanisms. Firm heterogeneity places firms in different minimum-wage, formality, and export regimes. Employer wage-setting power creates underemployment and hence a possible efficiency rationale for a positive minimum wage. Informal production gives firms an alternative when formal regulation becomes costly.

3.1 Environment

The economy is a one-sector small open economy. A representative household consumes a CES final good and supplies labor to differentiated jobs. Firms take as given the household's aggregate labor supply, N_w, and the wage index, $\widetilde{W}_w$, which summarizes the wages offered by all other jobs. Labor supplied to a job paying wage w is[18]

$$n^S(w) = N_w \left(\frac{w}{\widetilde{W}_w} \right)^{\eta}, \qquad \eta > 0. \tag{3}$$

The elasticity η measures how easily workers substitute across jobs. Because η is finite, each firm faces an upward-sloping labor-supply curve. Hiring another worker requires raising the wage paid to all of its workers. This is the source of employer wage-setting power.[19]

Final demand has elasticity of substitution $\sigma > 1$. Let E_D denote domestic real absorption and use the final-good price index as numeraire. A firm's market access, A, is the demand shifter it faces. With productivity z and employment n, the firm earns gross revenue

$$R(z, n; A) = A^{1/\sigma}(zn)^{(\sigma-1)/\sigma}.$$

A firm that serves only the domestic market has $A = E_D$. Exporting requires the fixed cost f^X and a bilateral iceberg cost $\tau > 1$, the same for imports and exports. The foreign demand shifter is

$$A_X = A^* e_x^{\sigma} \tau^{1-\sigma}, \tag{4}$$

where A^* collects foreign expenditure and the foreign price index and e_x is the domestic-currency price of the foreign numeraire. An exporter allocates output across the two markets

and nonexporters across different minimum-wage regimes.

[18] The household problem in Appendix B.1 allows a job-specific amenity. It is set to zero throughout this section. Section 4 introduces a formal-job amenity.

[19] The household problem that delivers equation (3), together with the aggregate labor-supply condition, is stated in Appendix B.1.

to equalize marginal revenue, so its market access is the sum of the two shifters:

$$E_X = E_D + A_X, \qquad \Gamma_X \equiv \frac{E_X}{E_D} > 1.$$

There are no tariffs in this analytical model, and all international trade barriers are bilateral iceberg costs.

After paying an entry cost, a firm draws productivity z from a Pareto distribution with cumulative distribution function $G(z) = 1 - (z_{\min}/z)^k$. Following Ulyssea (2018), formality is an endogenous production decision. An informal firm has effective productivity $(1-\delta^I)z$, where $\delta^I \in [0,1)$ is a productivity penalty,[20] pays operating cost f^I, is not subject to the minimum wage, and cannot export. Its value is

$$V_I(z) = \max_{w, n \le n^S(w)} \left\{ R((1-\delta^I)z, n; E_D) - wn - f^I \right\}. \tag{5}$$

A formal firm pays operating cost f^F, faces the minimum wage, and may export. For sales status $a \in \{D, X\}$, its value is

$$V_F^a(z) = \max_{w \ge w^{mw},\, n \le n^S(w)} \left\{ R(z, n; E_a) - wn - f^F - \mathbf{1}\{a = X\} f^X \right\}. \tag{6}$$

The firm chooses the largest of exit, with value zero, $V_I(z)$, $V_F^D(z)$, and $V_F^X(z)$. Informality therefore has two costs, the productivity penalty and the loss of export access, and one benefit, exemption from the minimum wage. Because the penalty costs more revenue at more productive firms, while the minimum wage binds less tightly there, informality is chosen by the least productive firms, and nonexporters above the formality cutoff remain formal even though they never export. Entry, operating, and export fixed costs are paid in units of the final good. Firms take aggregate labor supply, domestic absorption, the wage index, and the real exchange rate as given. These and the mass of entrants are determined in general equilibrium.[21]

3.2 Minimum-Wage Regimes and Efficiency

Consider a formal firm. Along its inverse labor-supply curve, the wage is $w(n) = bn^{1/\eta}$, where $b \equiv \widetilde{W}_w N_w^{-1/\eta}$. Hiring one more worker raises the wage paid to all workers, so the

[20] The penalty captures, in reduced form, costs of informal operation that rise with scale. In Ulyssea (2018), informal firms face an expected cost of detection that increases with their size, so informality is chosen by small, low-productivity firms. A proportional penalty delivers the same selection with closed-form regime boundaries.

[21] See Appendix B.1 for the equilibrium conditions.

marginal cost of labor is $\mu w(n)$, where $\mu \equiv 1 + 1/\eta$ is the wage markdown.[22] The marginal revenue product of labor is

$$\text{MRPL}(n; z, A) \equiv \frac{\partial R(z, n; A)}{\partial n} = \rho_\sigma A^{1/\sigma} z^{\rho_\sigma} n^{-1/\sigma},$$

where $\rho_\sigma \equiv (\sigma - 1)/\sigma$. Without a minimum wage, the firm chooses employment n^0 to equate the two. Suppressing z and A,

$$\text{MRPL}(n^0) = \mu w(n^0).$$

Because $\mu > 1$, the firm pays a wage equal to the marginal revenue product of labor divided by μ and employs fewer workers than it would if it took the wage as given.

Let n^c solve $\text{MRPL}(n^c) = w(n^c)$. This employment level maximizes employer-worker surplus, defined as gross revenue minus the area under the firm's inverse labor-supply curve. It is constrained efficient. It removes the employer wage-setting wedge while leaving the product-market markup in place. Because the marginal revenue product is decreasing in employment and $\mu > 1$, $n^0 < n^c$.

A formal job that pays the minimum wage attracts

$$\bar{n} = N_w \left(\frac{w^{mw}}{\widetilde{W}_w} \right)^{\eta} \tag{7}$$

applicants. Depending on its productivity and market access, a formal firm lies in one of three regimes, listed from most to least productive:

- **R1: unconstrained monopsony.** The firm's unconstrained wage is at least w^{mw}, so the policy does not affect its wage or employment.
- **R2: binding without rationing.** The unconstrained wage is below w^{mw}, but labor demand at the minimum wage is at least $\bar{n}$. The firm hires every applicant and employment equals $\bar{n}$.
- **R3: binding with rationing.** Labor demand at the minimum wage is below $\bar{n}$. The firm hires fewer workers than apply and sets marginal revenue product equal to w^{mw}.

[22] Because each job is small relative to the labor market and job-level labor supply is isoelastic, μ is a constant and does not vary with how many workers are employed formally or informally. Informal employment affects a firm's labor supply only through the wage index and aggregate labor supply. In models with granular employers, such as Berger, Herkenhoff, and Mongey (2022), the markdown varies with labor-market concentration.

Conditional on market access, employment in the three regimes is

$$\begin{aligned} n_1(z;A) &= \left[\frac{\rho_\sigma A^{1/\sigma} z^{\rho_\sigma}}{\mu b}\right]^{\sigma\eta/(\sigma+\eta)}, \\ n_2(z;A) &= \bar{n}, \\ n_3(z;A) &= \left[\frac{\rho_\sigma A^{1/\sigma} z^{\rho_\sigma}}{w^{mw}}\right]^{\sigma}. \end{aligned} \tag{8}$$

For sales status $a \in \{D, X\}$, let z_a^s denote the lower boundary of R2, which separates it from R3, and z_a^u the upper boundary, which separates it from R1. The boundaries satisfy

$$(z_a^u)^{\sigma-1} = \left(\frac{\mu}{\rho_\sigma}\right)^{\sigma} \frac{\bar{n}\,(w^{mw})^{\sigma}}{E_a}, \tag{9}$$

$$z_a^s = \mu^{-\sigma/(\sigma-1)} z_a^u, \tag{10}$$

where E_D applies to nonexporters and E_X to exporters. A higher minimum wage, a larger applicant pool, or lower market access moves both boundaries right, so more firms are constrained. By equation (10), the ratio z_a^s/z_a^u depends only on η and σ. A change in market access therefore moves the two boundaries of R2 in the same proportion.[23]

The three regimes and the employment result below follow Ahlfeldt, Roth, and Seidel (2023) and Berger, Herkenhoff, and Mongey (2025). Here a firm's regime also depends on its export status, through market access.

Proposition 1 (The minimum wage raises employment in R2, conditional on export status)**.** *Hold the aggregates fixed and let $a \in \{D, X\}$ be a formal firm's sales status. The firm is in R2 if and only if $z \in [z_a^s, z_a^u)$, and at every such firm*

$$n^0 < \bar{n} \leq n^c. \tag{11}$$

Relative to the firm's unconstrained choice, the minimum wage therefore raises employment, worker earnings, and employer-worker surplus. Firms with $z \geq z_a^u$ are in R1 and are unaffected. Firms with $z < z_a^s$ are in R3 and ration applicants. Their employment is set by labor demand at the minimum wage. It exceeds n^0 only at firms close to z_a^s and falls below n^0 at less productive firms.

At the minimum wage the firm can hire up to $\bar{n}$ workers without raising its wage, so its marginal cost of labor falls from μw to w^{mw} and it no longer gains from restricting

[23] Appendix B.2 derives the boundaries.

employment.[24] The proposition is a firm-level statement, conditional on formal operation. It does not imply that the policy raises aggregate welfare. Whenever the constraint binds, profit falls, and entry, formality, prices, and redistribution may change. The quantitative model evaluates those margins.

3.3 Trade and the Scope of the Minimum-Wage Rationale

Let $m \equiv w^{mw}/\widetilde{W}_w$ denote the relative minimum wage, its level relative to the wage index. Equation (9) can be written

$$(z_a^u)^{\sigma-1} = \left(\frac{\mu}{\rho_\sigma}\right)^{\sigma} m^{\sigma+\eta} \frac{N_w \widetilde{W}_w^{\sigma}}{E_a}. \tag{12}$$

The term in m is the relative-minimum-wage channel. It depends on how the minimum wage is adjusted when market wages change, and it is the same for every firm. The last term is the market-access channel. It measures a firm's market access relative to labor costs, which differs between exporters and nonexporters.

The market-access channel separates exporters from nonexporters. An exporter has market access $E_X = \Gamma_X E_D$, so equation (12) implies, for both boundaries,

$$z_X^r = \Gamma_X^{-1/(\sigma-1)} z_D^r, \qquad r \in \{s, u\}. \tag{13}$$

Both exporter boundaries lie below the corresponding nonexporter boundaries, whatever the minimum-wage rule. A firm with productivity between the two upper boundaries is unconstrained if it exports and constrained if it does not. A firm between the two lower boundaries rations applicants only if it does not export. Which firms export is a separate question. A firm exports if and only if its productivity exceeds a cutoff, z_x^*, at which the gain from foreign sales equals the export fixed cost.[25]

Proposition 2 (Trade moves the regime boundaries through two channels)**.** *Let $\ell \equiv -\log \tau$, so that a higher ℓ means lower bilateral iceberg costs. For each sales status $a \in \{D, X\}$ and each boundary $r \in \{s, u\}$,*

$$\frac{d \log z_a^r}{d\ell} = \underbrace{\frac{\sigma+\eta}{\sigma-1}\frac{d \log m}{d\ell}}_{\text{relative minimum wage}} - \underbrace{\frac{1}{\sigma-1}\frac{d}{d\ell}\log\left(\frac{E_a}{N_w \widetilde{W}_w^{\sigma}}\right)}_{\text{market access}}. \tag{14}$$

[24] The proof is in Appendix B.3.
[25] See Appendix B.5.

The relative-minimum-wage channel is common to all firms. The market-access channels of exporters and nonexporters differ by $(\sigma-1)^{-1}\, d\log\Gamma_X/\, d\ell$*, so the exporter boundaries fall relative to the nonexporter boundaries if and only if liberalization raises* Γ_X*.*

How the relative minimum wage moves is a matter of policy. I consider two rules. Under *wage indexation* the minimum wage is a fixed fraction of the wage index, so m is constant, trade acts only through market access, and the economy is compared before and after liberalization at the same policy stance. Under *price indexation* the minimum wage is fixed in units of the final good, so m falls whenever liberalization raises the wage index. Colombia's minimum wage is reset annually rather than mechanically tied to either rule in the model. Section 5 therefore states the counterfactual rule explicitly and holds the minimum wage fixed relative to average earnings when comparing tariff regimes.

Consider first wage indexation. The relevant object is the set of formal firms at which the minimum wage raises employment. By Proposition 1, these are the constrained firms with productivity of at least $z_a^e \equiv \mu^{-\eta/(\sigma-1)} z_a^s$.[26] These are every R2 firm and the R3 firms closest to R2. A firm that would belong to this set but operates informally is beyond the reach of the minimum wage. The set is therefore truncated from below by the formality cutoff, z_F, the productivity above which a firm operates formally.

Assumption 1 (The marginal exporter is constrained)**.** $z_D^s \le z_x^* < z_D^u$*. The firm that is indifferent about exporting is in R2 when it does not export.*

Under Assumption 1, the set of formal firms at which the minimum wage raises employment is the interval

$$\mathcal{C} = [\underline{z}, \bar{z}), \qquad \underline{z} \equiv \max\{z_F, z_D^e\}, \qquad \bar{z} \equiv \max\{z_x^*, z_X^u\}.^{27} \tag{15}$$

Its lower edge is set by nonexporters and by the formality margin, and its upper edge by exporters.

Proposition 3 (Export access shrinks $\mathcal{C}$ at a constant relative minimum wage)**.** *Suppose the minimum wage is indexed to the wage index, so that* m *is constant, and hold* $N_w\widetilde{W}_w$ *fixed. Suppose the marginal exporter does not ration as a nonexporter,* $z_x^* \ge z_D^s$*, and the least productive formal firm does not export. Then a rise in* Γ_X *weakly lowers the share of formal firms in* $\mathcal{C}$*.*

[26] See Appendix B.3.

[27] If exporters are already unconstrained, $z_x^* \ge z_D^u$, every firm in the set is a nonexporter and its upper edge is z_D^u.

(a) Under Assumption 1 the share falls strictly, because the upper edge of $\mathcal{C}$ falls relative to its lower edge and to the formality cutoff.

(b) If exporters are already unconstrained, $z_x^* \geq z_D^u$, the share is unchanged.

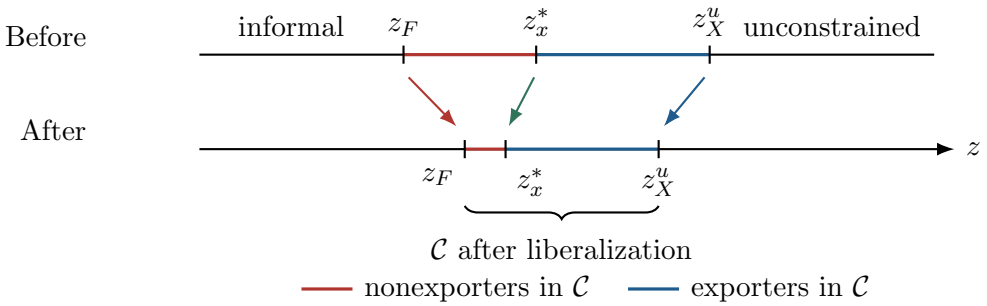
(c) When $\underline{z} = z_D^e$ and $\bar{z} = z_X^u$,

$$\frac{\bar{z}}{\underline{z}} = \mu^{(\sigma+\eta)/(\sigma-1)} \, \Gamma_X^{-1/(\sigma-1)}, \tag{16}$$

under any minimum-wage rule.

Exporters gain market access, so the most productive firms in $\mathcal{C}$ become unconstrained and leave it at the upper edge.[28] The proposition holds $N_w \widetilde{W}_w$ fixed. A cut in trade costs also changes it, which moves the formality cutoff. If the formality cutoff moves right, the least productive firms in $\mathcal{C}$ become informal, where the minimum wage does not reach them, and $\mathcal{C}$ shrinks from both ends. Figure 6 illustrates this case.

Figure 6: Bilateral liberalization under a wage-indexed minimum wage



Notes: The colored interval is $\mathcal{C}$, the set of formal firms at which the minimum wage raises employment. Nonexporters are in red and exporters in blue. Firms to its left are informal and firms to its right are unconstrained. The figure illustrates the case under Assumption 1 in which the lower edge of $\mathcal{C}$ is the formality cutoff and that cutoff rises. The exporter upper boundary falls relative to it, and the export cutoff falls. The figure is schematic.

Price indexation changes the outcome, because trade then moves the relative minimum wage. When liberalization raises the wage index, m falls and the relative-minimum-wage channel moves every boundary left. For nonexporters it outweighs the market-access channel. The nonexporter and the exporter boundaries both move left. A minimum wage that is fixed in goods binds on fewer firms after liberalization, because market wages rise past it. When absorption moves with the wage bill, the nonexporter boundaries move right only if the minimum wage rises by more than a fraction $(1+\eta)/(\sigma+\eta)$ of the increase in the wage index.[29] The same liberalization therefore narrows the reach of the minimum wage under one

[28] The proof is in Appendix B.5.

[29] See Appendix B.4.

rule and lowers the relative minimum wage under the other. How trade interacts with the minimum wage depends on how the minimum wage is set, which is why Section 5 evaluates the two policies jointly.

Three qualifications apply. First, Proposition 3 excludes the case in which the marginal exporter rations applicants when it does not export ($z_x^* < z_D^s$). There exporting eases its rationing and $\mathcal{C}$ can grow. Second, a firm that leaves $\mathcal{C}$ at its upper edge is not harmed. Market access has raised its wage above the minimum wage, and it no longer needs the correction. The result concerns the scope of the minimum wage's efficiency role. It does not imply that national informality rises, aggregate welfare falls, or the optimal minimum wage declines, because employment in the rest of R3, entry, consumer prices, fiscal effects, and redistribution also adjust. Third, the analytical exercise changes bilateral iceberg costs, whereas the Colombian reform primarily changes import tariffs.[30] Their complete aggregate effects require the quantitative model.

3.4 Connection to Informality and the Reduced-Form Analysis

The reduced-form analysis compares industries with larger and smaller tariff cuts within the same local labor market. Firms in a labor market face the same wage index, the same labor supply, and the same minimum wage, so the relative-minimum-wage channel is common to them and drops out of the comparison, whatever the indexation rule. What differs across industries is the market-access channel. A larger tariff cut lowers the market access of an industry's nonexporters relative to labor costs. By Proposition 2, a larger tariff cut raises its nonexporters' regime cutoffs, so firms move from R1 toward R2 and from R2 into R3, and the least productive formal firms may become informal. Because every worker at an R2 or R3 firm is paid the minimum wage, the empirical exposure measure, the share of formal salaried workers paid at or near the minimum wage, approximates the share of a labor market's formal employment at firms the minimum wage constrains. The model yields three implications for how the response varies with the local bite of the minimum wage.

1. Minimum-wage exposure can amplify the informality response to import competition. Where the minimum wage does not bind, formal firms are in R1 and respond to lower market access by cutting wages and employment. Informal firms and formal nonexporters contract in the same proportion, so the informal share is unchanged. Where the minimum wage binds, two further margins operate. R3 firms cannot lower their wage and cut employment by more, and firms near the formality cutoff switch to informal production. The

[30] Appendix B.7 connects the import-cost component of the two exercises.

model therefore predicts a larger rise in informality after a tariff cut in labor markets with high minimum-wage exposure when rationing firms account for enough of the employment bound by the minimum wage. Formal employment does not rise in any regime. Whether it falls by proportionally more than informal employment, so that the informal share rises, depends on the mix of constrained firms. Rationing firms cut employment by more than informal firms do, and R2 firms by less.[31] Section 2 documented this pattern.

2. Formal employment losses are concentrated at the minimum wage. The three regimes respond differently to lower market access. R1 firms cut both wages and employment. R2 firms keep employment fixed. R3 firms keep the wage at the minimum and cut employment one for one with market access, which is more than an R1 firm does. Where R3 firms account for most minimum-wage employment, formal job losses after a tariff cut are therefore concentrated among workers paid the minimum wage. Section 2 documented this in the distribution of formal employment around the minimum wage.

3. Employment appears below the minimum wage. Jobs lost at the minimum wage reappear below it in two ways. A firm that switches to informal operation pays its unconstrained wage, which is below the minimum. Workers released by rationing firms lower the wage index, which raises employment at firms that were already informal.[32]

Higher measured minimum-wage exposure indicates that more workers are employed near the minimum wage, but it does not reveal how those workers are divided between R2 and R3. These implications therefore concern relative responses across labor markets with different initial minimum-wage exposure. They do not sign the national change in informality after trade liberalization. Aggregate welfare and the optimal minimum wage remain questions for the quantitative spatial model in Section 4.

4 Spatial General-Equilibrium Model

Section 3 shows, in a one-sector small open economy, when a minimum wage can raise employment by limiting employer wage-setting power and how trade liberalization changes the set of firms at which it does so. It interprets the relative responses documented in Section 2, but it cannot aggregate heterogeneous responses into national informality, welfare, redistribu-

[31] The informal share rises when rationing firms account for more than a share $\eta/(\sigma+\eta)$ of employment at nonexporters bound by the minimum wage. See Appendix B.6.

[32] In this model the only informal activity is informal wage employment. In the data, the rise below the minimum wage is in self-employment rather than informal wage employment (Section 2). Section 4 adds self-employment, which does not require another employer to create the job.

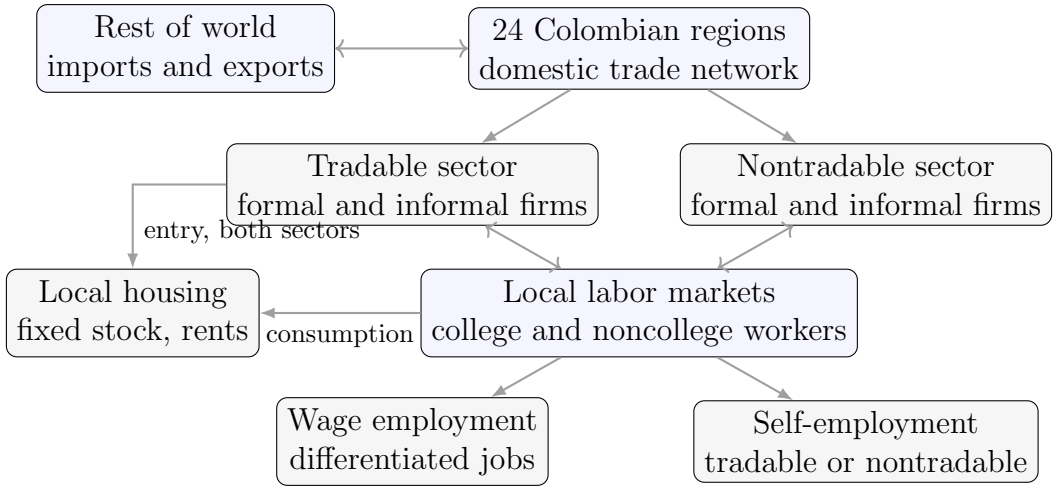
tion, or optimal policy. This section develops the quantitative model used for those purposes without re-deriving the firm-level mechanism. The model connects 24 regions through domestic trade, and tradable and nontradable sectors share local labor markets. College and noncollege workers face the same statutory minimum wage but differ in productivity, earnings, and exposure. Workers allocate hours between wage employment and sector-specific self-employment. The model also adds import tariffs to the iceberg trade costs of Section 3, with tariff revenue rebated to workers. Housing, firm entry, government transfers, and external balance close the spatial equilibrium. These additions are necessary because tariff exposure and the bite of the minimum wage differ across regions, sectors, and workers, while local responses propagate through domestic trade, prices, and common labor markets. The model is organized by region rather than by industry. The bite of the national minimum wage, the model counterpart of the minimum-wage exposure in Section 2, depends on the local wage distribution, and in the model a worker who loses a formal job moves to another employer, to informal work, or to self-employment within the same region. Following Kovak (2013), a region's tariff change is the average of the tariff changes in its tradable industries, weighted by their pre-reform employment, so the model keeps the industry composition that makes regions differ in their exposure to the reform. It abstracts from reallocation across tradable industries within a region, from differences in the bite of the minimum wage across those industries, and from imported intermediate inputs. Figure 7 summarizes the structure. The national minimum wage applies to formal wage jobs in every region, and international trade shocks enter through the tradable sector before spreading across regions and sectors. Section 5 estimates the model and uses it to simulate the tariff reform, alternative minimum wages, and joint trade and minimum-wage policies.

4.1 Geography, Sectors, and Regional Market Access

The economy contains Colombian regions $d, k \in \mathcal{D}$ and the rest of the world. The quantitative application uses 24 regions. Region d has fixed populations L_{hd} of college and noncollege workers, indexed by $h \in \{C, N\}$, and a fixed housing stock $\overline{H}_d$. I hold populations fixed to focus on the within-location adjustment margins measured by the reduced-form analysis. Wages, employment, self-employment, prices, rents, firm entry, formality, and trade nevertheless adjust endogenously.[33] Every region produces a tradable composite, T, and a nontradable composite, N. Both composites contain differentiated varieties produced across Colombian regions. Thus "nontradable" means that the sector does not trade internationally. Its varieties can still be shipped between Colombian regions. Only tradable varieties

[33] Appendix C.3 describes an extension with migration.

Figure 7: Spatial, sectoral, and labor-market structure of the quantitative model



Notes: Tradable firms buy and sell across Colombian regions and may trade internationally; nontradable firms trade only within Colombia. Both sectors hire from the same region–education labor markets. Workers allocate hours between wage employment and tradable or nontradable self-employment. Households spend a fixed share of consumption on local housing, and entrants in both sectors use housing, so rents link household purchasing power and firm entry.

are imported from and exported to the rest of the world. This two-sector structure matters because tariffs directly affect tradable goods, while workers and expenditure can reallocate toward the nontradable sector.

Households combine the two sectoral composites using the region-specific tradable expenditure share $\alpha_{T,d}$ and combine the resulting goods aggregate with housing. The goods and consumption price indexes are

$$P_{G,d} = \left(\frac{P_{T,d}}{\alpha_{T,d}}\right)^{\alpha_{T,d}} \left(\frac{P_{N,d}}{1-\alpha_{T,d}}\right)^{1-\alpha_{T,d}}, \qquad P_{C,d} = P_{G,d}^{1-\alpha_H} R_d^{\alpha_H},$$

where R_d is the regional rent and α_H is the housing expenditure share.[34] A common national minimum wage can therefore have different purchasing power across regions even before accounting for regional differences in wages.

Within sector $s \in \{T, N\}$, destination k spends $X_{s,k}$ on differentiated varieties. Let τ_{dk}^s denote the iceberg cost of shipping from origin d to destination k, and let $\sigma > 1$ be the elasticity of substitution across varieties. Domestic market access for a producer in region d is

$$\Pi_{sd}^D = \sum_{k \in \mathcal{D}} X_{s,k} P_{s,k}^{\sigma-1} \left(\tau_{dk}^s\right)^{1-\sigma}. \tag{17}$$

[34]The consumption aggregate is $(G/(1-\alpha_H))^{1-\alpha_H}(H/\alpha_H)^{\alpha_H}$, so $P_{C,d}$ carries no constant. Because α_H is common to all regions, this is a choice of units.

Equation (17) is the central spatial object in the model. A firm's demand depends on expenditure, prices, and trade costs in every domestic destination it serves, not only in its home region. A tariff shock in one region can therefore change market access and labor demand in other regions through both expenditure and competition effects. Tradable producers may also pay a fixed cost to access foreign market demand, $\Pi_d^X = X^*(\tau_d^M)^{1-\sigma} e_x^\sigma$, where X^* is foreign expenditure. In the notation of Section 3, Π_{sd}^D takes the place of domestic absorption E_D and Π_d^X takes the place of the foreign demand shifter A_X. Imports enter region d at tariff-inclusive cost

$$p_d^M = (1 + t_d)\tau_d^M e_x, \tag{18}$$

where t_d is the effective import tariff, τ_d^M is the international iceberg cost, and e_x is the domestic-currency price of the foreign good, whose foreign price is the numeraire abroad. A tariff reduction directly lowers the import price. It also changes domestic sectoral price indexes, regional expenditure, export profitability, and the exchange rate. The domestic trade network in equation (17) transmits these responses across regions and into the nontradable sector.

4.2 Worker Types, Wage Employment, and Self-Employment

For each education group h and region d, a representative household pools the income of its members and chooses consumption and hours. Pooling delivers one consumption level for each region–education cell while preserving welfare differences across cells. Let c_{hd} denote consumption per worker and n_{hd} the per-worker composite of hours. Preferences are

$$u_{hd}(c_{hd}, n_{hd}) = \frac{c_{hd}^{1-\sigma_c} - 1}{1 - \sigma_c} - \bar{\phi}_{hd}^{-1/\psi} \frac{n_{hd}^{1+1/\psi}}{1 + 1/\psi}, \qquad \sigma_c > 0,\ \psi > 0, \tag{19}$$

with the usual logarithmic limit when $\sigma_c = 1$. The parameter ψ is the Frisch elasticity and $\bar{\phi}_{hd}$ shifts labor supply.

Workers allocate hours among wage employment, tradable self-employment, and nontradable self-employment. Write these activities as $a \in \{W, ST, SN\}$. Their CES aggregate is

$$n_{hd} = \left[\sum_{a \in \{W,ST,SN\}} \chi_{a,hd}^{-1/\theta} n_{a,hd}^{(1+\theta)/\theta} \right]^{\theta/(1+\theta)}, \tag{20}$$

where θ governs substitution across activities and $\chi_{a,hd}$ shifts their relative attractiveness or

productivity. If $W_{a,hd}$ is the return to activity a, the corresponding inclusive return is

$$\mathcal{W}_{hd} = \left[\sum_{a \in \{W,ST,SN\}} \chi_{a,hd} W_{a,hd}^{1+\theta} \right]^{1/(1+\theta)}. \tag{21}$$

The household first-order conditions give

$$n_{hd} = \bar{\phi}_{hd} \left(\frac{\mathcal{W}_{hd}}{P_{C,d}} \right)^{\psi} c_{hd}^{-\sigma_c \psi}, \tag{22}$$

$$n_{a,hd} = \chi_{a,hd} \left(\frac{W_{a,hd}}{\mathcal{W}_{hd}} \right)^{\theta} n_{hd}, \qquad a \in \{W, ST, SN\}. \tag{23}$$

Thus a policy shock changes both total hours and their allocation across activities. Self-employment is not combined with informal wage employment. It is a separate competitive activity in each sector, with its return determined by sectoral prices and self-employment productivity. This distinction allows workers displaced from formal jobs to become own-account workers, the margin on which the additional informality response in Section 2 is concentrated, and allows the welfare calculation to value changes in their earnings separately (Amodio, Medina, and Morlacco, 2025).

The household budget is

$$P_{C,d} c_{hd} = y_{hd}^{W} + W_{ST,hd} n_{ST,hd} + W_{SN,hd} n_{SN,hd} + T, \tag{24}$$

where y_{hd}^{W} is realized wage income and T is the national per-worker government transfer. Rents accrue to housing owners who are outside the worker households. They do not enter the budget and are spent on local goods.[35] Realized rather than potential wage income enters because applicants can be rationed at minimum-wage jobs.

Conditional on wage employment, workers choose among differentiated employers in their region–education labor market. Labor offered to employer j at wage w_{jhd} is

$$n_{jhd}^{S} = D_{hd} \left([1 + \xi(j)] w_{jhd} \right)^{\eta}, \qquad \eta > 0, \tag{25}$$

where $D_{hd} \equiv L_{hd}\, n_{W,hd}\, W_{W,hd}^{-\eta}$ combines the cell's supply of wage work with its wage index $W_{W,hd}$, as in Section 3. Both worker types have the same elasticity η. Their supply curves differ only through D_{hd}, which reflects the size of each type's pool and the wages that other employers pay that type. Formal jobs have amenity ξ_F and informal jobs have amenity zero.

[35] See Appendix C.2.

In standard heterogeneous-firm trade models such as Melitz (2003), every firm pays the same wage. Here each employer faces an upward-sloping supply curve, so more productive employers pay higher wages, and identical workers earn different wages at different firms in the same labor market. Firms bound by the minimum wage pay exactly the minimum wage. As in Section 3, a formal firm that rations at the minimum wage hires fewer workers than apply. Unhired workers reallocate toward other wage jobs or either self-employment activity through equations (20)–(23).[36]

Worker types differ in productivity. An hour of type-h labor supplies s_h efficiency units in production, with $s_N = 1$ and $s_C > 1$. Labor supply in equation (25), the wage w_{jhd}, and the minimum wage are all per hour of work, not per efficiency unit, and a firm sets a separate wage for each type. An unconstrained formal firm pays every efficiency unit the same, so it pays a college worker s_C times the hourly wage of a noncollege worker. The statutory minimum wage is the same per hour for both types and therefore costs more per efficiency unit for noncollege workers. It can bind for noncollege workers but not for college workers within the same firm. College and noncollege workers also differ in activity returns and regional distributions, so they face different outside options and different wage indexes within the same region. The model can consequently generate different employment and welfare responses across education groups even though the statutory policy is national.

4.3 Firms, Formality, and Exporting

Each region and sector has a pool of potential entrants. An entrant in sector s pays $R_d f_E^s$ for an entry cost of f_E^s units of local housing and draws permanent productivity φ. Based on expected profits, it exits or commits to formal or informal operation. It then draws transitory productivity ε, giving realized productivity $z = \varphi\varepsilon$, and chooses wages, employment, output, and prices.[37] A formal tradable firm also chooses whether to export after observing its realized productivity. Formality is therefore an ex ante organizational choice, while operating decisions respond to current productivity and market access. Trade nevertheless moves the formality margin, because the expected value of each choice depends on domestic market access and, for formal tradable firms, on the option to export after a favorable transitory draw. This two-stage structure allows formal and informal firms to overlap in realized size and productivity. Only entry uses local housing. Formal and informal operating costs and export costs are paid in units of the firm's own sector composite in its region, at price $P_{s,d}$.

[36] Appendix C.1 constructs the wage-work index when firms ration.

[37] Permanent productivity is Pareto with shape k, as in Section 3, bounded above (Appendix D.2). The transitory draw is log-normal, $\log \varepsilon \sim \mathcal{N}(-\sigma_\varepsilon^2/2, \sigma_\varepsilon^2)$, so that ε has mean one.

Production uses effective labor. A formal firm j employs

$$n_j^{\text{eff}} = \sum_h s_h n_{jh}.$$

After the firm sets its CES price, let $\mathcal{R}_{sd}(z, n^{\text{eff}}; \Pi)$ denote revenue for a firm in sector s and region d with market access Π. Revenue is increasing and concave in effective labor, and it includes a region- and sector-specific productivity shifter, A_d^s.

Conditional on realized productivity and export status $x \in \{0, 1\}$, a formal firm solves

$$\begin{aligned} \pi_{sd}^F(z, x) = \max_{\{w_h, n_h\}_h} \; & \mathcal{R}_{sd}\left(z, \sum_h s_h n_h; \Pi_{sd}^D + x\Pi_d^X\right) - \Xi_F \sum_h w_h n_h - P_{s,d}\left(f_s^F + x f_s^X\right) \\ \text{s.t.} \quad & 0 \le n_h \le D_{hd}\left[(1+\xi_F) w_h\right]^\eta, \qquad w_h \ge w^{mw} \quad \text{if } n_h > 0. \end{aligned} \tag{26}$$

The factor $\Xi_F \equiv 1 + \tau_p + \xi_F$ collects the payroll tax rate τ_p and the cost of the formal benefits that workers value as the amenity ξ_F.

An informal firm is not subject to the payroll tax or minimum wage and cannot export. It instead produces with region-specific productivity penalty δ_d^I and solves

$$\begin{aligned} \pi_{sd}^I(z) = \max_{\{w_h, n_h\}_h} \; & \mathcal{R}_{sd}\left((1-\delta_d^I) z, \sum_h s_h^{\lambda_I} n_h; \Pi_{sd}^D\right) - \sum_h w_h n_h - P_{s,d} f_s^I \\ \text{s.t.} \quad & n_h = D_{hd} w_h^\eta. \end{aligned} \tag{27}$$

Informal firms use skill less intensively. An hour of type-h labor supplies $s_h^{\lambda_I}$ efficiency units there, with $\lambda_I \in (0, 1]$. The productivity penalty summarizes the costs of operating outside the formal sector, including restricted access to customers, finance, public inputs, and contract enforcement, and the risk of detection. Formality follows from the comparison of expected values before ε is realized, as in Ulyssea (2018).[38]

The wage and employment policies inside equations (26) and (27) are those characterized in Section 3, applied to each worker type, and are taken as given here. The spatial model integrates those policies over two productivity draws, both sectors, two worker types, and all regions, while allowing wage indexes and market access to adjust in general equilibrium. The relative minimum wage of Section 3 is now specific to a region and worker type, because one national minimum wage meets a different wage index in each local labor market. The sectoral and spatial environment changes the interpretation of the simple mechanism. Import competition directly changes tradable demand, but it also affects nontradable firms through

[38] Appendix C.4 gives these expected values and the free-entry condition.

regional expenditure and common labor markets. Export access is available only to formal tradable firms. Informal firms and self-employed workers supply domestic markets, while interregional trade exposes every origin to demand and competition elsewhere. Consequently, the mass of workers in each minimum-wage regime is an equilibrium outcome of firm selection, sectoral reallocation, and spatial propagation rather than a direct function of the local tariff alone.

4.4 Spatial Equilibrium

Housing links worker purchasing power and firm entry. Households spend the fixed share α_H on housing, while entrants use f_E^s units of local housing in sector s. Market clearing is

$$\alpha_H \frac{P_{C,d}}{R_d} \sum_h c_{hd} L_{hd} + M_{T,d}^E f_E^T + M_{N,d}^E f_E^N = \overline{H}_d,$$

where $M_{s,d}^E$ is entrant mass. Rents therefore respond to both household income and entry. They affect real consumption and the cost of entry, but not the operating fixed costs in equations (26) and (27). Labor markets clear separately by region and education. Wage-work supply equals employment at formal and informal firms across both sectors, and the remaining activity supply is allocated to tradable or nontradable self-employment. Because the two sectors share these labor markets, a contraction in tradable employment changes the outside options and labor costs of nontradable firms. Aggregate adjustment therefore cannot be recovered by summing the reduced-form coefficients for tradable workers.

Goods-market clearing equates output from firms and self-employment with household demand, the spending of housing owners, operating expenditures, domestic shipments, and, for tradables, imports and exports. Free entry equates the expected value of a productivity draw with the sector-specific housing-denominated entry cost $R_d f_E^s$ in each region and sector.

The government rebates payroll-tax and tariff revenue uniformly per worker. Its national budget is

$$\sum_d \sum_h L_{hd} T = \mathcal{T}^{\text{payroll}} + \mathcal{T}^{\text{tariff}}.$$

External balance determines the exchange rate e_x. Trade and minimum-wage policies therefore affect households through wages and employment, self-employment returns, consumer prices, rents, fiscal transfers, and the terms of trade. A spatial equilibrium consists of household choices, firm policies and status choices, entry, sectoral prices, wage indexes, rents, transfers, and an exchange rate. In equilibrium, households and firms optimize, the labor, goods, and housing markets clear, free entry holds, and the government and external accounts

balance. The equilibrium determines the regional allocation of formal wage employment, informal wage employment, tradable and nontradable self-employment, firm entry, exporting, consumption, and trade.[39]

4.5 Normalization, Welfare, and Policy Experiments

Following Ahlfeldt, Roth, and Seidel (2023), I choose employment-weighted average earnings as the numeraire and set $\overline{w} = 1$ in every equilibrium. Nominal wages and prices are otherwise identified only up to a common multiplicative constant. This choice fixes their units without changing the real wages or consumption allocations relevant for welfare. The minimum-wage policy is therefore entered in units of average earnings. For computation, write

$$\kappa \equiv \frac{w^{mw}}{\overline{w}}, \tag{28}$$

so that setting κ sets the minimum wage directly in the normalized units. Holding κ fixed across equilibria is the wage-indexed rule of Section 3.3, with average earnings in place of the wage index. Each counterfactual stores both κ and the regional purchasing power of the minimum wage, $w^{mw}/P_{C,d}$.[40] Section 5 states the reference equilibrium for every welfare comparison.

Welfare is evaluated for every region–education cell using utility in equation (19). Following Berger, Herkenhoff, and Mongey (2025), let CEV_{hd} be the proportional increase in the consumption of cell (h, d) in a stated reference equilibrium that delivers its utility in the counterfactual, $u_{hd}\big((1 + CEV_{hd})c^0_{hd}, n^0_{hd}\big) = u_{hd}\big(c^1_{hd}, n^1_{hd}\big)$, where superscripts 0 and 1 denote the reference and counterfactual equilibria. Aggregate worker welfare is the common proportional increase in every cell's reference consumption that delivers the counterfactual level of utilitarian social welfare:

$$\sum_d \sum_h \omega_{hd}\, u_{hd}\big((1 + \overline{CEV})\, c^0_{hd}, n^0_{hd}\big) = \sum_d \sum_h \omega_{hd}\, u_{hd}\big(c^1_{hd}, n^1_{hd}\big), \tag{29}$$

where ω_{hd} is the population share of cell (h, d). The change for one education group applies the same definition to that group's cells. Because utility is separable and of the power form in consumption, $1 + \overline{CEV} = \big[\sum_d \sum_h \tilde{\omega}_{hd}(1 + CEV_{hd})^{1-\sigma_c}\big]^{1/(1-\sigma_c)}$ with weights $\tilde{\omega}_{hd} \propto \omega_{hd}\,(c^0_{hd})^{1-\sigma_c}$. Because $\sigma_c = 2$ in the calibration, cells with lower consumption receive more weight than their population share. The welfare-maximizing minimum wage in Section 5 maximizes $\overline{CEV}$. It is therefore a utilitarian optimum for workers, which val-

[39] Appendix C.5 lists the outer unknowns and equilibrium conditions.

[40] Appendix C.6 gives the reporting details.

ues redistribution toward low-consumption cells, and it does not count the rent income of housing owners. Reporting the cell-level changes alongside $\overline{CEV}$ makes redistribution across education groups and regions explicit.

5 Estimation and Quantitative Analysis

I now take the spatial model to Colombian labor-market, firm, trade, housing, and regional expenditure data and use it to ask how the minimum wage and trade policy jointly determine informality and welfare. After estimating the model, I run four counterfactual exercises. First, I simulate the 2010–2011 tariff cuts to measure their national effect on informality, which the reduced-form estimates cannot identify. Second, I compute the gains from lowering international trade costs at different levels of the minimum wage, to ask whether the minimum wage changes what trade does to workers. Third, I repeat this exercise in an economy without informal firms and in one without informal firms or self-employment, to isolate the role of informal employment in those gains. Fourth, I solve for the minimum wage that maximizes worker welfare under current tariffs and compare it with Colombia's current minimum wage. I then ask whether the trade regime changes this optimum.

5.1 Taking the Model to the Data

I estimate the model on 2015 data. The household survey measures the number of college and noncollege workers in each region and how each region–education group divides its work among formal wage work, informal wage work, and self-employment. It also measures earnings, minimum-wage exposure, employer size, and regional rents. Customs records measure trade and exporter participation, and the 2015 interregional input–output table measures expenditure and trade flows among the 24 regions.[41]

Regional endowments, expenditure shares, tariffs, and spatial trade costs are constructed from data before the simulated method of moments (SMM) estimation. Bilateral iceberg costs are recovered from domestic expenditure shares and interregional trade flows using the Head–Ries procedure (Head and Ries, 2001). International trade costs come from input–output and world output data, and tariffs from customs data. Table 3 classifies the remaining parameters as externally set, estimated in separate empirical exercises,[42] mechanically recovered from observed quantities, or estimated internally by SMM, the only group searched over in the objective.

[41] Appendix D.1 documents the source, unit, and construction of every model object.
[42] See Appendix D.3.

Table 3: Parameters

Parameter	Value	Source or main identifying moments
Set externally		
σ_c, ψ	2.0, 0.40	Maintained calibration values
α_H, τ_p, ξ_F	0.252, 0.25, 0.264	Household budget survey 2016–17; statutory rate; benefits data
f_s^I	0	Restriction
w^{mw}, L_{hd}, $\alpha_{T,d}$, t_d	Data	Household survey, input–output table, customs
τ_{dk}^T, τ_{dk}^N, τ_d^M	Data	Head–Ries inversion of 2015 regional and world flows
Estimated outside the SMM objective		
η	2.50	Amodio and de Roux (2024)
θ	1.50	Tariff-IV estimate of substitution between wage work and self-employment
σ	4.00	Gravity estimate on 2015 customs data
Recovered by inversion		
A_d^S, $\chi_{a,hd}$, $\bar{\phi}_{hd}$	Data-implied	Self-employment earnings, activity shares, and hours
$\overline{H}_d$	Data-implied	Rents relative to Bogotá
Estimated by simulated method of moments (estimates in Appendix D.2)		
A_d^T, A_d^N		Region-by-sector wage levels and sectoral employment shares
δ_d^I		Formality and formal–informal wage gaps
f_E^T, f_E^N, f_T^F, f_N^F		Firm counts, sectoral formality, and regional employment
f_T^X, X^*		Exporter share and export-sales ratio
σ_ε, k, s_C, λ_I		Wage dispersion, wage gaps, bite by type, and type premia

Let m_ℓ^{data} and $m_\ell^{model}(\Theta)$ denote moment ℓ in the data and model and let se_ℓ put the discrepancy on a common sampling-error scale. The estimator minimizes

$$\widehat{\Theta} = \arg\min_{\Theta} \sum_{\ell=1}^{L} \left[\frac{m_\ell^{model}(\Theta) - m_\ell^{data}}{se_\ell} \right]^2 .$$

Scaling by se_ℓ prevents moments measured in pesos or large employment counts from mechanically dominating moments measured as shares. Because several parameters affect the same outcomes, identification is joint rather than one moment per parameter.

The moment set uses variation across regions, education groups, sectors, activities, and exporters. Region-by-sector wage levels inform sectoral productivity, and housing stocks are recovered so that the model matches regional rents exactly. Formality and the minimum-wage

bite by region, sector, and education, and formal–informal wage gaps by region and sector, inform the local formality wedge, the dispersion of realized productivity, and the incidence of the wage constraint. Regional tradable employment shares identify comparative advantage across the two sectors. The national formal-firm count informs entry costs, while exporter participation and the export-sales ratio inform the fixed cost and value of foreign market access. Wage dispersion and the college–noncollege wage gap help determine productivity dispersion and the skill parameters.[43]

Table 4 reports selected moments.[44] The model reproduces the geography of formality and sectoral specialization, and it matches the number of formal firms, while its share of tradable firms that export is about 10 percent above the data. Two misses matter for what follows. The model places fewer formal workers near the minimum wage than the data in tradables and among college workers. The noncollege tradable cells average 0.42 against 0.50, and the median college cell is 0.125 against 0.163. It also trades less than the data. Exports relative to GDP are about 70 percent of their value in the data, and imports, which the external-balance closure sets equal to exports, less than half of theirs. This scales down the direct effects of tariffs and the tariff revenue rebated to workers. Region-by-sector wage levels and formal–informal wage gaps account for most of the remaining misfit.

Table 4: Selected moments in the data and the estimated model

	Data	Model	Correlation
Targeted			
Formal share of wage employment, noncollege			0.91
Formal share of wage employment, college			0.77
Tradable share of wage employment, by region			0.89
Minimum-wage bite, noncollege (median)	0.469	0.455	0.49
Minimum-wage bite, college (median)	0.163	0.125	0.48
Formal–informal log wage gap (mean over region–sector cells)	0.529	0.683	
Share of tradable firms that export	0.102	0.112	
Formal firm count (model units)	0.218	0.217	
Export sales ratio, value-added basis	0.300	0.249	
Untargeted			
Tradable share of wage employment, national	0.277	0.345	
Informal share of wage jobs, nontradables minus tradables	−0.130	−0.079	
Exports / GDP	0.143	0.1025	

Notes: The last column is the correlation between model and data across regions, or across region–sector cells within an education group for the minimum-wage bite (the share of formal wage workers paid at most 1.05 times the hourly minimum wage). Appendix D.5 reports every block of the objective and plots the regional fit.

[43] Appendix D.4 lists the 318 moments by block.

[44] Appendix D.5 reports every block of the objective and plots the regional fit.

5.2 The Tariff Reform in the Estimated Model

I next simulate the tariff reductions implemented in 2010–2011. This is an untargeted assessment of the model's central transmission mechanism. The reduced-form analysis assigns industry-level tariff changes to workers in tradable industries, whereas the quantitative model has one tradable composite in each region. I map the reform into regional tariff changes with pre-reform employment weights,

$$\Delta t_d = \sum_i \omega_{di,0} \Delta t_i, \tag{30}$$

where $\omega_{di,0}$ is region d's pre-reform employment share in tradable industry i and Δt_i is the industry's 2009–2011 tariff reduction. The regional reductions average 2.66 percentage points. I treat the estimated 2015 equilibrium as the post-reform economy and compare it with a counterfactual in which each region's tariff is higher by Δt_d, holding the minimum wage fixed relative to average earnings.

The simulation gives two results, and they answer different questions. One is the national change in informality, across both sectors and all regions. The other is how the change in a region's informality varies with the size of its tariff cut and with how tightly the minimum wage binds there. This second result is the one closest to Section 2, but it is not the same comparison. Section 2's estimate is not informative about the national change, and the national change is not a test of Section 2's estimate.

The simulated reform raises exports by 6.5 percent and lowers national informality by 0.08 percentage points. The effect is small for two reasons. The regional tariff reductions are about 2.7 percentage points on average, and the model trades less than the data (Section 5.1). Informality falls slightly more among noncollege workers (-0.103 percentage points) than among college workers (-0.064). The national decline is equivalent to reducing the current minimum wage by about 0.4 percent.

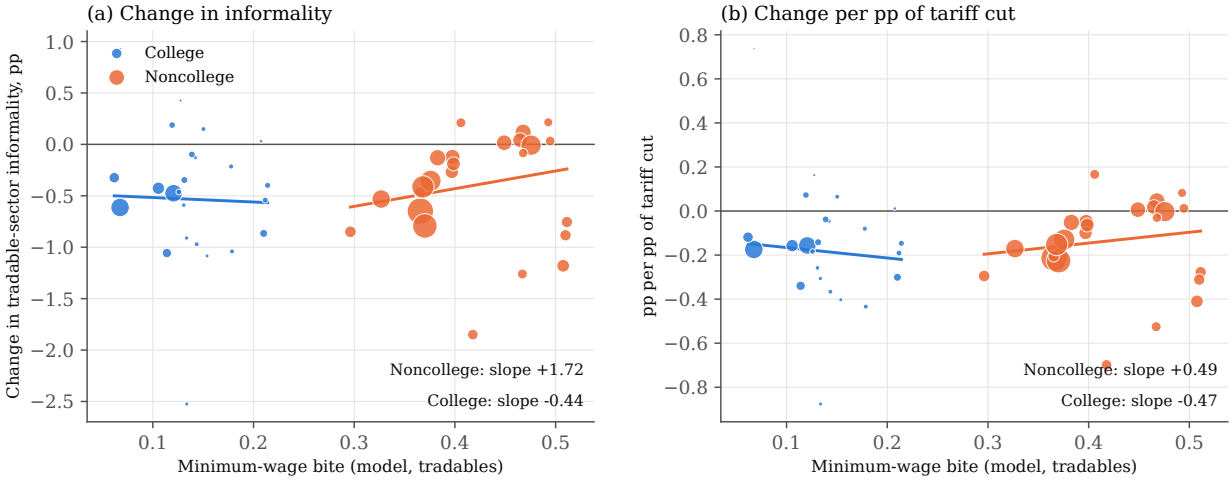
The changes are small in every region, between -0.41 and $+0.02$ percentage points, and slightly more negative where the tariff cut is larger.[45] The slope is -0.053 percentage points per percentage point of tariff reduction. The pooled numbers mix two sectors that move in opposite directions. Within the tradable sector, the reform lowers informality by 0.42 percentage points among noncollege workers and by 0.52 among college workers (employment-weighted means across regions), mostly through fewer informal wage workers. The declines are larger where the tariff cut is larger. The slope is -0.45 percentage points per percentage point of tariff reduction for noncollege workers and -0.85 for college workers.

[45] Appendix Figure D.2 displays the regional distribution, pooling both sectors and both education groups.

In the nontradable sector, informality rises slightly, by 0.08 percentage points for noncollege and 0.16 for college workers. The model therefore predicts formalization in the sector exposed to import competition and a small spillover of informality into nontradables.

The comparison with Section 2 asks whether tariff reductions lower informality by less where the minimum wage binds more tightly. Qualitatively, the spatial model reproduces this pattern. Figure 8 plots the change in tradable-sector informality, and that change per percentage point of tariff reduction, against the model's minimum-wage bite in each region and education group. Among noncollege workers the relationship is positive. In regions where the bite is higher, the reform lowers informality by less. Among college workers, whose bite is low everywhere, it is negative. The model's relationship is weaker than the one in the data, and it does not appear when cells are split at the median exposure of Section 2.[46] The main reason is aggregation. Section 2 compares industries with different tariff cuts within the same region and education group, whereas the model pools all tradable industries into one sector per region, so each region receives an average cut. These regional averages vary much less than the industry cuts. Their standard deviation is 0.40 percentage points, against 2.7 for the industry cuts.

Figure 8: The 2010–2011 tariff reform and the minimum wage within the tradable sector



Notes: One point per region and education group. Panel (a): change in tradable-sector informality (informal wage workers and the self-employed); panel (b): the same change divided by the region's tariff cut. The horizontal axis is the model's minimum-wage bite, the share of the group's formal tradable-sector wage workers paid at or below 1.05 times the minimum wage in the pre-reform equilibrium. Marker area is proportional to pre-reform tradable-sector employment; lines are employment-weighted least-squares fits by education group, with slopes reported in each panel. Changes are the 2015 equilibrium minus the pre-reform counterfactual at the current minimum wage.

[46] See Appendix D.6.

5.3 Trade and the Minimum Wage

I next ask whether the minimum wage changes what trade does to workers. As in Dix-Carneiro et al. (2026), I lower international iceberg trade costs, here by 10 percent. I do so at seven levels of the minimum wage, from none to 1.25 times the current one, each held fixed relative to average earnings. At each level, I compare the economy after the cut with the same economy before it and measure worker welfare by the consumption-equivalent change in equation (29).

The minimum wage barely changes the gains from trade (Table 5). The cut raises worker welfare by 0.97 percent without a minimum wage and by 0.92 percent at the current one, and across the seven levels the gain of all workers varies by at most 0.07 percentage points. Because the minimum wage is held fixed relative to average earnings, the cut barely changes how tightly it binds. At the current minimum wage, the noncollege bite falls by less than 1 percentage point.

The gains differ by education. The cut lowers the price of imported goods, which raises workers' real income, but it also lowers earnings in tradable self-employment, which competes with imports, by about 2 percent relative to wage earnings. Tradable self-employment accounts for 22 percent of noncollege work and 4 percent of college work, so college workers gain more, 1.20 against 0.90 percent at the current minimum wage. The cut also lowers informality by 0.67 percentage points.

To isolate the role of informal employment, I repeat the exercise in two economies that keep the estimated parameters. In the first, informal firms cannot operate. In the second, self-employment is also shut down, which corresponds to the no-informality scenario of Dix-Carneiro et al. (2026), who count the self-employed as informal firms with one worker. Removing informal firms lowers the gain slightly, to 0.84 percent at the current minimum wage, and the minimum wage still matters little. Removing self-employment as well raises the gain to 1.30 percent. The increase goes to noncollege workers, whose gain rises to 1.32 percent because they no longer hold the tradable self-employment jobs whose earnings fall. This is the opposite of Dix-Carneiro et al. (2026), who find that informality more than doubles the real income gains from trade, because trade reallocates resources from informal firms, which escape taxes and regulations, to formal firms. In this model, informality includes tradable self-employment, which competes with imports, so informality lowers workers' gains instead. Without self-employment, the minimum wage also matters more, and the gain of all workers varies by up to 0.16 percentage points across its levels.[47]

[47] Appendix D.7 reports the move to free trade and the same exercise in a re-estimated model that also counts formal benefits as worker income.

Table 5: Worker gains from a 10 percent cut in international trade costs

	No minimum wage	Current minimum wage			Range
	All workers	All workers	Noncollege	College	All workers
Baseline	0.97	0.92	0.90	1.20	0.07
No informal firms	0.88	0.84	0.82	0.99	0.11
No informal firms or self-employment	1.26	1.30	1.32	1.04	0.16

Notes: Consumption-equivalent change in worker welfare, in percent, from cutting international iceberg trade costs by 10 percent, relative to the same economy at 2015 trade costs and the same minimum wage, held fixed relative to average earnings. Range is the largest minus the smallest gain for all workers over minimum wages of 0, 0.25, 0.50, 0.56, 0.75, 1.00, and 1.25 times the current level (Appendix Figure D.3). In the economy without informal firms or self-employment, the self-employment shifters are set to zero. Every economy keeps the estimated parameters.

5.4 Minimum-Wage Policy

This subsection asks which national minimum wage maximizes worker welfare under current tariffs, and how the current policy compares with it. I hold the 2015 tariff schedule fixed and solve the model for a range of minimum wages, from none to levels above the current policy. I compare each policy with the equilibrium without a minimum wage under the same tariffs, following Berger, Herkenhoff, and Mongey (2025), with welfare measured as in equation (29), and I also report the change from the current policy to the optimum.[48]

The welfare-maximizing minimum wage is about 0.56 times the current Colombian minimum wage.[49] In wage-relative terms, the optimum sets κ at 0.38, against 0.67 for the current policy. At the current policy, aggregate worker welfare is 0.65 percent lower than without a minimum wage. At the optimum, it is 0.14 percent higher. Moving directly from the current policy to the optimum raises aggregate worker welfare by 0.80 percent. Panel (a) of Figure 9 plots aggregate and education-specific welfare against the minimum wage. A minimum wage raises welfare only at low levels, and the current policy lies well beyond the point where it turns negative.

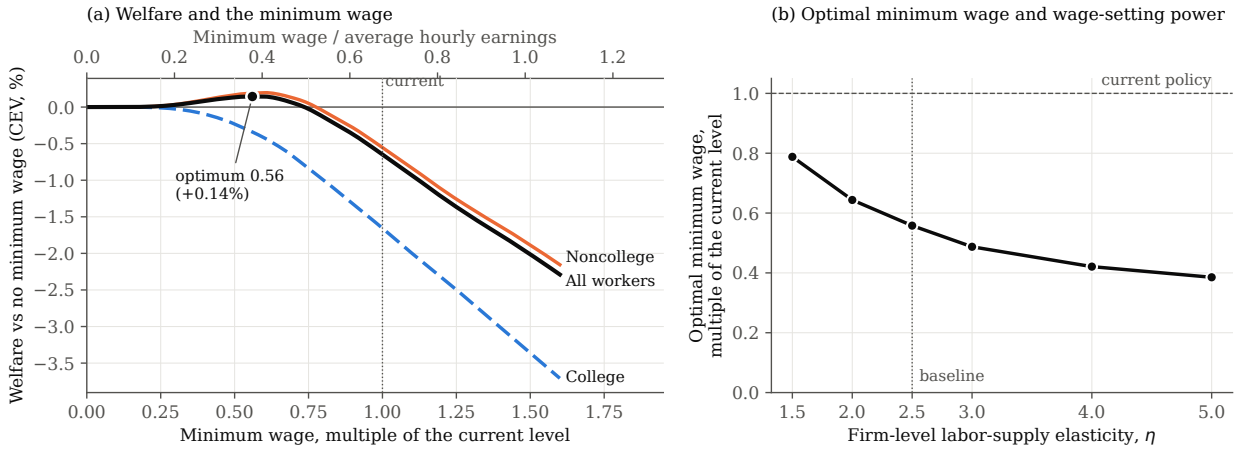
At the current minimum wage, 65.8 percent of workers are informal. Moving from the current policy to the optimum lowers informality by 7.2 percentage points, and removing

[48] Appendix D.8 reports the minimum-wage levels considered, the solution method, tolerances, and initialization checks.

[49] I solve the model for minimum wages from 0 to 1.6 times the current level in steps of 0.01 and report outcomes at the best solved policy, 0.56. Worker welfare is flat near the top. Every solved policy from 0.54 to 0.58 is within 0.001 percentage points of the maximum. Welfare falls faster above this range than below it. It is 0.042 percentage points below the maximum at 0.66 and 0.028 below at 0.46.

the minimum wage altogether lowers it by 8.6. Total employment barely changes. The employment rate is 58.3 percent at the current policy and 58.1 percent both at the optimum and without a minimum wage. The minimum wage therefore acts mainly on the composition of employment between formal wage work, informal wage work, and self-employment rather than on its level.

Figure 9: Welfare and the minimum wage



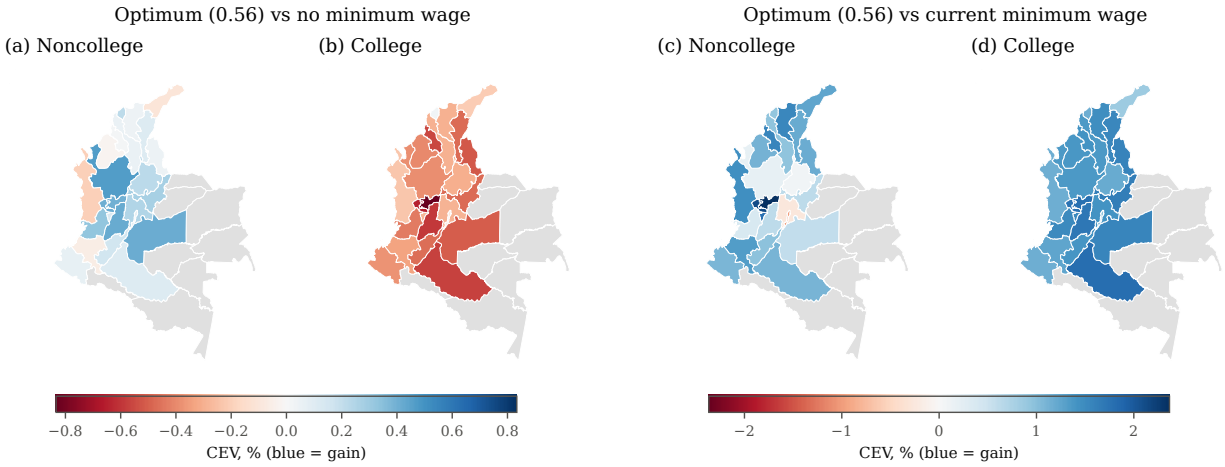
Notes: Panel (a): consumption-equivalent welfare of workers relative to the no-minimum-wage equilibrium under the 2015 tariff schedule, for all workers (utilitarian aggregate), noncollege workers, and college workers. The bottom axis expresses the minimum wage as a multiple of its current level; the top axis reports the same policy relative to average hourly earnings, which are normalized to one in every equilibrium. The minimum wage varies from 0 to 1.6 times its current level in steps of 0.01. The dot marks the optimum (0.56) and the dotted line the current policy. Panel (b): the welfare-maximizing minimum wage, as a multiple of its current level, when the model is solved with the firm-level labor-supply elasticity η changed and all other parameters at their estimates. The dotted line marks the baseline, $\eta = 2.5$.

Relative to no minimum wage, the optimum raises the welfare of noncollege workers by 0.19 percent and lowers that of college workers by 0.34 percent. Noncollege workers receive the direct wage gain because the same statutory policy binds more often for them. At the current policy, both groups lose relative to no minimum wage, noncollege workers by 0.55 percent and college workers by 1.66 percent.

Reducing the minimum wage from its current level to the optimum raises the welfare of noncollege workers by 0.75 percent and of college workers by 1.34 percent. Both groups gain from the reduction, even though college workers would lose from introducing the optimum starting from no minimum wage. Figure 10 reports both comparisons by region and education. Relative to no minimum wage, college workers lose from the optimum in 23 of the 24 regions, while noncollege workers gain in 20. Relative to the current policy, college workers gain in every region, and noncollege workers gain in 22 of the 24.

The positive optimum can reflect both employer wage-setting power and redistribution.

Figure 10: Regional and worker-type incidence of minimum-wage policy



Notes: Each panel reports the consumption-equivalent welfare change of the indicated education group in each region when the minimum wage is set at the national optimum (0.56 times its current level), under the 2015 tariff schedule. Panels (a) and (b) use the no-minimum-wage equilibrium as the reference; panels (c) and (d) use the current minimum wage. Each pair of panels shares one color scale. Gray regions are not in the model.

To gauge the role of wage-setting power, I solve the model again at other values of the firm-level labor-supply elasticity η, holding all other parameters at their estimates. A lower η means more wage-setting power, a larger markdown $\mu = 1 + 1/\eta$. At the baseline $\eta = 2.5$, $\mu = 1.4$, so, before payroll taxes and nonwage benefits, an unconstrained firm pays workers about 71 centavos in wages for each peso of marginal revenue product. Panel (b) of Figure 9 shows the result (Table D.10). The optimal minimum wage falls steadily as wage-setting power weakens, and the welfare gain it delivers falls much faster. At the highest elasticities the optimum is still positive but worth almost nothing. The current policy improves on no minimum wage only at the lowest elasticity. A positive minimum wage is therefore valuable in the model because employers have wage-setting power. Given that power, redistribution toward noncollege workers sets its level. The efficiency terms of the welfare change alone would put the optimum at about 0.35 times the current minimum wage, for a gain of only about 0.007 percent (Table D.11).

Table D.12 reports how the optimum changes under alternative assumptions. In a re-estimated version of the model that also counts formal benefits as worker income, the optimum is about 0.54, and the two near-best ranges overlap. Allowing workers to migrate across regions moves the best policy to about 0.60 to 0.61, but welfare at 0.56 stays within about 0.003 percentage points of the best.

5.5 Trade Policy and the Optimal Minimum Wage

This subsection asks whether the trade regime changes the welfare-maximizing minimum wage. I repeat the exercise of the previous subsection under free trade, the 2015 schedule, the 2009 schedule, and a higher-protection schedule. In each, the minimum wage is defined relative to average earnings, the wage-indexed rule of Section 3.3. The welfare-maximizing ratio is almost the same in all four regimes. κ lies between 0.37 and 0.38 (Figure 11).[50]

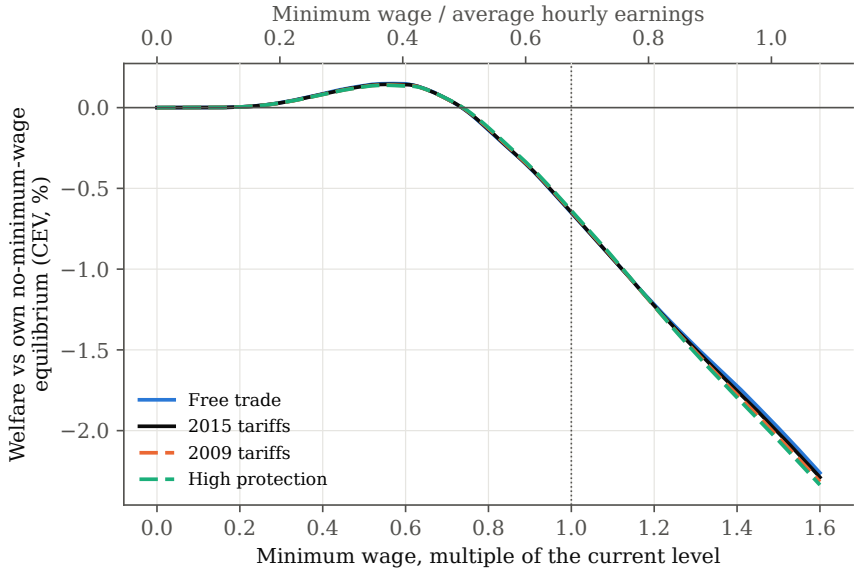
Tariffs matter much more for welfare levels than for the optimal minimum wage. Measured against the 2015 equilibrium with the current minimum wage, worker welfare at the optimal minimum wage is 0.28 percent higher under free trade, 0.80 percent higher under 2015 tariffs, 1.35 percent higher under 2009 tariffs, and 1.80 percent higher under high protection. This ranking comes from the lump-sum rebate of tariff revenue to workers. The trade regime therefore changes how much workers gain, mainly through fiscal transfers, but it does not change which minimum wage maximizes their welfare.

Section 3 shows that trade can change the efficiency case for a minimum wage. The welfare decomposition separates this efficiency effect from redistribution. At a given minimum wage, the labor-wedge term, the gain from limiting wage-setting power, is almost the same in all four regimes. The remaining terms make a slightly higher minimum wage a little more valuable at lower tariffs, but the difference moves the optimum by at most 0.01 times the current minimum wage. The same holds when wage-setting power changes. At every value of η, the four regimes' optima differ by at most 0.02, and their near-best ranges overlap (Table D.14).

International trade costs also move the optimum little. Changing every region's international trade cost by up to 10 percent in either direction leaves the best policy at 0.56. Larger increases raise it to about 0.60, and larger cuts lower it, to 0.54 at a 15 percent cut.[51] Larger cuts make Bogotá's nontradable sector begin to informalize at a lower minimum wage, which shrinks the payroll-tax base that funds transfers to every worker.

[50] Appendix D.9 reports the four regimes side by side.
[51] See Appendix Figure D.5.

Figure 11: Trade policy and the welfare-maximizing minimum wage



Notes: Consumption-equivalent welfare of workers under each tariff regime, relative to the no-minimum-wage equilibrium under the same regime. Mean tariffs are 0 (free trade), 3.5 percent (2015), 7.9 percent (2009), and 12.3 percent (high protection). The bottom axis is the minimum wage as a multiple of its current level; the top axis is the same policy relative to average hourly earnings. The dotted line marks the current policy.

6 Conclusion

This paper studies how the minimum wage shapes informality responses to trade liberalization and how the two policies jointly determine welfare. Combining Colombia's 2010–2011 tariff reform with differences in minimum-wage exposure across local labor markets, I find that a one-percentage-point larger tariff reduction raises informality by 0.8 percentage points where the minimum wage binds more tightly, and has no detectable effect where it binds less tightly. Formal employment near the statutory minimum declines, and informal employment below it rises.

A moderate minimum wage can raise employment by limiting employers' wage-setting power, but when import competition lowers market access, informality rises where enough minimum-wage employment is at firms that ration jobs. An estimated spatial general-equilibrium model reproduces the direction of this heterogeneity, though much more weakly. The welfare-maximizing minimum wage is about 56 percent of the current one, and moving to it raises worker welfare by 0.80 percent. It is nearly the same across tariff regimes, and the minimum wage barely changes the gains from cutting international trade costs by 10 percent, 0.97 percent without a minimum wage and 0.92 percent at the current one. Removing informal firms and self-employment raises that gain to 1.30 percent, because tradable

self-employment competes with imports.

The implication is not that trade liberalization necessarily raises informality or that openness calls for a lower minimum wage. When the minimum wage moves with average earnings, trade barely changes how tightly it binds, so the optimal minimum wage and the gains from trade are nearly separable, even though their local incidence is not. Informality matters for the gains from trade in its own right. Where much of it is self-employment that competes with imports, it lowers workers' gains, the opposite of what models in which informal firms escape taxes predict.

A Appendix to Section 2

A.1 Data and Sample Construction

The GEIH is a monthly repeated cross-section rather than a worker panel. Pooling 2007–2019 produces approximately 3.8 million employed person-month observations before restrictions. January interviews are excluded because the earnings question refers to the previous calendar month. A January interview reports December earnings but would otherwise be compared with the newly adjusted January minimum wage. The analysis then restricts the sample to workers aged 18–65 in the 29 two-digit agriculture, mining, and manufacturing industries carrying tariff lines and drops observations with missing employment status, region, education, industry, or survey weight. Electricity, gas, and water supply (ISIC 40) also carries a tariff line but is excluded, because the tradable sector of the quantitative model covers ISIC 01–37. These steps yield 689,898 worker-month observations in the estimating sample after singleton cells are removed. All descriptive statistics and regressions use the GEIH survey weights, and all regressions include calendar-month fixed effects.

The baseline sample contains agriculture, mining, and manufacturing industries for which a tariff measure can be assigned. It is consequently less urban and more male than either the Colombian workforce as a whole or a manufacturing-only sample. The same industries, plus recycling (ISIC 37), make up the tradable sector of the quantitative model. Appendix A.11 reports manufacturing-only estimates. The customs records and interregional input-output accounts used to estimate the spatial model are described in Section 5.

A.2 Tariff Construction and Industry-Level Changes

UNCTAD-TRAINS reports effectively applied and statutory Most Favored Nation tariff rates at the HS6 product level. Products are mapped to four- and two-digit ISIC Rev. 3 industries, and industry rates are constructed as simple averages over the mapped tariff lines. Figure 2 uses the four-digit measures, while the reduced-form regressions use the two-digit measures. The baseline treatment is the change in the effectively applied industry rate between 2009 and 2011. Appendix A.6 uses the statutory tariff reduction as an instrument. Table A.1 reports all three two-digit measures for each industry.

Three industries experience small increases in effectively applied tariffs between 2009 and 2011. They remain in the sample because they are part of the same tariff-schedule adjustment. Dropping them would select industries on the realized treatment. Electricity, gas, and water supply, whose applied rate rose by 4.9 percentage points, is excluded for a different reason. It lies outside the tradable sector of the quantitative model. The av-

Table A.1: Tariff rates and reductions by industry

ISIC rev.3	Industry	Applied rate (%)		Reduction (pp)			Worker share (%)
		2009	2011	2009–11	2010–11	MFN	
1	Agriculture, hunting, related services	9.28	6.51	2.77	1.66	1.18	49.54
2	Forestry and logging	7.60	5.95	1.65	1.36	1.77	0.71
5	Fishing	5.40	5.35	0.05	1.18	4.33	2.17
10	Mining of coal and lignite	4.94	4.90	0.04	0.07	0.00	0.81
11	Crude petroleum and natural gas	9.99	4.92	5.07	4.26	5.00	0.91
13	Mining of metal ores†	2.41	3.34	-0.93	-0.93	0.00	1.49
14	Other mining and quarrying†	3.62	3.65	-0.03	0.36	0.02	0.60
15	Food products and beverages†	7.84	7.91	-0.07	3.19	1.53	11.06
16	Tobacco products	3.65	1.68	1.97	1.41	5.00	0.12
17	Textiles	14.27	7.20	7.07	8.41	9.14	2.27
18	Wearing apparel and fur	17.83	13.89	3.94	5.41	5.00	7.56
19	Leather, luggage and footwear	16.65	13.04	3.61	5.64	5.04	2.91
20	Wood and wood products	5.87	2.45	3.42	4.05	9.07	0.78
21	Paper and paper products	8.24	4.83	3.41	3.64	4.34	0.70
22	Publishing and printing	6.27	4.75	1.52	4.68	3.05	1.43
23	Coke and refined petroleum	7.46	3.73	3.73	4.02	4.16	0.08
24	Chemicals and chemical products	5.52	4.00	1.52	1.48	2.07	2.45
25	Rubber and plastics	11.73	5.50	6.23	7.51	7.87	1.56
26	Other non-metallic minerals	8.99	4.56	4.43	6.30	7.35	1.73
27	Basic metals	4.94	2.59	2.35	2.54	4.24	0.38
28	Fabricated metal products	11.48	5.36	6.12	6.20	7.05	3.36
29	Machinery and equipment	8.15	5.03	3.12	3.41	3.54	0.85
30	Office and computing machinery	5.01	4.46	0.55	0.39	0.47	0.04
31	Electrical machinery	10.23	5.23	5.00	5.30	5.58	0.36
32	Radio, TV and communication equipment	3.83	3.34	0.49	0.67	2.40	0.04
33	Medical and optical instruments	6.12	4.88	1.24	1.39	1.30	0.33
34	Motor vehicles and trailers	22.01	10.74	11.27	10.43	2.07	0.63
35	Other transport equipment	1.79	1.68	0.11	1.03	0.12	0.26
36	Furniture and manufacturing n.e.c.	15.56	11.33	4.23	4.78	4.17	4.88
Mean		8.51	5.61	2.89	3.44	3.68	3.45
Std. dev.		4.90	3.09	2.68	2.75	2.70	
Worker-weighted mean		10.09	7.29	2.80	2.97	2.62	

Notes: The 29 two-digit ISIC rev.3 tradable industries of the estimation sample. Electricity, gas and water supply (ISIC 40) also carries a tariff line but is excluded, because the quantitative model's tradable sector covers ISIC 01–37. Applied (AHS) and statutory (MFN) rates from UNCTAD TRAINS, simple averages over the HS6 lines mapped to each industry. Reduction is the 2009 rate minus the 2011 rate, so a positive entry is liberalization; the 2010–11 column is the alternative window and the MFN column the instrument. Worker share is the share of the estimation sample's survey-weighted person-months. † marks the three industries whose applied rate rose. The correlation between the 2009–11 reduction and the 2009 level is 0.82.

erage applied reduction is 2.89 percentage points, its standard deviation is 2.68, and the employment-weighted average is 2.80. The statutory and applied reductions share the same broad compression pattern but differ because effectively applied rates incorporate preferential agreements and exemptions.

Table A.2 examines whether the applied tariff reduction is related to pre-reform indus-

try characteristics. The reduction is strongly related to the 2009 tariff level, as expected from the compression of the tariff schedule. By contrast, it is not significantly related to pre-reform informality, its 2007–2009 change, average wages, or the industry's female and tertiary-educated worker shares. The relationship with the urban worker share is positive and marginally significant. The balance evidence therefore indicates little systematic selection on the observed pre-reform characteristics.

Table A.2: Tariff reductions and pre-reform industry characteristics

Pre-reform industry characteristic	Correlation	Permutation p-value
Applied tariff rate, 2009	0.823	< 0.001
Average informality, 2007–2009	−0.246	0.205
Change in informality, 2007–2009	0.031	0.866
Log average hourly wage, 2007–2009	0.187	0.332
Female worker share, 2007–2009	0.100	0.604
Tertiary-educated worker share, 2007–2009	0.046	0.811
Urban worker share, 2007–2009	0.354	0.053

Notes: Each row reports the industry-level correlation between the stated characteristic and the 2009–2011 reduction in the effectively applied tariff, measured in percentage points. Pre-reform worker characteristics are survey-weighted within the 29 two-digit tradable industries of the estimation sample; January interviews are excluded. The change in informality is the 2009 value minus the 2007 value. Permutation p-values are based on 5,000 reassignments of the tariff reduction across industries.

A.3 Labor-Market Statistics for All Sectors

The main-text statistics describe workers in tariff-exposed tradable industries because that is the estimation sample. In the all-sector 2015 data used to construct the quantitative moments, the survey-weighted national informality rate is 69.9 percent and self-employment accounts for 51.3 percent of employment. Table A.3 instead summarizes the cross-sectional distribution of region × education cells in 2009 using workers from all Colombian sectors, including nontradables. Its mean is unweighted across the 96 cells and is therefore not the national aggregate. The table is descriptive and does not extend the tariff specification to nontradable industries, for which no direct industry tariff can be assigned.

A.4 The Minimum Wage, 2007–2019

Figure A.1 deflates the statutory monthly minimum wage by the annual average of the monthly consumer price index. The real minimum wage rose by 18.8 percent between 2007 and 2019. It fell slightly in 2008, 2015, and 2016, when inflation exceeded the nominal

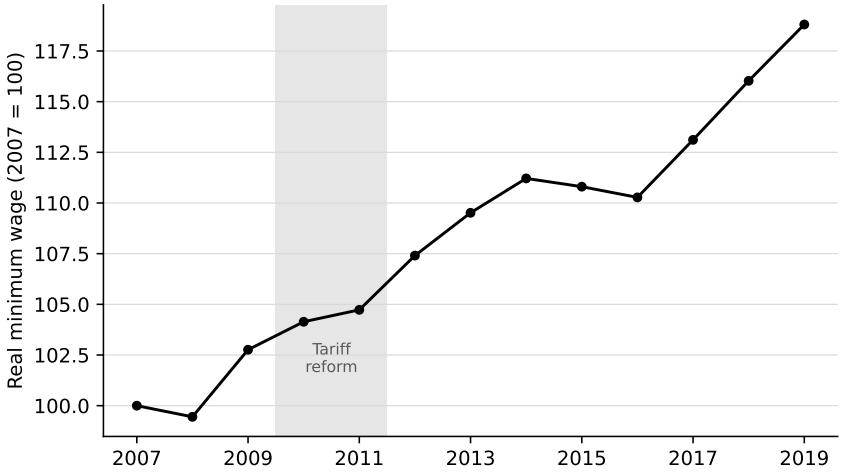
Table A.3: Labor-market statistics across all Colombian sectors, 2009

	Mean	Std. dev.	P10	Median	P90	Cells
Informality rate	0.734	0.190	0.440	0.778	0.951	96
Self-employment rate	0.514	0.159	0.319	0.494	0.726	96
Informal wage employment rate	0.220	0.094	0.105	0.209	0.340	96
Share of formal salaried workers within 1.10 MW	0.447	0.206	0.146	0.502	0.680	96
Kaitz index (MW / median formal salaried wage)	0.820	0.208	0.488	0.911	1.000	96
Mean formal salaried hourly wage / MW	1.733	0.905	1.067	1.293	3.053	96
Mean informal wage-worker hourly wage / MW	1.048	0.392	0.700	0.871	1.682	96

Notes: The unit of observation is a region × education cell. The table reports the unweighted distribution across 96 cells in the 2009 GEIH, February–December, with survey-weighted cell means. Wages are hourly and expressed relative to the hourly minimum wage. The Kaitz index is the hourly minimum wage over the cell's survey-weighted median hourly wage of formal salaried workers. Informality is the sum of self-employment and informal wage employment. The table includes workers in tradable and nontradable sectors.

adjustment, and rose by about 2.5 percent per year in 2017–2019. Around the tariff reform, it rose by 1.3 percent in 2010 and 0.6 percent in 2011.

Figure A.1: Real minimum wage, 2007–2019



Notes: The statutory monthly minimum wage deflated by the annual average of the monthly DANE consumer price index, indexed to 2007 = 100. The shaded area marks the 2010–2011 tariff reform.

A.5 Construction and Variation of Minimum-Wage Exposure

For each region d and education group g, baseline exposure is

$$\text{Exposure}_{dg} = \frac{\sum_{i\in(d,g)} \omega_i \mathbf{1}\{i \text{ is a formal salaried worker and } w_i \leq 1.10\, MW_t\}}{\sum_{i\in(d,g)} \omega_i \mathbf{1}\{i \text{ is a formal salaried worker}\}},$$

where ω_i is the survey weight and the sums pool 2007–2009. Hourly earnings and the hourly equivalent of the contemporaneous minimum wage are used. Averaging over three years reduces sampling noise and ensures that the measure precedes the tariff reform.

The baseline cell groups workers by region and education but not by industry. Workers can move across industries within a region, while education captures differences in skills and wage distributions that determine how tightly the national minimum wage binds. Excluding industry from the baseline cell also keeps minimum-wage exposure distinct from the industry-level tariff treatment. Appendix A.10 alternatively measures exposure at the industry × region and industry × region × education levels. These finer definitions allow exposure to vary by industry but are estimated from substantially smaller samples.

The formal-salaried denominator is chosen to measure the concentration of compliant jobs near the statutory minimum wage. A denominator containing informal workers would also vary with the prevalence of informal employment. Appendix A.10 evaluates that alternative denominator, thresholds at 1.00 and 1.05 times the minimum wage, continuous exposure, terciles, Kaitz-index measures, and the alternative cell definitions.

The baseline exposure measure is estimated precisely at the region × education level. A split-sample calculation produces a correlation of 0.96 between exposure computed on two random halves of the pre-reform observations, and the sampling-variance reliability is approximately 0.99. The within-education component is also precisely measured, with an estimated reliability of approximately 0.97. For this reason, the empirical analysis reports both the pooled comparison and specifications that examine variation within education groups.

A.6 Interaction Specification, Instruments, and Inference

The dynamic version of equation (1) is

$$\begin{aligned}\text{Informality}_{ijdgt} = &\sum_{h\neq 2010} \beta_{L,h}\mathbf{1}[t=h]\Delta\tau_j + \sum_{h\neq 2010} \beta_{H,h}\mathbf{1}[t=h]\Delta\tau_j \text{High}_{dg} \\ &+ \mathbf{X}_i'\mathbf{\Gamma}_t + \alpha_{jdg} + \delta_{dgt} + \varepsilon_{ijdgt}. \end{aligned} \tag{A.1}$$

The two paths plotted in Figure 4 are $\beta_{L,h}$ and $\beta_{L,h} + \beta_{H,h}$. The differential is $\beta_{H,h}$. The outcome and tariff reduction are expressed in percentage points, so each coefficient is the percentage-point change in informality associated with a one-percentage-point tariff reduction.

The fixed-effect ladder in Table A.4 shows why the comparison must hold fixed the type of worker employed in a particular industry and region. With only exposure-group × year effects and the tariff-reduction levels, the static differential is negative. Adding industry ×

region × education effects changes its sign and moves it to 0.64. It remains between 0.64 and 0.71 as region × year and education × year and then region × education × year effects are added. The result therefore requires controls for persistent education composition within industries, but it does not depend on the final, most saturated local-year effects.

Table A.4: Fixed-effect ladder

	Low path		High path		Gap		Static	
	2007–09	2011–19	2007–09	2011–19	2007–09	2011–19	differential	N
A. Region × education exposure (headline split)								
L0	−0.494	0.508	0.615	1.203	1.109	0.695	−0.152	690,030
	(0.286)	(0.309)	(0.516)	(0.363)	(0.552)	(0.483)	(0.560)	
L1	−0.315	−0.048	0.338	1.086	0.652	1.134	0.636	689,898
	(0.202)	(0.254)	(0.334)	(0.352)	(0.336)	(0.450)	(0.298)	
L2	−0.289	−0.082	0.376	1.087	0.665	1.169	0.661	689,898
	(0.186)	(0.230)	(0.280)	(0.348)	(0.298)	(0.431)	(0.275)	
L3 (headline)	−0.229	−0.076	0.405	1.116	0.634	1.193	0.706	689,898
	(0.230)	(0.261)	(0.261)	(0.326)	(0.280)	(0.420)	(0.273)	
B. Education split: primary or lower secondary								
L0	−0.309	0.207	0.456	1.302	0.765	1.095	0.492	690,030
	(0.257)	(0.200)	(0.487)	(0.339)	(0.400)	(0.310)	(0.294)	
L1	−0.224	−0.045	0.259	1.093	0.484	1.138	0.763	689,898
	(0.231)	(0.221)	(0.273)	(0.376)	(0.205)	(0.436)	(0.325)	
L2	−0.218	−0.087	0.318	1.137	0.536	1.224	0.809	689,898
	(0.203)	(0.199)	(0.226)	(0.377)	(0.185)	(0.428)	(0.308)	
L3 (headline)	−0.152	−0.080	0.340	1.173	0.492	1.253	0.869	689,898
	(0.249)	(0.228)	(0.232)	(0.355)	(0.174)	(0.400)	(0.293)	

Notes: Two-path OLS event studies and the matching static regression under four fixed-effect sets, from loose to tight. L0: High × year cells, with the dose level and the dose × High level entered as regressors. L1: L0 plus industry × region × education cells. L2: L1 plus region × year and education × year cells (for the education split High × year is then spanned). L3: industry × region × education and region × education × year cells, the headline specification of Table 2. Every rung includes the demographic controls × year and calendar-month fixed effects, survey weights and two-way clustering by industry and region; base year 2010. Entries: percentage points of informality per percentage point of applied cut; standard errors in parentheses. The static differential is the coefficient on dose × post × High.

The statutory MFN reduction provides an alternative source of tariff variation. Both the applied and statutory reductions arise from the compression of the tariff schedule, but the applied tariff also incorporates preferential agreements and exemptions. At the industry level, the first-stage regression of the applied reduction on the statutory reduction has a slope of 0.60, a standard error of 0.15, and an F-statistic of 15.1. In the worker-level static regression, however, the Kleibergen–Paap F-statistic is 4.79, so the instrument is weak by conventional standards. The two instruments are individually strong, with Sanderson–Windmeijer F-statistics of 30 and 41, but nearly collinear, because the second is the first interacted with High. Appendix Table A.5 reports the static coefficients for the headline OLS specification alongside the corresponding instrumented estimates. The static instrumented differential is 0.69, close to the OLS estimate of 0.71.

Table A.5: OLS and instrumented static interaction estimates

	OLS (1)	Statutory IV (2SLS) (2)
Static interaction specification, post = 2011–2019		
Low exposure: dose × post (β_L, slack)	0.088	0.154
	(0.145)	(0.198)
Differential: dose × post × High (β_H)	0.706	0.691
	(0.273)	(0.311)
High exposure: $\beta_L + \beta_H$ (binds)	0.795	0.844
	(0.250)	(0.341)
Observations	689,898	689,898
Kleibergen–Paap rk Wald F, static		4.79
Industry × region × education FE	Yes	Yes
Region × education × year FE	Yes	Yes
Demographic controls × year; calendar-month FE	Yes	Yes
Sample years	2007–2019	2007–2019

Notes: Worker-level regressions of the informality indicator on the applied 2009–2011 tariff cut of the worker's industry (dose, percentage points) interacted with post (2011–2019) and with High, the indicator for high minimum-wage-exposure region × education cells (share of formal salaried workers paid at or below 1.10 times the hourly minimum wage, 2007–2009). The sample contains 29 tradable industries. Entries: percentage points of informality per percentage point of cut; high exposure = $\beta_L + \beta_H$. Column (2) instruments the applied cut with the statutory MFN cut. Controls: year-specific age, age squared, male, age × male, literacy, household head, married and urban, plus calendar-month fixed effects; survey weights. Parentheses contain standard errors two-way clustered by 29 industries and 24 regions. Small-cluster inference based on a collapsed-cell analogue is reported separately in the text.

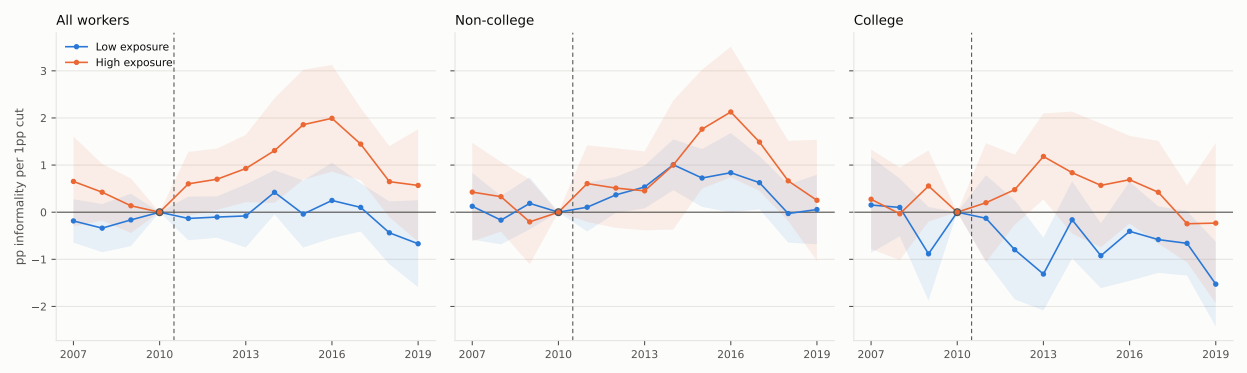
Inference is difficult because treatment varies across only 29 industries and the specification includes 24 regions. The conventional standard errors in Table A.5 are two-way clustered by industry and region, but cluster asymptotics may be optimistic with so few clusters. As additional checks, I estimate a covariate-free analogue on an industry × region × education × year panel of cell means, using the same fixed effects and cell-size weights. Its static differential is 0.75, compared with 0.71 in the worker-level regression. For this collapsed-cell estimate, the wild-cluster-bootstrap p-value is 0.083. Permutation p-values are 0.017 unconditionally and 0.006 conditional on the 2009 applied tariff. These values are reported separately because they do not correspond exactly to the worker-level coefficient in Table A.5.

A.7 Dynamics and Pre-Reform Diagnostics

Figures A.2–A.4 report the interaction event study for the full sample and the noncollege and college subsamples under three functional forms, with exposure measured at the region × education level. In every sample and functional form, informality rises after the reform in the most exposed group. The high-minus-low differential is positive in every case except college workers under terciles, where the least exposed tercile also rises and its path is imprecisely

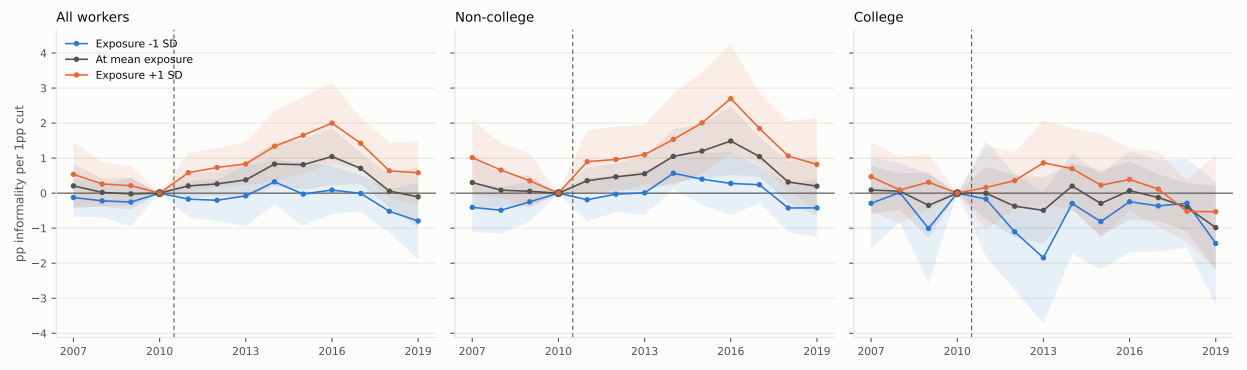
estimated.

Figure A.2: Event studies using a binary exposure split



Notes: Panels show the full sample, noncollege workers, and college workers, with exposure measured at the region × education level. Each panel plots the informality response to a one-percentage-point applied tariff reduction in each year, with 2010 omitted. Shaded bands are 90 percent confidence intervals based on standard errors two-way clustered by industry and region.

Figure A.3: Event studies using continuous exposure, standardized to one standard deviation

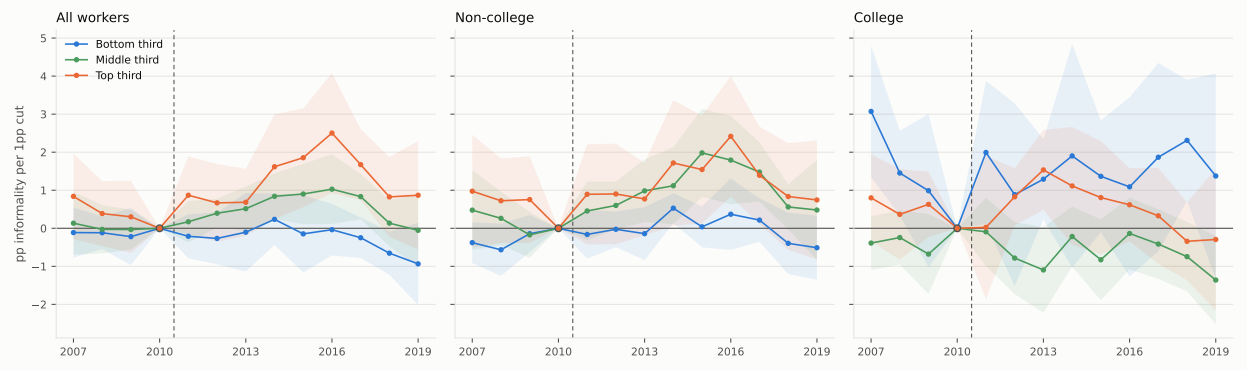


Notes: Conventions follow Figure A.2.

The baseline high-minus-low differential averages 0.634 over 2007–2009 and 1.193 over 2011–2019. The pre-reform average is 53 percent of the post-reform average. Under conventional clustered inference the 2007 coefficient (0.84) and the pre-reform average are significant at the 5 percent level, whereas the 2008 and 2009 coefficients are not. The wild-bootstrap p-value of the collapsed-cell pre-reform average is 0.274. Subtracting the pre-reform average from the post-reform average gives 0.559, a conservative magnitude. A linear extrapolation of the three pre-reform coefficients is not informative here because normalization at zero in 2010 converts a persistent level difference into an apparent slope. I therefore treat the pre-period as evidence of a possible level difference and do not use a linear trend adjustment.

Minimum-wage exposure is primarily an education measure with a smaller regional component. Of its variance, 82.2 percent lies between education groups. Estimating the original

Figure A.4: Event studies comparing exposure terciles



Notes: Conventions follow Figure A.2.

split separately by education produces positive post-reform differentials for both college and noncollege workers. A stricter specification that standardizes exposure within education and therefore uses only regional differences gives a positive estimate for college workers but not for noncollege workers. Pooling those strict within-education estimates yields −0.16 with a p-value of 0.325 because the two subsamples have opposite signs. The data therefore support heterogeneity across region × education labor markets, but do not establish that the interaction is driven by regional variation independently of education.

A.8 Self-Employment and Informal Wage Employment

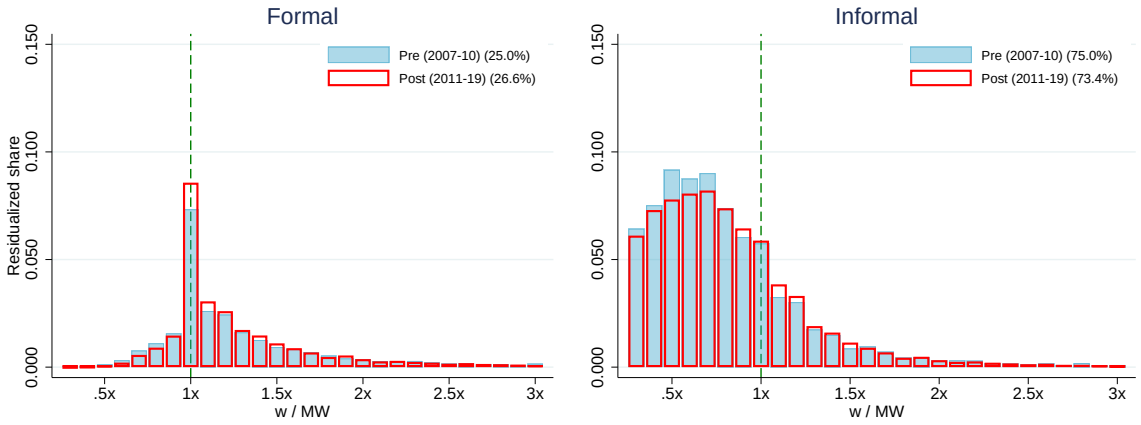
Informality is the sum of self-employment and informal wage employment, so re-estimating equation (1) for the two components decomposes the headline estimate exactly. Under OLS, the self-employment differential is 0.89 with a conventional p-value of 0.001, while the informal-wage-employment differential is −0.19 with a p-value of 0.301. The self-employment event study has a pre-reform average of 0.15 and a post-reform average of 1.04. The additional informal employment in high-exposure labor markets is therefore concentrated in own-account work.

A.9 Wage-Distribution Evidence

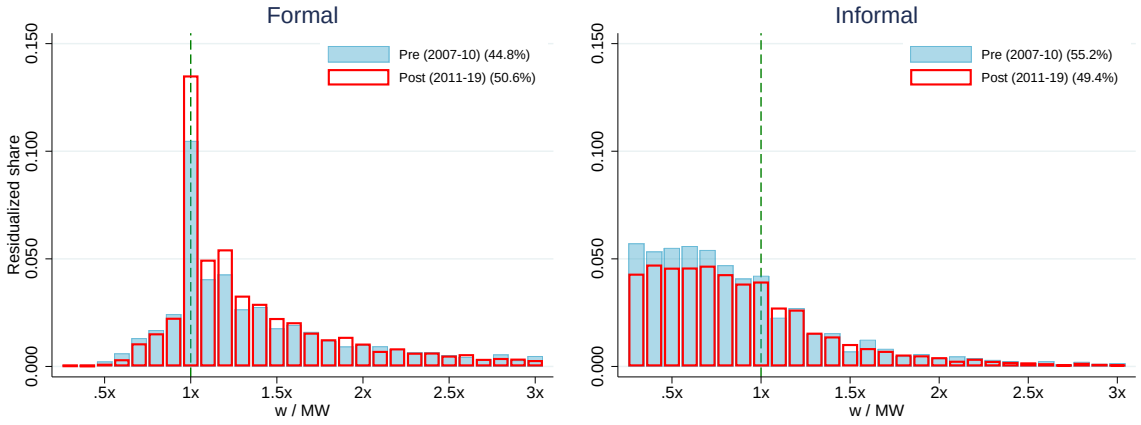
The main text estimates equation (2) separately for 28 earnings bins of width 0.10 covering $0.25 \leq w/\text{MW} < 3.05$. Table A.6 reports the same specification for broader wage intervals, pooled and separately by education group. Table A.7 compares alternative fixed effects and a fixed-denominator outcome, and Figure A.6 and Table A.8 report dynamic versions of the broader-band regressions.

Figure A.5 plots the levels behind the coefficient estimates. Both groups become more formal over time. The identifying variation is therefore not the change within the group receiving larger reductions alone, but its change relative to the group receiving smaller reductions.

Figure A.5: Residualized wage distributions before and after the reform, by size of the tariff reduction



(a) Industries receiving larger tariff reductions



(b) Industries receiving smaller tariff reductions

Notes: Each panel plots the share of all workers in the relevant industries located in each earnings bin, separately for formal and informal status. The dashed line marks the minimum wage.

Table A.6 reports the wage-band estimates. Formal employment falls by 0.28 percentage points per percentage point of tariff reduction in the band containing the minimum wage and by 0.14 in the band immediately above it. Neither is significant on its own under the wild bootstrap. Total formal employment falls by 0.63 ($p = 0.10$), and informal employment below the minimum wage rises by 0.64 ($p = 0.02$). The responses come from workers with primary

and lower-secondary education. In these groups, formal employment in the minimum-wage band falls by 0.33 and 0.70, and informal employment below the minimum wage rises by 0.94 and 0.74. Both responses are small for workers with upper-secondary or tertiary education.

Table A.6: Differential employment changes by wage band

	Pooled	Primary	Lower secondary	Upper secondary	Tertiary
A. Applied tariff cut × post (percentage points per percentage point of cut)					
Formal, $w/\text{MW} < 0.95$	−0.10 {0.164}	−0.23 {0.050}	−0.03 {0.655}	−0.13 {0.386}	0.10 {0.230}
Formal, $0.95 \leq w/\text{MW} < 1.15$ (minimum wage)	−0.28 {0.234}	−0.33 {0.295}	−0.70 {0.065}	−0.22 {0.207}	0.05 {0.739}
Formal, $1.15 \leq w/\text{MW} < 1.65$	−0.14 {0.599}	−0.21 {0.388}	−0.25 {0.482}	0.26 {0.562}	−0.28 {0.190}
Formal, $1.65 \leq w/\text{MW} < 2.15$	−0.10 {0.013}	−0.10 {0.177}	0.01 {0.965}	−0.09 {0.467}	0.09 {0.288}
Formal, $w/\text{MW} \geq 2.15$	−0.01 {0.783}	0.00 {0.967}	−0.06 {0.480}	0.07 {0.371}	0.02 {0.914}
Formal, all wages	−0.63 {0.104}	−0.87 {0.125}	−1.04 {0.088}	−0.10 {0.521}	−0.02 {0.862}
Self-employed, $w/\text{MW} < 0.95$	0.72 {0.383}	1.42 {0.144}	0.54 {0.749}	0.21 {0.847}	0.30 {0.286}
Informal wage worker, $w/\text{MW} < 0.95$	−0.07 {0.849}	−0.48 {0.384}	0.20 {0.681}	0.08 {0.812}	−0.08 {0.553}
Informal (wage or self-employed), $w/\text{MW} < 0.95$	0.64 {0.023}	0.94 {0.018}	0.74 {0.083}	0.30 {0.091}	0.22 {0.284}
B. Pre-reform level, 2007–2010 (percent of the group's workers)					
Formal, $w/\text{MW} < 0.95$	4.5	3.4	5.3	6.5	5.0
Formal, $0.95 \leq w/\text{MW} < 1.15$ (minimum wage)	10.9	5.8	12.9	20.5	14.9
Formal, $1.15 \leq w/\text{MW} < 1.65$	8.4	3.8	7.9	15.3	20.4
Formal, $1.65 \leq w/\text{MW} < 2.15$	3.0	0.9	2.4	4.6	12.9
Formal, $w/\text{MW} \geq 2.15$	2.3	0.4	1.2	2.9	14.7
Formal, all wages	29.1	14.3	29.7	49.9	67.9
Self-employed, $w/\text{MW} < 0.95$	29.9	39.4	26.8	17.5	8.3
Informal wage worker, $w/\text{MW} < 0.95$	20.6	26.7	19.6	13.0	4.5
Informal (wage or self-employed), $w/\text{MW} < 0.95$	50.5	66.1	46.4	30.5	12.8
Observations	572,274	229,522	93,045	165,082	84,625
Industry clusters	29	29	29	29	29

Notes: Worker-level linear probability models for membership of a status × wage band, pooled and by education group, with the specification of equation (2): the applied 2009–2011 tariff cut of the worker's industry interacted with post (2011–2019); industry × region and region × year fixed effects; age, age squared, sex, age × sex, literacy, household head, marital status, urban status and education, each interacted with year; calendar-month fixed effects. Survey weights; standard errors clustered by industry. Braces: wild cluster bootstrap p-value (Webb weights, 9,999 draws, industry clusters). The minimum-wage band $[0.95, 1.15)$ and the three formal bands above it, $[1.15, 1.65)$, $[1.65, 2.15)$ and $[2.15, 3.05)$, are fixed by rule as unions of 0.10-wide bins rather than chosen from the estimates, and the five formal bands partition formal employment. Panel B: weighted means over 2007–2010. Sample: GEIH 2007–2019, tradable industries, January excluded, $0.25 \leq w/\text{MW} < 3.05$.

Table A.7 shows that the signs of these estimates do not depend on the fixed effects. With industry, year, region and calendar-month fixed effects and level controls, total formal employment falls by 0.87 and informal employment below the minimum wage rises by 0.98, both more precisely estimated. Adding region × year, region × education × year or industry × region fixed effects moves the estimates toward the main specification. Columns (6) and (7) follow Cengiz et al. (2019) and measure jobs relative to the industry's pre-reform employment, so that changes in industry size are not absorbed by the denominator. The estimates keep their signs and total employment changes little, but they are less precise and, under the main specification, not statistically significant.

Figure A.6 and Table A.8 report the dynamic estimates. Total formal employment and informal employment below the minimum wage have small pre-reform averages, −0.02 and

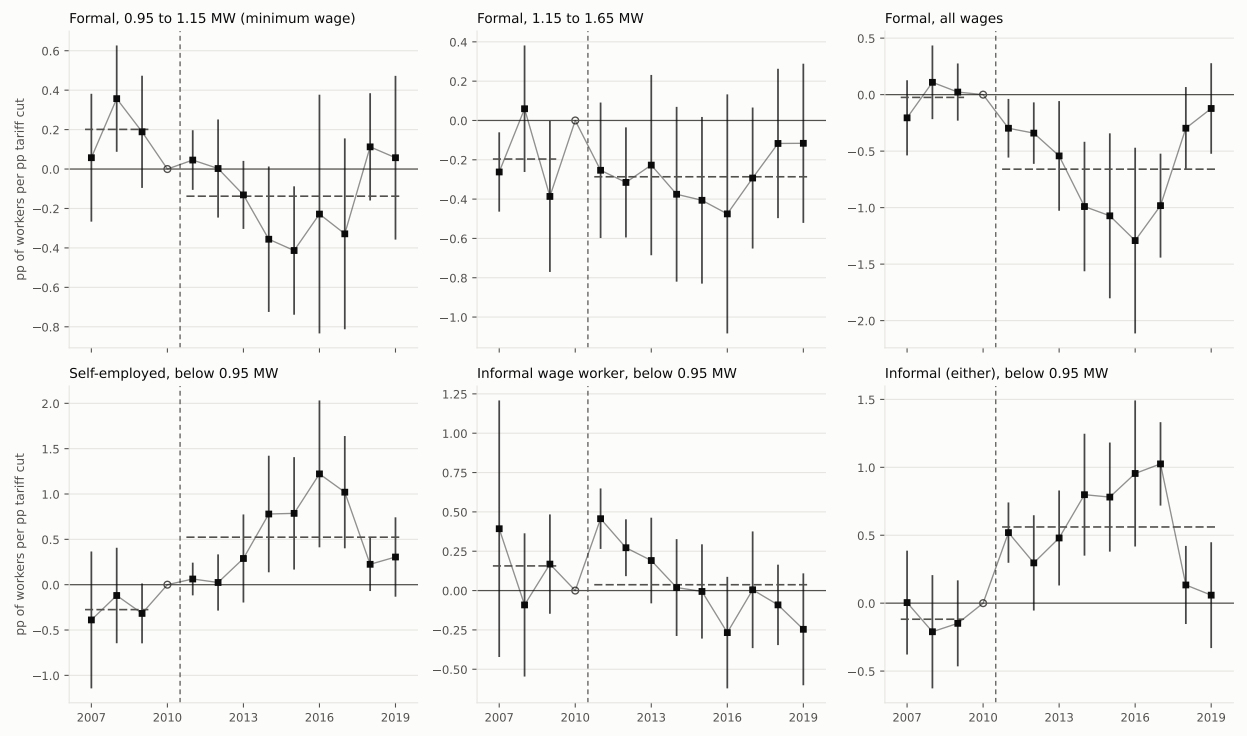
Table A.7: Wage-distribution estimates under alternative specifications

	Share of the industry's workers (pp)					Percent of 2007–2009 employment	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Formal, all wages	−0.63 (0.22) {0.104}	−0.60 (0.21) {0.087}	−0.77 (0.25) {0.052}	−0.82 (0.23) {0.025}	−0.87 (0.26) {0.018}	−0.99 (0.48) {0.195}	−0.94 (0.59) {0.279}
Formal, $0.95 \leq w/\text{MW} < 1.05$ (minimum-wage bin)	−0.25 (0.09) {0.121}	−0.23 (0.12) {0.237}	−0.29 (0.10) {0.074}	−0.28 (0.11) {0.118}	−0.23 (0.15) {0.340}	−0.34 (0.18) {0.206}	−0.23 (0.26) {0.547}
Formal, $0.95 \leq w/\text{MW} < 1.65$	−0.42 (0.27) {0.482}	−0.37 (0.31) {0.591}	−0.54 (0.29) {0.287}	−0.53 (0.31) {0.318}	−0.52 (0.34) {0.329}		
Informal, $w/\text{MW} < 0.95$	0.64 (0.16) {0.022}	0.67 (0.15) {0.001}	0.78 (0.17) {0.001}	0.84 (0.19) {0.002}	0.98 (0.24) {0.002}	0.47 (0.30) {0.348}	0.69 (0.30) {0.050}
Total employment						−0.49 (0.76) {0.572}	−0.28 (0.85) {0.767}
Industry × region FE	Yes	Yes				Yes	
Industry FE			Yes	Yes	Yes		Yes
Region × year FE	Yes	Yes		Yes		Yes	
Region × education × year FE			Yes				
Region and year FE					Yes		Yes
Demographics and education × year	Yes					Yes	

Notes: Worker-level regressions on the applied 2009–2011 tariff cut interacted with post. Column (1) is the specification of equation (2) (industry × region and region × year fixed effects; age, age squared, sex, age × sex, literacy, household head, marital status, urban status and education, each interacted with year; calendar-month fixed effects). Columns (2)–(5) and (7) use age, age squared, sex, urban status and education as level controls and calendar-month fixed effects. Columns (1)–(5): the outcome is an indicator for the status and wage range, so coefficients are percentage points of the industry's workers per percentage point of cut. Columns (6)–(7): following Cengiz et al. (2019), the outcome counts jobs relative to the industry's average 2007–2009 employment, so coefficients are percent of pre-reform employment and the rows need not sum to zero; the effect on total employment is reported in the last row. Standard errors clustered by industry in parentheses; braces: wild cluster bootstrap p-value (Webb weights, 9,999 draws, industry clusters). Sample as in Table A.6.

−0.12. After 2010 they move to −0.66 and 0.56 (wild-bootstrap $p = 0.10$ and 0.02). The formal minimum-wage band has a positive pre-reform average of 0.20 ($p = 0.12$) and turns negative after the reform. The split of below-minimum-wage informality is imprecise. Self-employment rises after the reform (0.52, $p = 0.24$) while informal wage work does not, but neither average is significant. The wage-bin analysis therefore does not identify the composition of informal adjustment.

Figure A.6: Event-study estimates by wage band



Notes: Each path reports the year-specific coefficient on the applied tariff reduction, with 2010 omitted. Whiskers are 90 percent confidence intervals clustered by industry. Dashed horizontal lines mark the 2007–2009 and 2011–2019 averages.

Table A.8: Pre- and post-reform averages by wage band

	2007–2009 average	2011–2019 average
Formal, $0.95 \leq w/\text{MW} < 1.15$ (minimum wage)	0.20 0.085 {0.124}	−0.14 0.389 {0.529}
Formal, $1.15 \leq w/\text{MW} < 1.65$	−0.20 0.174 {0.233}	−0.29 0.217 {0.367}
Formal, all wages	−0.02 0.855 {0.843}	−0.66 0.009 {0.098}
Self-employed, $w/\text{MW} < 0.95$	−0.27 0.375 {0.758}	0.52 0.045 {0.241}
Informal wage worker, $w/\text{MW} < 0.95$	0.16 0.584 {0.667}	0.04 0.795 {0.806}
Informal (wage or self-employed), $w/\text{MW} < 0.95$	−0.12 0.537 {0.588}	0.56 0.001 {0.023}
Observations	572,274	
Industry clusters	29	

Notes: Event-study versions of the pooled band regressions of Table A.6: the applied tariff cut interacted with year indicators for 2007–2019, 2010 omitted, with the same fixed effects, controls, survey weights and industry clustering. Entries are averages of the yearly coefficients relative to 2010, in percentage points per percentage point of cut. Second line of each row: cluster-robust p-value, and in braces the wild cluster bootstrap p-value (Webb weights, 9,999 draws, industry clusters) for the same average. Sample as in Table A.6.

A.10 Alternative Measures of Minimum-Wage Exposure

Table A.9 reports estimates under alternative definitions of exposure. Changing the threshold from 1.10 to 1.05 times the minimum wage does not change the median classification of the 96 cells and therefore leaves the binary estimate unchanged. A threshold equal to the minimum wage produces a somewhat smaller differential. The result remains positive when exposure is entered continuously, divided into terciles, or replaced with the minimum wage relative to the cell's median formal wage. Measuring exposure in finer industry × region × education cells gives a differential of 0.71, the same as the baseline. Measuring it in industry × region cells, which pool education groups, gives 0.06, statistically indistinguishable from zero. Both sets of cell means are estimated from substantially smaller pre-reform samples.

Table A.9: Alternative definitions of minimum-wage exposure

Definition of exposure	Differential	Std. err.	p
A. Threshold, formal salaried denominator			
1.10× MW (baseline)	0.706	(0.273)	0.017
1.05× MW	0.706	(0.273)	0.017
1.00× MW	0.551	(0.240)	0.031
B. Threshold, all salaried in the denominator			
1.10× MW	0.258	(0.310)	0.414
1.05× MW	0.258	(0.310)	0.414
1.00× MW	0.301	(0.353)	0.403
C. Alternative measure			
Kaitz: MW / cell median formal wage	0.829	(0.241)	0.002
D. Functional form (threshold 1.10*, formal denominator)*			
Continuous, per standard deviation	0.409	(0.159)	0.017
Terciles, top minus bottom	1.022	(0.415)	0.022
E. Cell at which exposure is measured			
Region × education (baseline)	0.706	(0.273)	0.017
Industry × region × education	0.712	(0.330)	0.042
Industry × region	0.059	(0.269)	0.827

Notes: Each row is the differential β_H from the static interaction specification, in percentage points of informality per percentage point of applied tariff reduction, re-estimated with the stated definition of minimum-wage exposure and with the demographic controls absorbed, so every row is comparable with Table 2. Standard errors two-way clustered by industry and region. Panels A to C use the median split. The 1.10 and 1.05 rows of panel A coincide because, although the two measures correlate 0.9975 rather than being equal, a median split assigns all 96 cells identically under both; their continuous counterparts differ (0.409 and 0.388 per standard deviation).

The finer measures are noisier. The median industry × region cell contains 50 person-month observations and the median industry × region × education cell contains 14. Their

split-half correlations are approximately 0.44 and 0.46. Measurement error attenuates the estimates based on these cells.

The denominator matters. When exposure is computed among all salaried workers, the differential falls to 0.26 and has a p-value of 0.414. The baseline formal-salaried denominator measures the concentration of legally covered jobs near the minimum wage. The alternative denominator also moves with the fraction of workers outside covered formal employment. Because informality is the outcome, this distinction is substantive rather than a mechanical robustness choice.

A.11 Alternative Samples and Industry Scope

The baseline analysis uses 29 tradable two-digit industries in agriculture, mining, and manufacturing. Restricting the sample to the 22 manufacturing industries produces a larger static differential, 0.93 rather than 0.71, and a larger post-reform average, 1.54 rather than 1.19. The manufacturing estimates also exhibit a more pronounced pre-reform differential. The pre-period average is 0.80 and has a wild-bootstrap p-value of 0.016. In the baseline sample, the corresponding pre-period average is 0.63, and the wild-bootstrap p-value of its collapsed-cell analogue is 0.274. I therefore use the broader tradable sample as the baseline because its identifying diagnostic is more favorable and because selecting manufacturing would lead with the larger estimate.

The conservative post-minus-pre differential is 0.74 in manufacturing and 0.56 in the baseline sample. Adding electricity, gas, and water supply (ISIC 40), the one remaining industry with a tariff line, lowers the static differential to 0.61 and the post-minus-pre differential to 0.47. The three scopes therefore remain similar in order of magnitude after subtracting the pre-reform gap. The four-digit tariff data display the same compression of the tariff schedule documented in Section 2, but the baseline worker-level treatment is aggregated to two-digit industries to limit sampling noise and preserve a sufficient number of observations in each industry $\times$ region $\times$ education cell.

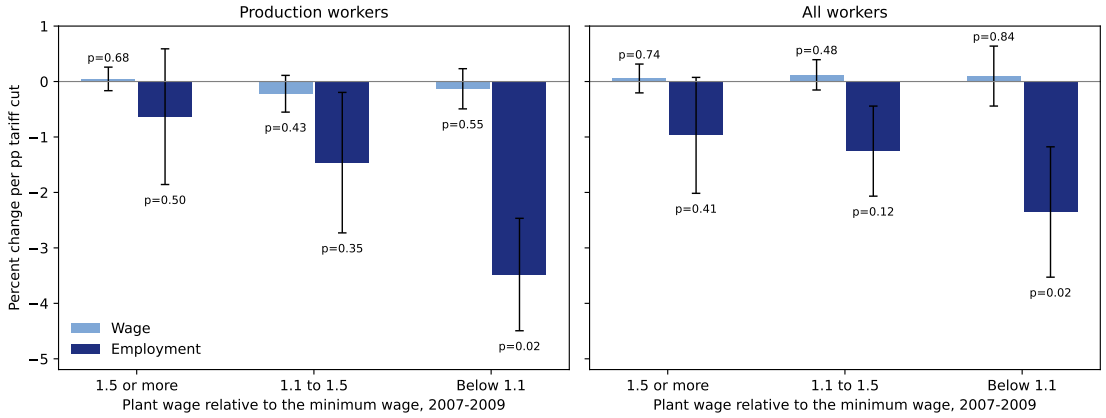
A.12 Plant-Level Evidence from Manufacturing

As a complement to the worker-level evidence, I use the Annual Manufacturing Survey (EAM), which covers formal manufacturing plants with ten or more workers and reports employment and pay separately for production and administrative workers. Tariffs faced by Colombian manufacturing exports averaged about 1 percent in 2010 and changed little through 2016, so the reform mainly changed plants' domestic market access. I classify plants into three groups by the average 2007–2009 monthly pay of their production workers relative

to the minimum wage. The groups pay below 1.1 times the minimum wage, between 1.1 and 1.5 times, and 1.5 times or more. For each group, I estimate the response of log production pay per worker and log production employment to the plant's two-digit tariff reduction interacted with Post, with plant and region × year fixed effects. Plants are assigned to their 2009 industry, and standard errors are clustered by the 22 manufacturing industries.[52]

Figure A.7 shows that the responses differ across groups as the rationing regime of Section 3 predicts. At plants paying production workers close to the minimum wage, a one-percentage-point larger tariff reduction lowers production employment by 3.5 percent (wild-bootstrap $p = 0.02$), while production pay is unchanged (-0.1 percent, $p = 0.55$). At plants paying 1.5 times the minimum wage or more, neither employment nor pay changes significantly. For all workers, employment falls by 2.4 percent at plants close to the minimum wage ($p = 0.02$).

Figure A.7: Wage and employment responses by plant pay relative to the minimum wage

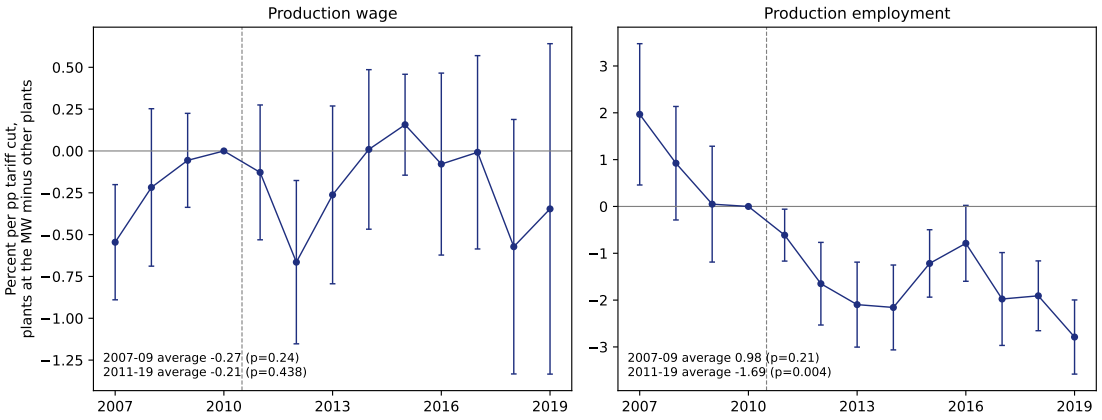


Notes: Each pair of bars reports the response of log pay per worker and log employment to a one-percentage-point larger two-digit tariff reduction after 2010, for plants grouped by their 2007–2009 average pay relative to the minimum wage. Whiskers are 90 percent confidence intervals clustered by industry, and p-values are from the wild cluster bootstrap.

Figure A.8 reports the event-study version. After 2010, production employment at plants close to the minimum wage falls by 1.7 percent per percentage point of tariff reduction relative to other plants ($p = 0.004$). Before the reform, the average difference is not statistically significant (0.98, $p = 0.21$). Production pay at these plants shows no differential change before or after 2010.

[52]The sample keeps plants observed in 2008 and 2009 with at least ten workers in 2009. Plant-years with production pay below half the minimum wage or above the 99.5th percentile are set to missing, as are one-year changes in log employment or pay larger than one in absolute value that reverse in the following year. Weighting plants by their 2007–2009 employment gives a similar employment decline at plants near the minimum wage for all workers (-3.7 percent, $p = 0.03$). In that specification, production pay also falls at plants paying 1.5 times the minimum wage or more (-0.3 percent, $p = 0.04$).

Figure A.8: Event-study estimates for plants close to the minimum wage



Notes: Each point is the year-specific difference in the response to the tariff reduction between plants paying production workers below 1.1 times the minimum wage and other plants, with 2010 omitted. Whiskers are 90 percent confidence intervals clustered by industry.

B Appendix to Section 3

B.1 Household Problem and General-Equilibrium Closure

The representative household consumes the final good and supplies labor to differentiated jobs. Utility is

$$U(C, N_w) = \frac{C^{1-\sigma_c}}{1-\sigma_c} - \bar{\phi}^{-1/\psi} \frac{N_w^{1+1/\psi}}{1+1/\psi}, \qquad \sigma_c > 0,\ \psi > 0.$$

The inner labor aggregator is

$$N_w = \left[\int_{j \in \Omega} (1+\xi_j)^{-1} n_j^{(\eta+1)/\eta} \, dj \right]^{\eta/(\eta+1)}.$$

Conditional on aggregate labor supply and the consumption marginal utility, cost minimization across jobs delivers equation (3) when $\xi_j = 0$ for every job. When a minimum-wage firm rations applicants, the wage index uses the shadow wage that rationalizes realized employment on the job-level labor-supply curve rather than the posted minimum wage.

For a given (τ, w^{mw}), the equilibrium solves for

$$\mathcal{Z}(\tau, w^{mw}) = \big(C, E_D, N_w, \widetilde{W}_w, M_E, e_x\big),$$

where M_E is the mass of entrants. The six outer conditions are top-level labor supply, wage-index consistency, the household budget, free entry, final-good absorption consistency,

and trade balance. The final-good price index is the numeraire rather than an additional unknown. In compact form,

$$\mathcal{F}(\mathcal{Z}; \tau, w^{mw}) = 0.$$

Entry, formality, and export fixed costs enter final-good demand. Household income contains wages, rationing rents, and net firm profits. There is no tariff revenue. Trade balance equates integrated export revenue to the value of imports in a common currency. It does not equate the foreign demand shifter A_X to aggregate exports.

If the Jacobian $\mathcal{F}_{\mathcal{Z}}$ is nonsingular, the local response to lower iceberg costs is

$$\frac{d\mathcal{Z}}{d\ell} = -\mathcal{F}_{\mathcal{Z}}^{-1}\mathcal{F}_{\ell}.$$

This expression emphasizes that the responses of E_D, E_X, N_w, and $\widetilde{W}_w$ in Proposition 2 are equilibrium outcomes. The proposition maps them into the regime boundaries but does not sign them.

B.2 Regime Cutoffs

The no-minimum-wage first-order condition is

$$\rho_\sigma A^{1/\sigma} z^{\rho_\sigma} n^{-1/\sigma} = \mu b n^{1/\eta}. \tag{B.1}$$

Solving equation (B.1) gives the first line of equation (8). The R2–R1 boundary sets the unconstrained wage equal to w^{mw}, equivalently $n_1 = \bar{n}$. This gives

$$(z_D^u)^{\sigma-1} = \left(\frac{\mu}{\rho_\sigma}\right)^{\sigma} \frac{\bar{n}\,(w^{mw})^{\sigma}}{E_D}.$$

At the R3–R2 boundary, labor demand at the minimum wage equals applicant supply. Using the third line of equation (8) and equation (7) yields

$$z_D^s = \mu^{-\sigma/(\sigma-1)} z_D^u.$$

Replacing E_D by E_X multiplies both boundaries by $\Gamma_X^{-1/(\sigma-1)}$, proving equation (13).

B.3 Proof of Proposition 1

Under unconstrained monopsony, marginal revenue product equals marginal factor cost:

$$\text{MRPL}(n^0) = \mu w(n^0) > w(n^0).$$

Because marginal revenue product is decreasing and inverse labor supply is increasing, $n^0 < n^c$. The argument below does not use the value of market access, so it applies to exporters and nonexporters alike.

At an R2 firm, the unconstrained wage is below the minimum wage, implying $n^0 < \bar{n}$. The absence of rationing requires

$$\text{MRPL}(\bar{n}) \geq w^{mw} = w(\bar{n}),$$

which implies $\bar{n} \leq n^c$. Employer-worker surplus is increasing between n^0 and $\bar{n}$, proving the R2 statement.

At an R3 firm, employment is demand determined at the minimum wage rather than pinned down by applicant supply. The cutoff at which R3 employment equals unconstrained monopsony employment is

$$z_a^e = \mu^{-\eta/(\sigma-1)} z_a^s, \qquad a \in \{D, X\}.$$

Thus $n_3 > n^0$ for $z_a^e < z < z_a^s$ and $n_3 < n^0$ for $z < z_a^e$. Throughout R3, $n_3 < n^c$.

B.4 The Relative Minimum Wage under Alternative Minimum-Wage Rules

Let

$$\varepsilon_{mw} \equiv \frac{d \log w^{mw}}{d\ell}, \qquad \varepsilon_N \equiv \frac{d \log N_w}{d\ell}, \qquad \varepsilon_W \equiv \frac{d \log \widetilde{W}_w}{d\ell}, \qquad \varepsilon_D \equiv \frac{d \log E_D}{d\ell}.$$

The relative minimum wage responds by $d \log m / d\ell = \varepsilon_{mw} - \varepsilon_W$. Under wage indexation $\varepsilon_{mw} = \varepsilon_W$. Because the final-good price index is the numeraire, w^{mw} is a real minimum wage, and price indexation is $\varepsilon_{mw} = 0$. For nonexporters, Proposition 2 gives

$$(\sigma - 1)\frac{d \log z_D^r}{d\ell} = (\sigma + \eta)(\varepsilon_{mw} - \varepsilon_W) + \varepsilon_N + \sigma\varepsilon_W - \varepsilon_D. \tag{B.2}$$

If no firm is constrained, every firm's wage bill is the share ρ_σ/μ of its revenue. With balanced trade, the revenue of domestic firms equals domestic absorption, so $E_D = (\mu/\rho_\sigma)\widetilde{W}_w N_w$ and $\varepsilon_D = \varepsilon_N + \varepsilon_W$. Equation (B.2) then becomes

$$(\sigma - 1)\frac{d\log z_D^r}{d\ell} = (\sigma + \eta)\varepsilon_{mw} - (1 + \eta)\varepsilon_W. \tag{B.3}$$

The nonexporter boundaries move right if and only if $\varepsilon_{mw} > [(1 + \eta)/(\sigma + \eta)]\varepsilon_W$. When liberalization raises the wage index, wage indexation satisfies this inequality and price indexation violates it. With constrained firms the proportionality between absorption and the wage bill is only approximate.

B.5 Proofs of Propositions 2 and 3

Taking logs of equation (12) and differentiating gives equation (14) for $r = u$, and equation (10) extends it to $r = s$. Because $E_X = \Gamma_X E_D$,

$$\frac{d}{d\ell}\log\left(\frac{z_X^r}{z_D^r}\right) = -\frac{1}{\sigma - 1}\frac{d\log\Gamma_X}{d\ell},$$

which does not depend on m. This proves Proposition 2.

Single crossing of the export decision follows from the envelope theorem. Let $\Pi(z, A)$ be optimized operating profit and define the gain from exporting as $\Delta\Pi(z) = \Pi(z, E_X) - \Pi(z, E_D) - f^X$. Then

$$\frac{d\Delta\Pi(z)}{d\log z} = \frac{\sigma - 1}{\sigma}\left[R(z; E_X) - R(z; E_D)\right] > 0. \tag{B.4}$$

Thus export status is selected by a productivity cutoff, z_x^*.

To obtain equation (15), note that formal nonexporters have $z_F \le z < z_x^*$. By Appendix B.3, the minimum wage raises a nonexporter's employment if and only if $z_D^e \le z < z_D^u$, and Assumption 1 gives $z_x^* < z_D^u$. The nonexporters in $\mathcal{C}$ are therefore those with $\max\{z_F, z_D^e\} \le z < z_x^*$. Exporters have $z \ge z_x^* \ge z_D^s > z_X^e$, so the minimum wage raises an exporter's employment if and only if $z < z_X^u$. The union is $[\underline{z}, \bar{z})$ with $\bar{z} = \max\{z_x^*, z_X^u\}$. Equation (16) follows from $z_X^u = \Gamma_X^{-1/(\sigma-1)} z_D^u$, $z_D^u = \mu^{\sigma/(\sigma-1)} z_D^s$, and $z_D^s = \mu^{\eta/(\sigma-1)} z_D^e$. It uses only the boundary formulas and therefore holds under any minimum-wage rule, which is part (c) of Proposition 3.

For the remainder of Proposition 3, normalize productivity by $\zeta \equiv (N_w \widetilde{W}_w^\sigma / E_D)^{1/(\sigma-1)}$ and write $u \equiv z/\zeta$ and $n = N_w \hat{n}$. A firm with sales status a earns revenue $N_w \widetilde{W}_w (\Gamma_a u^{\sigma-1})^{1/\sigma} \hat{n}^{\rho_\sigma}$, with $\Gamma_D = 1$, and an informal firm earns the same expression with u replaced by $(1 - \delta^I)u$.

Attracting $\hat{n}$ requires the wage $\widetilde{W}_w \hat{n}^{1/\eta}$, so a formal firm pays $\widetilde{W}_w \max\{m, \hat{n}^{1/\eta}\}$ per worker. Profit divided by $N_w \widetilde{W}_w$ is therefore a function of (u, m, Γ_a) minus the relevant fixed cost divided by $N_w \widetilde{W}_w$. Every status cutoff compares these normalized profits, and every regime boundary is a condition in (u, m, Γ_a). Hold m and $N_w \widetilde{W}_w$ fixed and raise Γ_X. Because firms at and below the formality cutoff do not export, the normalized values of z_F, z_D^e, z_D^s, and z_D^u do not change. The normalized exporter boundary $z_X^u = \Gamma_X^{-1/(\sigma-1)} z_D^u$ falls. The normalized export cutoff falls as well, because the gain from exporting is increasing in productivity by equation (B.4) and increasing in Γ_X.

When $z_x^* \geq z_D^s$, the argument above gives $\mathcal{C} = [\underline{z}, \bar{z})$ with $\bar{z} = \min\{\max\{z_x^*, z_X^u\}, z_D^u\}$. No exporter rations, and nonexporters are constrained only below z_D^u. With Pareto productivity, the share of formal firms in $\mathcal{C}$ is $(z_F/\underline{z})^k - (z_F/\bar{z})^k$. The first term does not change. If $z_x^* < z_D^u$, then $\bar{z} = \max\{z_x^*, z_X^u\}$ falls relative to z_F and $\underline{z}$, and the share falls strictly, which is part (a). If $z_x^* \geq z_D^u$, then $\bar{z} = z_D^u$ and the share is unchanged, which is part (b).

B.6 Derivation of the Informality Condition

Define $q \equiv \eta/(\sigma + \eta)$, the employment elasticity of an unconstrained firm with respect to its own market access. R2 employment has elasticity zero, while R3 employment has elasticity one.

The informal wage-employment share is

$$s_{I,c} = \frac{N_{I,c}}{N_{I,c} + N_{F,c}}.$$

Differentiation gives the exact identity

$$\frac{ds_{I,c}}{d\ell} = s_{I,c}(1 - s_{I,c}) \left(\frac{d \log N_{I,c}}{d\ell} - \frac{d \log N_{F,c}}{d\ell} \right).$$

Let $\omega_{r,x,c}$ be the initial share of formal wage employment in regime $r \in \{1, 2, 3\}$ and export status $x \in \{0, 1\}$ in cell c, and let $\omega_{X,c} = \sum_r \omega_{r,1,c}$. Define $a_{D,c} \equiv d \log E_{D,c}/d\ell$ and $a_{X,c} \equiv d \log E_{X,c}/d\ell$ as the local elasticities of domestic-only and exporter market access with respect to bilateral liberalization. The decomposition below focuses on the local case $a_{D,c} < 0 < a_{X,c}$, in which domestic-only market access falls while exporter market access rises. A first-order decomposition of formal wage employment is

$$\frac{d \log N_{F,c}}{d\ell} = (q\omega_{1,0,c} + \omega_{3,0,c})\, a_{D,c} + (q\omega_{1,1,c} + \omega_{3,1,c})\, a_{X,c} + b_{F,c}^{\ell}, \tag{B.5}$$

where $b_{F,c}^{\ell}$ collects entry, exit, formality switching, export switching, and equilibrium wage

responses. If informal wage firms remain unconstrained,

$$\frac{d\log N_{I,c}}{d\ell} = qa_{D,c} + b^{\ell}_{I,c}. \tag{B.6}$$

Substituting equations (B.5) and (B.6), using

$$\sum_{r=1}^{3}\sum_{x=0}^{1}\omega_{r,x,c} = 1,$$

and collecting the exporter terms gives

$$\begin{aligned}\frac{1}{s_{I,c}(1-s_{I,c})}\frac{ds_{I,c}}{d\ell} = {} & |a_{D,c}|\left[(1-q)\omega_{3,0,c} - q\omega_{2,0,c} - q\omega_{X,c}\right] \\ & - a_{X,c}\left(q\omega_{1,1,c} + \omega_{3,1,c}\right) + (b^{\ell}_{I,c} - b^{\ell}_{F,c}).\end{aligned} \tag{B.7}$$

Among nonexporters and with the b terms set to zero, the informal share rises if and only if $(1-q)\omega_{3,0,c} > q\omega_{2,0,c}$. An R3 firm cuts employment by $1-q$ more than an informal firm does, and an R2 firm by q less. The condition is equivalent to $\omega_{3,0,c}/(\omega_{2,0,c}+\omega_{3,0,c}) > q$. Rationing firms must account for more than a share $q = \eta/(\sigma+\eta)$ of the employment of constrained nonexporters. The b terms are first-order objects. Formality switching can occur in both high- and low-exposure cells, and informal exit can offset an increase generated by contracting formal employment.

B.7 From Bilateral Iceberg Costs to the Colombian Tariff Reform

The delivered import-cost change in industry i can be written

$$\Delta\ell^{M}_{i} = -\Delta\log\left[(1+t_i)\tau^{M}_{i}\right].$$

If physical import costs are unchanged, the observed tariff component is $-\Delta\log(1+t_i)$. This shock does not directly reduce Colombian exporters' foreign iceberg costs. It can nevertheless affect export access through the real exchange rate. A depreciation raises e_x in equation (4). The actual reform also changes tariff revenue. The reduced-form analysis isolates the import-side variation, while the estimated spatial model evaluates the complete unilateral reform.

C Appendix to Section 4

C.1 Household Allocation and Wage-Work Subnest

Given $P_{C,d}$ and activity returns, the household first allocates total hours using equation (22) and then allocates those hours using equation (23). The wage-work return $W_{W,hd}$ is the inclusive value of differentiated formal and informal jobs. In a rationing equilibrium it is constructed from shadow wages that rationalize realized employment on the job-level labor-supply curve rather than mechanically treating every applicant at the posted minimum wage as employed.

Let Ω_{hd} be the set of employers available to type h in region d. Abstracting from the formal amenity in the notation, the wage-work index takes the CES form

$$W_{W,hd} = \left[\int_{j \in \Omega_{hd}} \widetilde{w}_{jhd}^{1+\eta} \, dj \right]^{1/(1+\eta)},$$

where $\widetilde{w}_{jhd} = w_{jhd}$ when all applicants are hired and equals the shadow wage associated with realized employment when the firm rations. This construction makes household labor supply consistent with firm employment and realized wage income.

Tradable and nontradable self-employment are competitive activities. Their returns equal the value of their marginal products and enter equations (21) and (24) separately. The estimation allows their productivities and activity shifters to differ by region and education so that the model can match both the level and composition of self-employment.

C.2 Ownership of Housing and Firms

Worker households own neither the fixed housing stock nor claims to firms. Both are held by owners outside the worker households, so rent and profit income do not enter equation (24). Housing owners spend their rent income on local goods. Free entry drives the expected net return to firm ownership to zero after paying the housing-denominated entry cost. The operating surplus used to cover that cost ultimately accrues to housing owners as rent. The main welfare criterion therefore measures worker welfare.

C.3 Migration Extension

The baseline fixes L_{hd} to focus on within-location adjustment. An extension allows workers of education type h to choose among regions after observing region-specific location tastes. Let V_{hd} denote indirect utility from the consumption and activity allocation in region d, and

let ζ_{hd} collect time-invariant location amenities. Under type-I extreme-value idiosyncratic tastes, the population share in region d is

$$\frac{L_{hd}}{L_h} = \frac{\exp(\nu V_{hd} + \zeta_{hd})}{\sum_{k \in \mathcal{D}} \exp(\nu V_{hk} + \zeta_{hk})}, \tag{C.1}$$

where ν governs the responsiveness of location choice to utility. Endogenizing L_{hd} adds these population-share equations to the equilibrium system and feeds migration back into labor supply, housing demand, local expenditure, and firm entry. The baseline quantitative results do not use this extension or estimate ν.

C.4 Two-Stage Firm Values

Permanent productivity φ is observed before the formality commitment, whereas transitory productivity ε is realized afterward. Let $V^F_{sd}(\varphi)$ and $V^I_{sd}(\varphi)$ denote the expected values of formal and informal operation before the transitory draw. They satisfy

$$V^F_{sd}(\varphi) = \mathbb{E}_\varepsilon \left[\max_{x \in \{0,1\}} \pi^F_{sd}(\varphi\varepsilon, x) \right],$$
$$V^I_{sd}(\varphi) = \mathbb{E}_\varepsilon \left[\pi^I_{sd}(\varphi\varepsilon) \right].$$

The entrant chooses the largest of zero, $V^F_{sd}(\varphi)$, and $V^I_{sd}(\varphi)$. Exporting is available only in the tradable sector, so $x = 0$ for $s = N$. Conditional on realized $z = \varphi\varepsilon$, equations (26) and (27) deliver the same firm-level minimum-wage regimes characterized in Section 3.

The expected entry value is

$$\mathcal{V}^E_{sd} = \int \max\left\{0, V^F_{sd}(\varphi), V^I_{sd}(\varphi)\right\} \, dG_s(\varphi).$$

Free entry requires $\mathcal{V}^E_{sd} = R_d f^s_E$.

C.5 Outer Equilibrium System

With J regions and two worker types, the outer equilibrium contains nine regional unknowns. They are two type-specific wage indexes $W_{W,hd}$, two sectoral price indexes $P_{T,d}$ and $P_{N,d}$, rent R_d, two sectoral entrant masses $M^E_{T,d}$ and $M^E_{N,d}$, and two type-specific consumption levels c_{hd}. The exchange rate e_x is the remaining national unknown. The model therefore has $9J + 1$ outer unknowns, or 217 when $J = 24$.

Table C.1: Spatial equilibrium system

Condition	Number	Economic content
Labor-market clearing	$2J$	Wage labor supplied equals formal and informal labor demanded by region and education.
Goods-market clearing	$2J-1$	Sectoral supply equals household and firm demand, self-employment output, domestic trade, and international trade. One condition is redundant.
Housing-market clearing	J	Household housing demand plus entry costs equals the fixed regional housing stock.
Free entry	$2J$	Expected firm value equals the housing-denominated entry cost in each region and sector.
Household budgets	$2J$	Consumption expenditure equals wage income, self-employment income, and transfers.
External balance	1	Tradable exports equal tariff-exclusive import expenditure, determining e_x.
Numeraire	1	Employment-weighted average earnings equal one.
Total	$9J+1$	217 conditions when $J=24$.

C.6 Normalization and Reported Minimum-Wage Units

The average-wage normalization is applied separately in each equilibrium. If a counterfactual specifies $\kappa' = w^{mw\prime}/\overline{w}'$, its consumption-real value in region d is

$$\frac{w^{mw\prime}}{P'_{C,d}} = \kappa' \frac{\overline{w}'}{P'_{C,d}}.$$

The computation therefore stores κ', $\overline{w}'/P'_{C,d}$, and $w^{mw\prime}/P'_{C,d}$ for every counterfactual. Recovering a peso value additionally requires a nominal anchor. Wage indexation and price indexation are also distinct. Holding $w^{mw}/\overline{w}$ fixed keeps the policy aligned with average earnings, whereas holding $w^{mw}/P_{C,d}$ fixed keeps its regional purchasing power constant. Trade can move average earnings and consumption prices differently through terms-of-trade and spatial-price changes. These reporting conventions prevent a stable wage-relative minimum wage from being described as either a stable nominal minimum wage or a constant-real-purchasing-power policy.

D Appendix to Section 5

D.1 Data Construction

The quantitative data are mapped to the model at the region, region–education, region–sector, exporter, and national levels. Worker activities differ from Section 2 in three ways. A wage worker is formal if the worker reports contributing to a pension fund; because formality in the model is a choice of the firm, wage workers without pension contributions who report an employer with 11 or more workers are also counted as formal in every quantitative moment; and the self-employed are all employed persons who are not wage workers, including employers and unpaid workers. Industry codes in household, customs, and tariff files are harmonized before aggregation to tradable and nontradable sectors. Regional trade flows form origin–destination matrices in which diagonal entries are local purchases and off-diagonal entries are interregional shipments.

The model's 24 regions are the departments covered by the GEIH; the nine remaining departments are outside the model. The tradable sector consists of ISIC Rev. 3 two-digit industries 01–37 (agriculture, mining, and manufacturing), corresponding to input–output sectors S1–S32; these are the industries of the reduced-form sample of Section 2 plus recycling (ISIC 37, which carries no tariff line and 0.4 percent of tradable employment), and electricity, gas, and water (ISIC 40) is excluded from both. The nontradable sector consists of services excluding domestic service (S33–S53); domestic service is dropped. College workers are those with any tertiary education. Unless stated otherwise, worker-side moments use the 2015 GEIH, February–December: January interviews are dropped because they report December earnings against the new minimum wage, and, in income totals (the average earnings behind the minimum-wage ratio and self-employment earnings), missing labor income of paid workers who report employment is filled with the survey-weighted mean of reported pay in the same year, region, schooling, sector, and employment-status cell; unpaid workers keep zero, and hourly-wage moments use reported pay only. Table D.1 lists every data object.

Table D.1: Data construction for the quantitative model

Object	Source and unit	Construction
Workers and wages		
Population L_{hd}	GEIH 2015; region × education	Survey-weighted working-age population, employed or not, net of domestic-service workers, scaled so that the 24 regions sum to 24.

Continued on next page

Table D.1 – continued from previous page

Object	Source and unit	Construction
Activity allocation	GEIH 2015; region × education × sector	Formal wage workers, informal wage workers, and self-employed. A wage worker is formal if the worker contributes to a pension fund or reports an employer with 11 or more workers; the self-employed are all employed persons who are not wage workers, including employers and unpaid workers.
Formality (targeted)	GEIH 2015; region × sector × education	Formal share of wage employment.
Wages and earnings	GEIH 2015; region × sector, region × education	Hourly earnings. Wage-level, wage-gap, premium, and dispersion moments use wage earners with positive hourly wages between 0.22 and 10.09 times the hourly minimum wage. Self-employment earnings are labor income per self-employed worker.
Minimum wage	Decree; national	644,350 pesos per month in 2015, converted to 3,356 pesos per hour at 192 hours per month. Relative to average hourly earnings of tradable and nontradable workers, it equals 0.674.
Minimum-wage bite (targeted)	GEIH 2015; region × sector × education	Share of formal wage workers paid at or below 1.05 times the hourly minimum wage.
Firms and housing		
Formal firm count (targeted)	GEIH 2015; national	Number of formal firms in the tradable and nontradable sectors implied by workers' reports of employer size: each formal wage worker is weighted by the survey weight divided by the midpoint of the reported firm-size bracket, and the total is divided by the 11 survey months; 332,728 firms, targeted relative to population (0.218 when the 24 regions' population sums to 24).
Rents (matched exactly)	GEIH 2015 housing module; region	Renter households' paid rent, regressed on region fixed effects and housing characteristics (urban and rural areas); the fixed effects are expressed relative to Bogotá.
Housing expenditure share α_H	ENPH 2016–17; national	0.252.
Trade		
Exporter share (targeted)	DIAN 2015 export records; GEIH 2015	Exporting firms (deduplicated tax identifiers with goods exports in the tradable scope, 8,193) divided by the GEIH formal tradable firm count (the same construction restricted to the tradable sector, 80,503); 0.102.
Export sales ratio (targeted)	Interregional input–output table 2015 (IIOAS; Haddad, Araujo, and Galvis); 24 regions	Tradable value added embodied in the tradable exports of the 24 regions, computed with the Leontief inverse of the domestic input–output matrix, divided by tradable value added of the same regions; 0.300.
Tradable expenditure share $\alpha_{T,d}$	IIOAS 2015; region	Use-side expenditure share on tradable sectors; mean 0.29, range 0.25–0.32.
Domestic trade costs τ^T_{dk}, τ^N_{dk}	IIOAS 2015; region pairs	Head–Ries index $\tau_{dk} = (X_{dk}X_{kd}/X_{dd}X_{kk})^{1/(2(1-\sigma))}$ with $\sigma = 4$, where flows include intermediate and final demand of region k for region d's output in the sector. Tradable flows use S1–S32 and leave zero-flow pairs at infinite cost; nontradable flows use S33–S53.

Continued on next page

Table D.1 – continued from previous page

Object	Source and unit	Construction
International trade costs τ_d^M	IIOAS 2015, WIOD 2014; region	Head–Ries index against one rest-of-world aggregate, using IIOAS foreign exports by region of production, IIOAS foreign imports by region of use, the region's own flow, and WIOD world gross output of the corresponding industries, at 2,741.88 pesos per dollar, with a correction for the region's realized tariff (Appendix D.3).
Regional tariffs t_d (2015)	DIAN 2015 import records; region	Realized duty over CIF value by input–output sector from 3.04 million import records, mapped from NANDINA product codes to sectors, then averaged across sectors with weights equal to the region's IIOAS imports of each sector.
Trade elasticity σ	DIAN 2015 import records	Gravity regression of duty-exclusive CIF imports on $\ln(1+t)$. The coefficient equals $-\sigma$ and gives 4.08, and σ is set to 4 (Appendix D.3).
Counterfactual tariff schedules		
2010–2011 reform	UNCTAD-TRAINS; GEIH 2007–2009	Applied (AHS) tariff reductions between 2009 and 2011 by two-digit industry, averaged within each region with 2007–2009 tradable employment weights; mean 2.66 percentage points.
2009 and high-protection regimes	WITS AHS 2009 and 2015; DIAN 2015; IIOAS 2015	The 2009–2015 change in national applied tariffs by sector, weighted by DIAN 2015 CIF values, is averaged within each region with its IIOAS import weights (mean 4.37 percentage points) and added to the 2015 regional tariff; high protection adds twice that change.

Notes: GEIH: Gran Encuesta Integrada de Hogares. ENPH: Encuesta Nacional de Presupuestos de los Hogares. DIAN: Colombian customs and tax authority. IIOAS: interregional input–output table for Colombia. WIOD: World Input–Output Database. Regional imports in the IIOAS are themselves estimated by applying national import coefficients to each region's sectoral output and final demand, so their regional variation reflects production and demand structure rather than observed customs flows.

D.2 Parameters and Estimation

This appendix records the parameterization behind Table 3. The classification is operational: an externally set parameter is held fixed throughout SMM; an externally estimated parameter is estimated in a separate data exercise and then held fixed; a recovered object is computed from an exact inversion and is not searched; and an internally estimated parameter is part of the SMM vector. The last category is the only one searched over; recovered objects are recomputed from each candidate equilibrium and therefore also change when the SMM objective is re-estimated.

Table D.2: Complete parameter inventory and identification

Parameter block	Classification	Value or status	Source, construction, or identifying moments
Preferences: σ_c, ψ	External set	2.0, 0.40	Consumption curvature and the Frisch elasticity are maintained calibration values.
Housing and institutions: α_H, τ_p, ξ_F	External set	0.252, 0.25, 0.264	Housing share from ENPH 2016–17; payroll tax from statutory rates; formal amenity from benefits data.
Support and normalization: $\varphi_{\min}$, $\varphi_{\max}$, f_s^I, P^*	External set	1, 500, 0, 1	Productivity support and informal operating cost are maintained restrictions; the foreign price is the external numeraire.
Observed policy and endowments: w^{mw}, L_{hd}, $\alpha_{T,d}$, t_d	External set/data	Observed	Household, national-accounts, and customs data provide the minimum wage, population by type, sectoral expenditure shares, and tariffs.
Trade costs: τ_{od}^T, τ_{od}^N, τ_d^M	External set/data	Data-based matrices	Domestic costs are recovered by Head–Ries from 2015 interregional input–output flows. International costs use the same flows and world output, with a customs-based tariff correction (Appendix D.3).
Firm labor supply: η	External estimate	2.50	Amodio and de Roux (2024), Colombian plants; held fixed in the spatial SMM.
Activity substitution: θ	External estimate	1.50	Shift-share tariff-IV estimate of substitution between wage work and self-employment, 1.49 (Appendix D.3). Held fixed in the spatial SMM.

Continued on next page

Table D.2 – continued from previous page

Parameter block	Classification	Value or status	Source, construction, or identifying moments
Trade elasticity: σ	External estimate	4.00	Gravity estimate using 2015 customs records, 4.08 (Appendix D.3), set to 4.00. The same value is used in demand and trade-cost construction. This is the elasticity denoted σ in Section 4.
Worker-side fundamentals: A_d^S, $\chi_{a,hd}$, $\bar{\phi}_{hd}$	Recovered	Data-implied	Closed-form inversion from observed self-employment earnings, activity shares, and labor supply conditional on the equilibrium and external elasticities.
Housing stocks: $\bar{H}_d$	Recovered	Table D.3	Set so that the model matches each region's rent relative to Bogotá exactly. Bogotá's stock fixes the housing unit.
Regional fundamentals: A_d^T, A_d^N, δ_d^I	Internal SMM	Table D.3	Region-by-sector mean log hourly wages, tradable employment shares, and sectoral formality, minimum-wage bite, and formal–informal wage gaps. Bogotá productivity levels are normalized.
Entry costs: f_E^T, f_E^N	Internal SMM	3.620, 5.655	Tradable entry costs are national, and nontradables have a separate entry cost. Formal firm counts and sectoral employment/formality moments provide the main information.
Formal operating costs: f_T^F, f_N^F	Internal SMM	1.760, 0.610	National sector-specific costs, identified jointly from sectoral formality, minimum-wage bite, and formal–informal wage moments.

Continued on next page

Table D.2 – continued from previous page

Parameter block	Classification	Value or status	Source, construction, or identifying moments
Foreign-market access: f_T^X, X^*	Internal SMM	17.703, 0.0564	Exporter participation and the all-tradables export-sales ratio identify the export fixed cost and foreign expenditure scale jointly.
Heterogeneity: σ_ε, s_C, λ_I, k	Internal SMM	1.0781, 2.2888, 0.2443, 2.2794	Wage dispersion, type-specific bite, formal/informal type premia, and wage gaps. k is estimated jointly with the other parameters.

Let m_ℓ^{data} and $m_\ell^{model}(\Theta)$ denote moment ℓ in the data and model, and let se_ℓ be its sampling standard error on the same scale. The formal firm count, the exporter share, and the export-sales ratio enter as log deviations. The exporter share is weighted by the sampling precision of its GEIH firm-count denominator (weight 1,397). The export-sales ratio is a national-accounts ratio without sampling error and receives a fixed weight of 4,748, set equal to the mean weight of the other targeted moments. The simulated-method-of-moments estimator minimizes

$$\widehat{\Theta} = \arg\min_{\Theta} \sum_{\ell=1}^{L} \left[\frac{m_\ell^{model}(\Theta) - m_\ell^{data}}{se_\ell} \right]^2 . \tag{D.1}$$

The estimation uses warm-started Levenberg–Marquardt steps with coordinate sweeps and parallel candidate evaluations. Housing stocks outside Bogotá are set at every evaluation of the objective so that the model matches relative rents exactly. The Pareto tail is searched jointly with the other national coordinates rather than profiled over an outer grid. The reported estimates start from the previous estimate of the model and are stationary after six accepted Levenberg–Marquardt steps, with an objective value of 11,216.8. Estimating the housing stocks jointly and targeting rents instead moves the welfare-maximizing minimum wage by less than 0.005 when both estimates are solved in steps of 0.05 (0.566 against 0.570), and welfare numbers by less than 0.02 percentage points.

D.3 Externally Estimated Parameters

I estimate the trade elasticity from 2015 Colombian customs records (DIAN), which report the CIF value of each import transaction and the duty paid. I aggregate the records to

Table D.3: Regional parameters

Region	Tradable productivity A_d^T	Nontradable productivity A_d^N	Informal penalty δ_d^I	Housing stock $\bar{H}_d$
Antioquia	1.332	0.538	0.111	0.658
Atlántico	1.210	0.629	0.129	0.176
Bogotá D.C.	1.637	0.700	0.000	1.000
Bolívar	1.566	0.714	0.102	0.139
Boyacá	1.511	0.651	0.125	0.117
Caldas	1.272	0.513	0.276	0.088
Caquetá	1.864	0.667	0.159	0.033
Cauca	1.739	0.652	0.160	0.094
Cesar	1.516	0.748	0.131	0.055
Chocó	2.367	0.934	0.169	0.038
Cundinamarca	1.628	0.751	0.122	0.230
Córdoba	1.669	0.642	0.039	0.075
Huila	1.634	0.759	0.111	0.062
La Guajira	1.657	0.961	0.050	0.053
Magdalena	1.667	0.732	0.264	0.074
Meta	1.566	0.839	0.172	0.063
Nariño	1.543	0.540	0.132	0.117
Norte de Santander	1.699	0.753	0.148	0.081
Quindío	1.752	0.713	0.299	0.062
Risaralda	1.304	0.545	0.271	0.088
Santander	1.611	0.780	0.023	0.179
Sucre	1.642	0.655	0.144	0.039
Tolima	1.563	0.605	0.152	0.101
Valle del Cauca	1.400	0.566	0.159	0.467

Notes: Productivities and the informal penalty are estimated by simulated method of moments. Housing stocks are set so that the model matches each region's rent relative to Bogotá exactly. Bogotá's productivity levels and housing stock are normalizations. Bogotá's estimated penalty δ_d^I is 9.9×10^{-7}: no bound is imposed, and the estimate runs to the boundary of the parameter space at zero.

partner-country × HS6-product cells and regress the log CIF import value on the log of one plus the realized tariff, defined as duties paid divided by CIF value within the cell, with country and product fixed effects. Because CIF values exclude duties, the coefficient equals $-\sigma$ under CES demand. Identification comes from differences in applied tariffs across origins within a product, mostly between preferential and most-favored-nation rates. The regression uses all 2015 goods imports. Table D.4 reports the estimates. The baseline estimate of σ is 4.08, with a standard error of 0.29 clustered by product, from 61,553 cells. Using declared rather than realized tariffs gives 3.60, and estimating on department-level cells gives 3.92

with additive department fixed effects and 3.90 with interacted fixed effects. I set $\sigma = 4.00$, a value commonly used in quantitative trade models.

Table D.4: Trade elasticity from 2015 customs records

Specification	$\widehat{\sigma}$	Std. err.	Cells
Country × HS6, realized tariff (baseline)	4.08	0.29	61,553
Country × HS6, declared tariff	3.60	0.32	61,553
Department × country × HS6, additive fixed effects	3.92	0.20	161,815
Department × country × HS6, interacted fixed effects	3.90	0.23	148,202

Notes: Regressions of the log CIF import value on $\ln(1+t)$, where t is duties paid over CIF value within the cell. The table reports minus the coefficient. All specifications include partner-country and HS6 fixed effects. The department rows add department fixed effects, additively or interacted with country and with product. Standard errors are clustered by HS6 product. *Source:* DIAN 2015 import records.

I estimate θ from the response of relative earnings to relative employment across Colombian local labor markets between 2007 and 2019. The unit is a department × demographic cell, defined by sex, three schooling groups, and three age groups. For each cell I compute the 2007–2019 change in the log ratio of wage workers, formal and informal, to self-employed workers, and in the log ratio of their mean hourly earnings, using GEIH survey weights. Under the activity CES of Section 4, the earnings ratio responds to the employment ratio with elasticity $1/\theta$. Because both ratios respond to local demand shocks, I instrument the change in relative employment with shift-share tariff exposure, the cell's 2007 industry employment shares times each industry's change in $\log(1 + \text{MFN tariff})$. The regression includes department fixed effects, clusters standard errors by department, and keeps cells with at least 50 workers in each group at both dates. The estimate of $1/\theta$ is 0.670 (standard error 0.267, 347 cells, first-stage F-statistic 18.2), which gives $\theta = 1.49$. Requiring at least 25 or 10 workers per cell gives 1.53 and 1.45. The weak-instrument-robust (Anderson–Rubin) 95 percent confidence interval for θ is [0.68, 5.30]. I set $\theta = 1.5$. The estimate concerns the margin between wage work and self-employment, and the model applies the same elasticity to the choice between tradable and nontradable self-employment.

Domestic iceberg costs are recovered from the 2015 interregional input–output table for Colombia (IIOAS; Haddad, Araújo, and Galvis), which reports flows among 33 departments and 54 sectors. For each pair of the 24 GEIH departments and each sector group, I sum the destination's intermediate and final demand for goods produced in the origin, with intermediate demand restricted to using sectors in the same group. Under symmetric costs, the Head–Ries index gives $\tau_{dk} = [X_{dk}X_{kd}/(X_{dd}X_{kk})]^{1/(2(1-\sigma))}$, with $\sigma = 4$. The tradable matrix uses sectors 1–32 and the nontradable matrix sectors 33–53, excluding household

services. Six department pairs have no tradable flows, and the model treats them as unable to trade tradable goods with each other. The median off-diagonal cost is 2.60 for tradables and 5.66 for nontradables.

International costs use the same table and world output. For each department d, τ_d^M satisfies $(\tau_d^M)^{2(1-\sigma)} = E_d M_d (1+t_d)^\sigma / (X_{dd} X_W)$, where E_d and M_d are the department's foreign exports and imports of tradable goods in the input–output table, X_{dd} its own tradable flow, X_W world gross output in agriculture, mining, and manufacturing from the 2014 World Input–Output Database, converted at 2,741.88 pesos per dollar, and t_d the department's realized import tariff. The tariff t_d weights national realized tariffs by sector, computed from 3.04 million 2015 customs records as duties paid over CIF value, by the department's import composition. The tariff term removes the part of the import barrier that the model represents separately as a tariff. The median τ_d^M is 6.10, ranging from 4.12 in La Guajira to 25.3 in Caquetá. World output is for 2014 rather than 2015, and Colombia is not removed from the world aggregate.

D.4 Moments and Identification

Table D.5 lists the 318 moments by block.

Domestic trade flows recover bilateral iceberg costs before estimation and therefore do not add rows to the 318-moment SMM vector. Regional rents are matched exactly through the housing stocks and are likewise not counted as moments. Aggregate exports and imports are not targeted; the model imposes balanced trade. The response to the 2010–2011 tariff reform remains untargeted.

D.5 Full Model Fit

Table D.6 summarizes the fit by block of the objective, and Figure D.1 shows the regional fit of three targeted blocks. Three blocks account for most of the remaining misfit: region-by-sector mean log wages, region-by-sector formal–informal wage gaps, and formality. The model's average wage level across cells is above the data, its formal–informal wage gap is larger than in the data (0.68 against 0.53 log points on average), and it places too small a share of formal workers at or below 1.05 times the minimum wage, especially among college workers and in tradables. Trade is under-fit as discussed in the main text.

Table D.5: Targeted moments and their role in estimation

Moment	Count	Variation used in estimation
Formality by region, type, and sector	96	Formal wage-work shares in the two sectors and two worker types.
Minimum-wage bite by region, type, and sector	96	Share of formal wage workers paid at or below 1.05 times the hourly minimum wage.
Mean log hourly wage by region and sector	48	Wage levels in the tradable and nontradable sectors.
Formal–informal mean log hourly wage gap	48	Status-specific wage differences by region and sector.
Formal and informal type premia	2	College–noncollege wage premia by activity.
Regional tradable wage-employment shares	24	Regional sectoral specialization.
Within-cell log-wage variance	1	Dispersion of hourly wages in the trimmed estimation window.
Formal firm count	1	Formal entry and the mass of observed firms.
Exporter participation	1	Share of formal tradable firms that export.
All-tradables export-sales ratio	1	Foreign sales relative to total tradable sales.
Total	318	

D.6 The Tariff Reform Simulation

Table D.7 reports the simulated 2010–2011 reform. The reform is the 2015 equilibrium minus a counterfactual in which each region's tariff is raised by its pre-reform-employment-weighted reduction, holding the minimum wage at its 2015 ratio to average earnings.

D.7 Gains from Trade

Table D.8 reports the worker gains of Section 5.3 by education group, Figure D.3 plots them against the minimum wage, and for the alternative estimate in which the formal-sector benefit ξ_F also counts as worker income. It also reports a second experiment, the move from 2015 tariffs to free trade (Figure D.4). Free trade lowers worker welfare in every economy and at every minimum wage, by 0.52 percent in the baseline at the current minimum wage. Removing tariffs worsens Colombia's terms of trade, as the peso depreciates by 1.3 percent, and removes the tariff revenue that the model rebates to workers. Noncollege workers lose

Table D.6: Fit of the estimated model by moment block

Moment block	Data	Model	Objective
Targeted			
Formality (correlation, noncollege / college)	0.91 / 0.77		1,700.7
Minimum-wage bite, median: noncollege	0.469	0.455	1,396.3
college	0.163	0.125	
Mean log hourly wage (mean over cells)	0.164	0.303	3,501.7
Formal–informal log wage gap (mean over cells)	0.529	0.683	3,068.6
College premium, formal / informal	0.570 / 0.178	0.483 / 0.221	245.5
Regional tradable share of wage jobs (correlation)	0.89		1,106.2
Within-cell variance of log wages	0.249	0.241	19.1
Export sales ratio, value-added basis	0.300	0.249	164.6
Share of tradable firms that export	0.102	0.112	13.9
Formal firm count (model units)	0.218	0.217	0.3
Total objective			11,216.8
Untargeted			
Tradable share of wage employment, national	0.277	0.345	
Informal share of wage jobs, nontradables minus tradables	−0.130	−0.079	
Exports / GDP	0.143	0.1025	
Imports / GDP	0.215	0.1025	

Notes: The last column is each block's sum of squared, weighted deviations; the blocks sum to the objective. Formality, bite, wage levels, and wage gaps are targeted by region, sector, and education group (wage levels and gaps by region and sector); correlations for formality are across region–education cells, pooled over sectors. The largest regional miss in the tradable share is 0.19. Wage moments use hourly wages on the estimation sample trimmed to 0.22–10.09 times the minimum wage. Imports equal exports in the model by the external-balance closure. Regional rents are matched exactly and are not part of the objective.

0.56 percent and college workers 0.05 percent, because the equal rebate is a larger share of noncollege income. As with the trade-cost cut, the minimum wage barely changes these effects, except in the economy without informal firms or self-employment, where the loss of noncollege workers reaches 1.09 percent at 1.25 times the current minimum wage.

D.8 Counterfactual Algorithm and Welfare

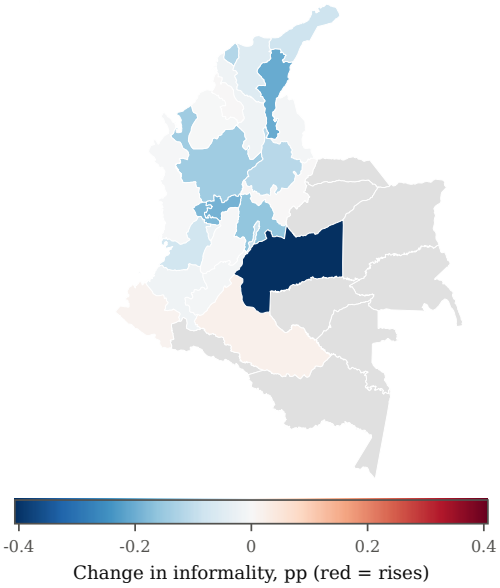
For each experiment, the algorithm starts from the estimated equilibrium, changes the tariff schedule or κ, and resolves the 217-equation system using the previous solution as an initial condition. A continuation path moves gradually between policies and stores regional wages, price indexes, rents, entry, employment, trade, transfers, and consumption. Welfare is computed from the utility of each region–education cell, following Berger, Herkenhoff, and Mongey (2025) (equation (29)). The aggregate, and each education group's, consumption-

Figure D.1: Spatial fit of the estimated model



Notes: Each point is a region–education cell (panel (a)), a region–sector–education cell (panel (b)), or a region (panel (c)); the dashed line is the forty-five-degree line. (a) Formal share of wage employment, pooled over sectors. (b) Share of formal wage workers paid at or below 1.05 times the hourly minimum wage. (c) Tradable share of wage employment. Panels (b) and (c) plot targeted 2015 moments; panel (a) aggregates the targeted formality cells over sectors. Model values are at 2015 tariffs and the current minimum wage.

Figure D.2: Regional incidence of the 2010–2011 tariff reform in the estimated model



Notes: Change in informality (informal wage workers and the self-employed, both sectors, all workers) by region. Each region's 2015 tariff is raised by its pre-reform-employment-weighted industry tariff reduction (equation (30)); the map reports the 2015 equilibrium minus that pre-reform counterfactual, holding the minimum wage at its 2015 ratio to average earnings. Gray regions are not in the model.

equivalent change is the common percentage change in the reference-equilibrium consumption of every worker in the group that yields the same population-weighted sum of utility as the

Table D.7: The 2010–2011 tariff reform in the estimated model

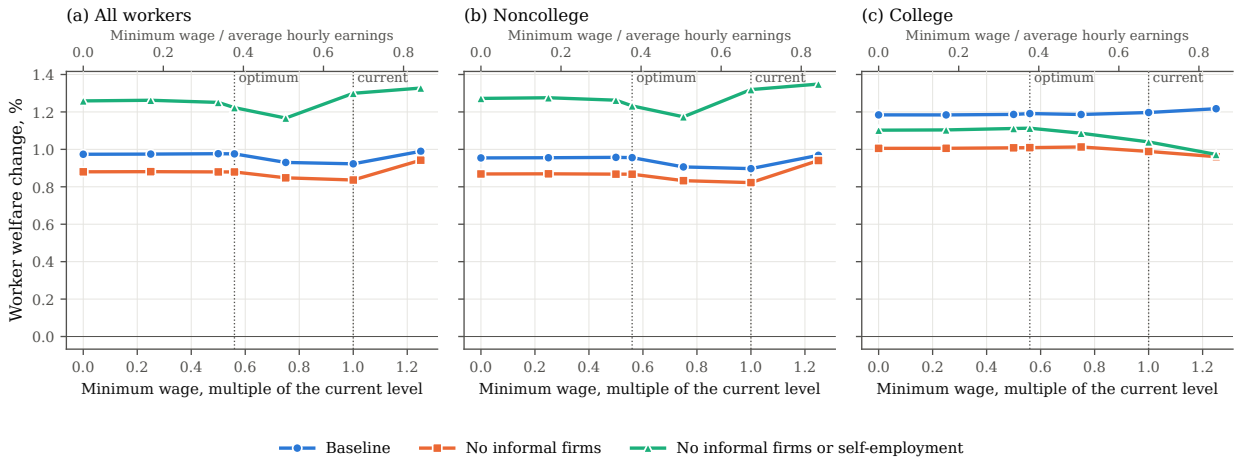
	Estimated model
Exports (percent change)	+6.51
Inclusive informality, all workers (pp)	−0.083
Noncollege (pp)	−0.103
College (pp)	−0.064
Tradable share of wage employment (pp)	+0.12
Within-sector informality, mean over regions weighted by pre-reform employment (pp)	
Tradables, noncollege	−0.420
Tradables, college	−0.524
Nontradables, noncollege	+0.078
Nontradables, college	+0.165
Tradables, slope on regional tariff reduction (pp per pp)	
Noncollege	−0.454
College	−0.851
Tradables, response per pp of cut: weighted slope on model minimum-wage bite	
Noncollege	+0.489
College	−0.473
Tradables, high- minus low-exposure response (pp per pp)	−0.012
Regions where informality rises	4 of 24
Largest regional decline (Meta, pp)	−0.407
Slope on regional tariff reduction (pp per pp)	−0.053
Informality response to a 10% higher minimum wage, 2015 tariffs (pp)	+1.933
Change in that response caused by the reform (pp)	−0.004

Notes: Changes are the 2015 equilibrium minus the pre-reform counterfactual at the current minimum wage. Informality includes informal wage workers and the self-employed in both sectors, pooled over education groups with employment weights. Regional tariff reductions average 2.66 percentage points (standard deviation 0.40, range 1.26 to 3.52). The slope is an unweighted least-squares fit across the 24 regions. The last row is the difference between the minimum-wage response of informality at 2015 tariffs and at pre-reform tariffs. Within-sector informality is the share of informal wage workers and the self-employed among workers in the sector, by region and education group. The high- minus low-exposure row splits the 48 tradable region-by-education cells at the median of the Section 2 exposure measure.

counterfactual.

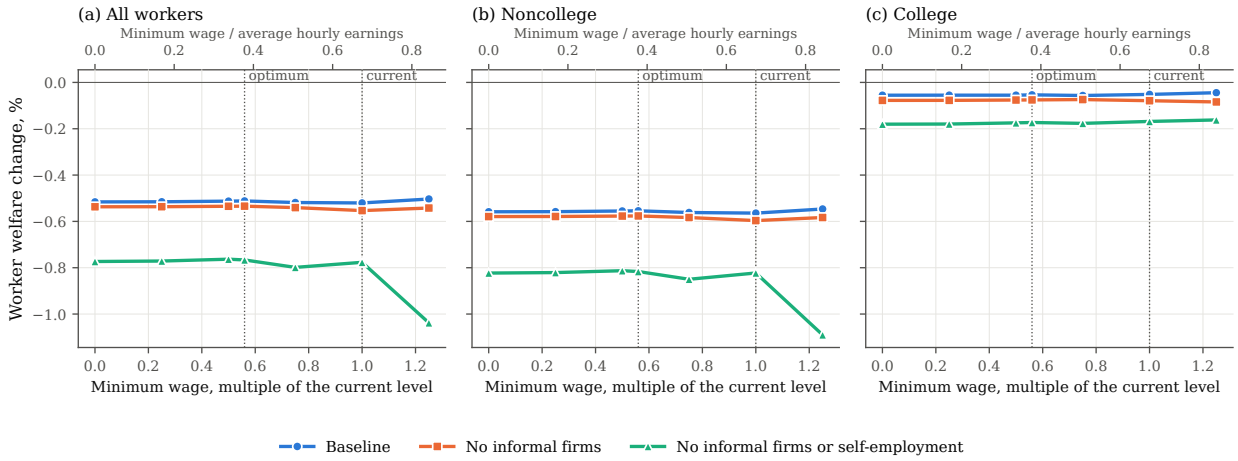
Minimum-wage comparisons within a tariff regime use that regime's no-minimum-wage equilibrium as the reference. Comparisons across tariff regimes use common real consumption units. The minimum wage runs from 0 to 1.6 times the current minimum wage in steps of 0.01, giving 161 policies in each of the four tariff regimes. The no-minimum-wage point

Figure D.3: Worker gains from a 10 percent cut in international trade costs, by minimum wage



Notes: Consumption-equivalent change in worker welfare, in percent, from cutting international iceberg trade costs by 10 percent, relative to the same economy at 2015 trade costs and the same minimum wage, held fixed relative to average earnings. In the economy without informal firms or self-employment, the self-employment shifters are set to zero. Every economy keeps the estimated parameters. Dotted lines mark the current minimum wage and the welfare-maximizing one (Section 5.4).

Figure D.4: Worker gains from moving to free trade, by minimum wage



Notes: Consumption-equivalent change in worker welfare, in percent, from removing all tariffs, relative to the same economy at 2015 tariffs and the same minimum wage, held fixed relative to average earnings. Economies as in Figure D.3. Dotted lines mark the current minimum wage and the welfare-maximizing one.

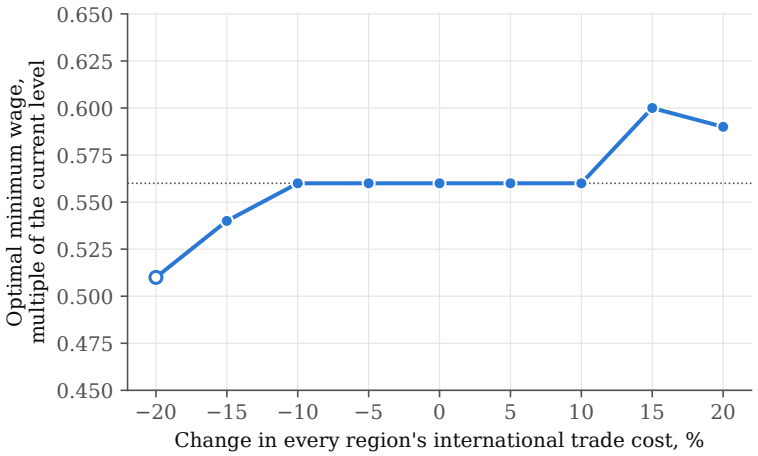
is solved at 10^{-4} times the current level, which is in effect no minimum wage. All 644 equilibria are solved, with a largest equilibrium residual of 6.4×10^{-12}. The outer loop over wage employment, self-employment, and self-employment earnings stops when no quantity changes by more than 10^{-8} in relative terms. The elasticity ladder solves 81 policies under the 2015 tariff schedule for $\eta \in \{1.5, 2, 3, 4, 5\}$, in steps of 0.01 between 0.30 and 0.90 and of

0.05 elsewhere, and all 405 of those equilibria also solve. The same 81 policies are also solved under the other three tariff regimes (Table D.14). Before the counterfactuals, each package re-solves the estimated equilibrium and reproduces it exactly, and a set of null experiments reproduces the baseline to machine precision. Reported optima are the best solved policy, and welfare at the optimum is evaluated there.

D.9 Supplementary Policy Results

Table D.9 reports welfare and informality at selected minimum wages under the 2015 tariff schedule. Table D.10 reports the elasticity ladder, Table D.11 decomposes the welfare change, Table D.12 reports the robustness checks, and Table D.13 the four tariff regimes. Table D.14 crosses the elasticity ladder with the tariff regimes, and Figure D.5 varies international trade costs.

Figure D.5: The optimal minimum wage and international trade costs



Notes: Best solved minimum wage for all workers, as a multiple of its current level, when every region's international iceberg trade cost, for exports and imports, is scaled by the same percentage, under 2015 tariffs. The dotted line marks 0.56. At 2015 trade costs the full range of minimum wages is solved in steps of 0.01. At the other levels the model is solved in steps of 0.01 on a band around the optimum, extended until welfare falls on both sides. The open marker at a 20 percent cut is an equilibrium in which Chocó's tradable sector does not operate. Cuts of 15 percent or more make Bogotá's nontradable sector begin to informalize at a lower minimum wage, which lowers the optimum.

Table D.8: Worker gains from trade by minimum wage

	Minimum wage, multiple of the current level				Range over
	None	Optimum	1.00	1.25	seven levels
International trade costs cut by 10 percent					
Headline estimate					
Baseline					
All workers	+0.97	+0.98	+0.92	+0.99	0.07
Noncollege	+0.95	+0.96	+0.90	+0.97	0.07
College	+1.18	+1.19	+1.20	+1.22	0.03
No informal firms					
All workers	+0.88	+0.88	+0.84	+0.94	0.11
Noncollege	+0.87	+0.87	+0.82	+0.94	0.12
College	+1.01	+1.01	+0.99	+0.96	0.05
No informal firms or self-employment					
All workers	+1.26	+1.22	+1.30	+1.33	0.16
Noncollege	+1.27	+1.23	+1.32	+1.35	0.18
College	+1.10	+1.11	+1.04	+0.97	0.14
Benefits also counted as worker income, all workers					
Baseline	+0.95	+0.95	+0.91	+0.96	0.05
No informal firms	+0.87	+0.86	+0.83	+0.91	0.08
No informal firms or self-employment	+1.26	+1.22	+1.27	+1.35	0.19
2015 tariffs to free trade					
Headline estimate					
Baseline					
All workers	−0.52	−0.51	−0.52	−0.50	0.02
Noncollege	−0.56	−0.55	−0.56	−0.55	0.02
College	−0.06	−0.05	−0.05	−0.04	0.01
No informal firms					
All workers	−0.54	−0.53	−0.55	−0.54	0.02
Noncollege	−0.58	−0.58	−0.60	−0.58	0.02
College	−0.08	−0.08	−0.08	−0.08	0.01
No informal firms or self-employment					
All workers	−0.77	−0.77	−0.78	−1.04	0.27
Noncollege	−0.82	−0.82	−0.82	−1.09	0.28
College	−0.18	−0.17	−0.17	−0.16	0.02
Benefits also counted as worker income, all workers					
Baseline	−0.47	−0.46	−0.47	−0.46	0.01
No informal firms	−0.48	−0.48	−0.50	−0.49	0.02
No informal firms or self-employment	−0.68	−0.67	−0.70	−0.95	0.28

Notes: Consumption-equivalent change in worker welfare, in percent, of the economy after the trade change relative to the same economy at 2015 tariffs and trade costs and the same minimum wage, held fixed relative to average earnings. The optimum is the welfare-maximizing minimum wage of each estimate, 0.56 for the headline estimate and 0.54 for the alternative. The range is the largest minus the smallest worker welfare change over the minimum-wage levels 0, 0.25, 0.50, the optimum, 0.75, 1.00, and 1.25. In the economy without informal firms or self-employment, the self-employment shifters are set to zero. Every economy keeps the estimated parameters.

Table D.9: Welfare and informality over the minimum-wage grid, 2015 tariffs

Minimum wage		Welfare vs no minimum wage (CEV, %)			Inclusive informality	
(multiple of current)	κ	All	Noncollege	College	All	Noncollege
0.00	0.00	0.000	0.000	0.000	0.572	0.653
0.20	0.13	+0.004	+0.004	−0.003	0.572	0.653
0.40	0.27	+0.083	+0.100	−0.097	0.575	0.657
0.56	0.38	+0.144	+0.190	−0.340	0.586	0.669
0.60	0.40	+0.143	+0.196	−0.421	0.590	0.674
0.80	0.54	−0.135	−0.054	−0.995	0.621	0.710
1.00	0.67	−0.648	−0.553	−1.656	0.658	0.754
1.20	0.81	−1.226	−1.122	−2.331	0.696	0.797
1.40	0.94	−1.751	−1.632	−3.016	0.730	0.835
1.60	1.08	−2.289	−2.155	−3.708	0.761	0.867

Notes: κ is the minimum wage relative to average hourly earnings. Welfare is the consumption-equivalent change for workers relative to the no-minimum-wage equilibrium under the same tariff schedule. Informality includes informal wage workers and the self-employed, pooled over education groups with employment weights. College informality rises from 0.358 without a minimum wage to 0.405 at the current policy.

Table D.10: The optimal minimum wage and employer wage-setting power

		Optimal minimum wage		Welfare vs no minimum wage (CEV, %)	
η	Wage/MRPL, $\eta/(1+\eta)$	Multiple of current	κ	At the optimum	At the current policy
1.5	0.60	0.79	0.53	+0.944	+0.699
2.0	0.67	0.64	0.43	+0.385	−0.176
2.5	0.71	0.56	0.38	+0.144	−0.648
3.0	0.75	0.49	0.33	+0.053	−0.782
4.0	0.80	0.42	0.28	+0.012	−0.787
5.0	0.83	0.39	0.26	+0.003	−0.766

Notes: Each row re-solves the minimum-wage grid under the 2015 tariff schedule with the firm-level labor-supply elasticity η changed and all other parameters at their estimates; the baseline is $\eta = 2.5$. The optimum is the best solved policy, and welfare at the optimum is evaluated there. Because η also governs wage dispersion, each row is a comparative static, not an alternative estimate.

Table D.11: Decomposition of the welfare effect of the minimum wage, 2015 tariffs

	Total	Relative prices and transfers	Labor wedges	Entry and composition	Efficiency (wedges + entry)
Optimum (0.56) vs no minimum wage					
All workers	+0.144	+0.222	+0.431	−0.509	−0.078
Noncollege	+0.190	+0.268	+0.432	−0.511	−0.078
College	−0.341	−0.269	+0.415	−0.487	−0.072
Current policy (1.00) vs no minimum wage					
All workers	−0.650	+0.282	+1.450	−2.382	−0.932
Noncollege	−0.555	+0.381	+1.453	−2.389	−0.936
College	−1.670	−0.782	+1.415	−2.303	−0.888

Notes: Cumulative change from the no-minimum-wage equilibrium, 100 times the change in log consumption-equivalent welfare, following the first-order decomposition in Atkin and Donaldson (2022). Relative prices and transfers collect the worker's own wage, rents, the payroll and tariff rebates, the foreign transfer, and the cost of living. Labor wedges are the employment effects of the gap between the marginal revenue product of labor and the wage, which the minimum wage narrows at constrained firms. Entry and composition collect firm entry, formality status, the productivity composition of firms, job premia, the composition of workers within cells, and self-employment. Each block is invariant to the choice of numeraire, while the individual terms within a block are not, so only blocks are reported. Totals differ slightly from the consumption-equivalent changes in the text because they are log changes. Maximizing the sum of the labor-wedge and entry-and-composition blocks alone for all workers gives 0.35 times the current minimum wage, worth 0.007 percent (every policy from 0.32 to 0.36 is within 0.001 percentage points of it).

Table D.12: Robustness of the welfare-maximizing minimum wage, 2015 tariffs

	Optimal minimum wage	Within 0.001 pp	Worker welfare vs no minimum wage (CEV, %)	
	(multiple of current)	of the best	At the optimum	At the current policy
Headline estimate	0.56	0.54–0.58	+0.144	−0.648
Formal benefits also counted as worker income (re-estimated)	0.54	0.53–0.55	+0.118	−0.676
Migration across regions, average elasticity 1	0.60	0.56–0.61	+0.150	−0.578
Migration across regions, average elasticity 2	0.60	0.59–0.61	+0.155	−0.530
Migration across regions, average elasticity 3	0.61	0.59–0.61	+0.158	−0.495
Earlier solution in steps of 0.05, quadratic vertex				
Headline estimate, same solution	0.570		+0.144	−0.648
Housing stocks estimated jointly, rents targeted, same solution	0.566		+0.146	−0.651

Notes: Each row solves the minimum-wage path under the 2015 tariff schedule, with the minimum wage held fixed relative to average earnings. The first five rows use steps of 0.01 (for migration, steps of 0.01 between 0.30 and 0.80 and of 0.10 elsewhere) and report the best solved policy. The re-estimated model has its own parameters and its own welfare units. With migration, workers choose a region with a logit choice whose average elasticity across regions is 1, 2, or 3, and welfare is the consumption equivalent of expected utility, amenities included. The last two rows come from an earlier solution of both estimates in steps of 0.05 and report the vertex of a quadratic through the best policy and its two neighbors.

Table D.13: The optimal minimum wage under alternative tariff regimes

	Free trade	2015 tariffs	2009 tariffs	High protection
Mean tariff (%)	0	3.54	7.91	12.29
Optimal minimum wage (multiple of current)	0.56	0.56	0.56	0.55
Optimal κ	0.38	0.38	0.38	0.37
Worker welfare relative to no minimum wage, same regime (CEV, %)				
At the optimum	+0.149	+0.144	+0.140	+0.138
Labor-wedge term at 0.56 times the current minimum wage (100 × log points)	+0.430	+0.431	+0.433	+0.434
At the current policy	−0.652	−0.648	−0.643	−0.637
Worker welfare at the optimum relative to 2015 tariffs and the current policy (CEV, %)				
All workers	+0.283	+0.801	+1.354	+1.802
Informality at the current policy	0.657	0.658	0.659	0.660
Informality response to a 10% higher minimum wage (pp)	1.928 to 1.940 across regimes			

Notes: Regional tariffs are import-weighted averages by region of use. The 2009 regime adds to each region's 2015 tariff the 2009–2015 change in applied national sector tariffs, weighted by the region's import composition (mean increase 4.37 percentage points); high protection adds twice that change. The optimum is the best solved policy in steps of 0.01, and welfare at the optimum is evaluated there. The labor-wedge term is the corresponding block of Table D.11, in 100 times log points. The last welfare row uses the 2015 equilibrium at the current minimum wage as the common reference, so it includes the effect of the tariff regime itself.

Table D.14: The optimal minimum wage by wage-setting power and tariff regime

η	Free trade	2015 tariffs	2009 tariffs	High protection
1.5	0.78 (0.78–0.79)	0.79 (0.78–0.79)	0.79 (0.79–0.79)	0.79 (0.79–0.80)
2.0	0.65 (0.64–0.67)	0.64 (0.63–0.67)	0.63 (0.63–0.66)	0.63 (0.63–0.65)
2.5	0.56 (0.55–0.59)	0.56 (0.54–0.58)	0.56 (0.54–0.57)	0.55 (0.54–0.56)
3.0	0.49 (0.48–0.50)	0.49 (0.48–0.50)	0.49 (0.48–0.49)	0.48 (0.47–0.49)
4.0	0.43 (0.39–0.45)	0.42 (0.38–0.45)	0.42 (0.38–0.44)	0.42 (0.38–0.44)
5.0	0.39 (0.34–0.42)	0.39 (0.34–0.42)	0.38 (0.33–0.42)	0.37 (0.33–0.42)

Notes: Best solved minimum wage for all workers, as a multiple of its current level, with the range of policies within 0.001 percentage points of it in parentheses. Each row re-solves the minimum-wage path with the firm-level labor-supply elasticity η changed and all other parameters at their estimates. The minimum wage is held fixed relative to average earnings.

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